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Working towards a comprehensive understanding of HPV and cervical cancer amongst Indigenous women: A qualitative systematic review

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Title: Working towards a comprehensive understanding of HPV and cervical cancer amongst Indigenous women: A qualitative systematic review

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Abstract:

Rationale: Indigenous peoples carry a disproportionate burden of infectious diseases and cancers and are over-represented among the socially disadvantaged of most countries. Human papillomavirus (HPV) is a risk factor and causative agent of cervical, oropharyngeal and other cancers. Recent literature shows evidence of Indigenous populations being at increased risk of HPV infections and its associated cancers.

Objective: This is a qualitative systematic review. The objective of this study was to explore the experiences and barriers Indigenous women face in relation to HPV awareness, knowledge and cervical screening, in order to better understand factors that may mitigate against or facilitate prevention efforts for HPV infection and associated cancers.

Methods: Two investigators independently searched MEDLINE, PubMed, SCOPUS and Web of Science databases using a pre-specified search strategy to identify qualitative studies on narratives of Indigenous women regarding HPV infection awareness, knowledge and cervical screening, across all geographic and income-level settings. Using a 'meta-study' approach, a social ecological model of cervical screening, infection and associated cancer prevention among Indigenous populations was formulated.

Results: Five core themes were identified and formulated within the Social Ecological Model; Intrapersonal factors, Interpersonal factors, Institutional/Organizational factors, Socio-cultural/Community factors and Public Policy. These collectively formed the proposed social ecological model of HPV infection awareness and cervical cancer prevention among Indigenous women. This model has been synthesized by taking into account personal stories of Indigenous women and health care workers, thus offering a more nuanced, organised, structured and culturally sensitive approach to policy translation.

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Conclusion: The social ecological model of HPV infection awareness and cervical cancer prevention among Indigenous women offers a holistic and practical approach for Indigenous health policy makers to address the high risk of Indigenous populations at a global level in experience of both HPV infection and HPV-related cancers.

Keywords: Indigenous women health, Human papillomavirus infections, cervical cancer, Social ecological model, qualitative systematic review, self-sampling strategies

Article Summary:

Highlights:

- This review seeks to better understand facilitators and barriers to HPV and associated cancer prevention, as experienced by Indigenous peoples on an international scale at each level of the social ecological model
- This review presents a social ecological model of HPV infection awareness and cervical cancer prevention among Indigenous women offering a holistic and practical approach for Indigenous health policy makers to address the high risk of Indigenous populations at a global level in experience of both HPV infection and HPV-related cancers.
- The reviews highlights the need for more research which incorporates the perspectives of Indigenous people; which would result in more acceptable and successful health prevention and policy implementation.

Strengths and Limitations:

- This systematic review is the first to address the qualitative aspects of HPV infection and associated cancers among Indigenous women at a global level. The review adhered to all protocols to ensure transparency and legitimacy.

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- Another strength is the summary of community-provided solutions, in the form of personal narratives, to help decrease transmission of HPV infection and to, in turn, prevent HPV-related cancers.
- Our study highlights the continuing impact of trauma at the public policy level, with our findings providing important evidence of the work required to address the resulting disparities across all health indicators that have resulted from this trauma
- Limitations include the decision to not include male participants, as the current statistics show a dramatic increase in the incidence of HPV associated oropharyngeal carcinoma among men.
- The community-identified solutions presented are mainly from one study; this demonstrates the need for community feedback to be embedded within Indigenous health research projects so that future policies can be derived from community suggestions that will have greater resonance and likely acceptance.

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Introduction

Indigenous peoples, as defined by the United Nations (2004), includes all "people with a historical continuity with pre-invasion and pre-colonial societies that developed on their territories, and who consider themselves distinct from other sectors of the societies now prevailing on those territories." Globally, Indigenous peoples experience disproportionate health inequalities, in comparison to non-Indigenous populations, due to a unique history of colonial settlement. This has resulted in environmental dispossession of traditional lands and resources, which has negatively impacted spiritual connections with the land and with each other[1-4]. Indigenous peoples are over-represented among the socially disadvantaged in almost every country, particularly in developed countries[5]. Public policy created by colonial settlers in many countries has been aimed at the assimilation and cultural annihilation of Indigenous peoples, resulting in mass loss of culture, language and community. Examples of assimilation policies include the Residential School System in Canada[6], the Stolen Generations of Australia[7] and racial amalgamation in New Zealand[8]. Adelson's (2005) notion of "the embodiment of inequity," and similar works[9,10] have demanded attention be paid to the ways in which the impacts of colonial legacies, marginalisation and discrimination manifest in Indigenous peoples' health outcomes[10]. Previous works[11,12] have explored the relationship between colonization and the intersectional nature of social determinants of health, such as race, demonstrating that health care patterns are rooted in historic economic, social and political circumstances and power relations[11]. This literature provides insight into the commonality of high-risk health profiles experienced by Indigenous peoples around the world and has created awareness of the need to take special measures to protect the rights of Indigenous peoples[13,14].

Human papillomavirus (HPV) infection and associated cancers have gained significant public and research attention in recent decades. WHO has launched a three step program for elimination of cervical cancer by 2050[15], and the International Society for Papillomavirus

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Research has demonstrated support[16] for the program with a call for equity in elimination strategies, particularly reinforcing best practices for Indigenous populations. While screening tools, such as the Papanicolaou test (Pap test), have been successful in reducing cervical cancer overall, and HPV vaccination programs are expected to reduce rates of HPV infection and all HPV-related cancers, Indigenous communities continue to experience significantly higher rates of cervical cancer and oncogenic HPV infection compared with non-Indigenous populations[17-21]. This discrepancy in cancer is related to Indigenous women being less likely to have been screened[22-24] and compounded by implications of colonial legacies[1-4]. HPV-related cancers are mediated by risk factors such as sexual behaviour, tobacco use and screening participation, but these relevant health behaviours of affected Indigenous Peoples co-exist within multidimensional contexts[25,26]. Bronfenbrenner's (1974) Social Ecological Model provides a theoretical framework in which to understand the micro and macro dimensions of a phenomenon, by simultaneously considering related political, social, physical and economic environments[27,28]. Many health promotion efforts focus on individual choice, ignoring the social context of health-related behaviours. The social ecological model acknowledges the importance of the social environmental impacts on human health, beyond the ideology of individual responsibility[28]. It may be that current HPV prevention programs are not successfully recognising and/or incorporating strategies targeted at factors beyond the individual. This is a particular shortcoming among initiatives targeting Indigenous communities who, at a global level, are much more collectivistic and group-oriented than their non-Indigenous peers[17].

This review therefore seeks to better understand facilitators and barriers to HPV and associated cancer prevention, as experienced by Indigenous peoples on an international scale at each level of the social ecological model[28]. Combining existing literature within the framework of the model acknowledges the significance of Indigenous culture and its interaction with

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socioeconomic conditions[17, 28]. The objective of this systematic review was to synthesise qualitative perspectives and experiences of Indigenous women and health care workers in the context of HPV infection, associated cervical cancer and screening methods to establish a foundation that might better facilitate targeted, holistic, and culturally relevant prevention strategies.

Materials and methods

This systematic review has been registered in PROSPERO (CRD42020207643) and the Joanna Briggs Systematic Reviews register. A prior search of the PROSPERO register revealed no similar studies. Both the PRISMA guidelines[29] and the Enhancing Transparency in Reporting the Synthesis of Qualitative Research (ENTREQ) statement (Table 1) were followed in the conduct of this systematic review.

Positionality

Acknowledging that personal experiences and beliefs heavily influence research perspectives, it is critical for researchers to self-situate. This review is a result of the desire to prioritize individual voices and stories of Indigenous women. After hearing first-hand accounts of various health disparities experienced by Indigenous women in South Australia, while conducting field work for a different HPV project, the primary reviewers (S.S and B.P) discussed the importance of the person behind a statistic and the desire to synthesise existing knowledge in HPV literature was established in order to identify future research steps. While both non-Indigenous researchers, S.S is an oral pathologist with experience working with Indigenous populations in Australia and B.P has qualitative experience with community-engaged scholarship in the context of Indigenous health in Canada and Australia. The supporting research team consists of Indigenous and non-Indigenous scholars with vast experience in the realm of Indigenous health research.

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Identifying studies for inclusion

The reviewers used a pre-established search strategy[30], which involved using terms (and their edited variants) describing the population of interest, the phenomenon being researched, as well as the included study designs (Supplementary file 1). Two investigators (S.S and B.P) independently screened the literature for eligible articles using MEDLINE, PubMed, SCOPUS and Web of Science databases from inception until June 2020. For example, the search strategy used for PubMed Database was as follows: First Nation/First Nations/Pacific Islander/Pacific Islanders/Torres Strait Islander/Torres Strait* Islanders/ Aborigin*/ Alaska*/ Aleut*/ Amerind*/ American Indian/ Arctic/ Aymara/ Bushmen/ Chukchi/ Chukotka*/ Circumpolar/ Eskimo*/ Greenland*/ Hmong/ Indian*/ Indigen*/ Inuit*/ Inupiaq/ Inupiat/ Khanty/ Maori*/ Mapuche/ Metis/ Native*/ Navaho*/ Navajo*/ Nenets/ Quechua/ Saami/ Sami/ Samoan*/ Siberia*/ Skolt/ Tribal/ Tribe*/ Xingu*/ Yup'ik/ Yupik/ Zuni/"African continental ancestry group"/"African continental ancestry group"/ "Asian continental ancestry group"/"Health Services, Indigenous"/"Oceanic ancestry group"/"arctic regions"/"ethnic groups", "HPV", "Human Papillomavirus", "Papillomavirus", "HPV 18", "HPV*", "Qualitative", "awareness", "barriers", "screening". The search was tailored as per the design of individual databases.

In the search for published studies, the reviewers made use of facilities where the option was given to run 'related' searches, where similar studies are automatically identified. The bibliography of each article was scanned manually for possible additions to the search. The titles and abstracts were screened by both reviewers independently to assess eligibility, with those considered relevant by either investigator advancing to a full-text review. The initial search prioritized first-hand experiences, however, a number of studies included perspectives from health care workers that provided valuable insights into patient experiences. The decision was made to include these studies. The investigator pair fully screened articles to identify studies that fulfilled the following criteria:

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- The study focused on the experiences of women and/or health care workers of Indigenous identity
- Findings contained personal illustrations or first-person accounts of HPV knowledge and cervical cancer screening
- The study was qualitative or mixed methods (with clear qualitative examples)
- Where cancer was the phenomenon of interest, studies only focused on cancers associated with HPV infection (mostly cervical cancer, but did not exclude on basis of oropharyngeal cancer)
- The study was available in English
- The study was available in hardcopy or in downloadable form
- The study was published prior to June 30, 2020

Exclusion criteria

- Based only on HPV vaccination
- Quantitative only studies
- Studies that included narratives from males only

While efforts were made to decrease publication bias, the reviewers recognize that limiting the search to the English language could result in loss of data in other native languages. Additionally, the inclusion of all grey literature could have provided additional findings for the study and decreased possible impacts of publication bias.

Critical Appraisal

There are various validated tools for appraisal of studies; this review employed the JBI (Joanna Briggs Institute) System for the Unified Management, Assessment and Review of Information

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(SUMARI) critical appraisal tool (Supplementary file 2). This tool includes questions regarding sample sizes, locations, methods used, techniques utilised and themes derived.

Data Extraction and Synthesis

Data were extracted in two phases. For the first phase, the reviewers utilized concepts of thematic analysis and comprehensively identified findings from each of the included studies. These findings were extracted via a piloted extraction form and then uploaded to JBI SUMARI. The second phase utilised the JBI data extraction tool for all studies. Each reviewer independently scored the findings within JBI SUMARI as "Credible", "Not Supported" and "Unequivocal"; the score for each finding was based on inter-reviewer agreement. The synthesis of findings was done manually by reviewers, which included writing all findings on a white board and identifying common phrases, themes and concepts. Common themes were grouped, with connections between other themes explored in the context of the lived experience of HPV prevention for an Indigenous woman. These categories were then transferred from the white board to the JBI SUMARI tool and each individual finding was placed within the appropriate category.

Social Ecological Model

Upon reviewing the categories and creating synthesized themes, it was apparent that findings were present at varying societal levels that are comprehensively identified in the social ecological model. The overarching themes were therefore generated to fit the social ecological model including: (1) intrapersonal factors; (2) interpersonal factors; (3) institutional and organizational factors; (4) socio-cultural and community factors; (5) public policy[27]. This was additional to the solutions provided by the community which included two main themes: self-sampling and community interventions.

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Results

The literature search returned 1377 records, of which 963 were duplicates and consequently excluded. After being screened against the pre-specified inclusion and exclusion criteria, 24 studies were shortlisted. Of the 24 potentially eligible studies reviewed in full, 10 fully satisfied the inclusion criteria (Figure 1). The average inter-reviewer score was 9.60, indicating a very high level of agreement between reviewers (Table 2). Although some studies (n = 4) did not perform well according to the criteria set by the Qualitative Assessment and Review Instrument of method (Supplementary file 3) of quality appraisal[31], the reviewers felt that studies should not be excluded on the basis of being 'less rigorous' according to one set of criteria. On the contrary, such studies were included precisely because of the richness of textual narrative provided. Other studies had minor methodological issues pertaining primarily to whether the researchers had adopted a reflexive stance by examining their own role in formulating and conducting the research[32].

After the literature search and article appraisal, ten studies were deemed eligible for inclusion in this review. Studies were conducted in four countries: with Māori communities in New Zealand, Aboriginal and Torres Strait Islander population in Australia, Inuit and First Nations communities in Canada, as well as Mam and Huichol communities in Mexico (Table 3). Reviewers identified 118 findings (Supplementary file 4) from the included ten studies, a chart with each finding, associated quotes and score was generated (Table 4). This collaborative process resulted in 36 distinct categories, which provided an appropriate template to meta-aggregate the findings. Five main themes were synthesized by the end of this process, with reviewers in complete agreement. After completion of this process, the JBI SUMARI tool generated a meta-aggregation flowchart of all findings, categories and themes.

Intrapersonal Factors

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A theme strongly reflected in all included studies was intrapersonal factors. Findings revealed marginal awareness of HPV and cervical cancer, including a lack of knowledge regarding HPV symptoms and disease progression. The lack of awareness of disease progression was highlighted in more than one paper[17,33-35]. A deficiency of empathetic services was identified as a potential barrier to seeking timely health support[34]. Feelings of embarrassment and shyness were the most commonly stated experiences when visiting a doctor for screening. One study suggested that younger women feel shy, especially individuals with a history of sexual abuse[36], while another suggested that older women are more hesitant to pursue screening options due to feeling shy[35]. Māori women have referred to this feeling as whakamā and that its tapu (a taboo) to let go of your body autonomy[34]. Feelings of vulnerability were common, and experiences of cervical screening were described as vulnerable[17]:

> "*I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this .... There's got to be a better, kinder, gentler, more humane way to do it."*

Associated with feelings of shyness were previous negative and painful healthcare experiences, which may serve to demotivate women from regular screenings. Specifically, concerns were expressed around sterilization, empathetic procedures and rushed appointments[33,37]. Another common intrapersonal factor was fear of prognosis; for example, some participants identified this as a barrier because they feared a cancer diagnosis[17,18,37]. However, other participants found fear to be a motivator, and the desire to avoid suffering or death from this disease ensured prioritization of prevention[18]. The financial burden associated with the Pap test and other costs including transport or childcare were substantial barriers among low income families[34,37]. Some studies suggested that health care as a whole was not identified as a high priority for many individuals as stated by health care workers[34]:

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> *"One in seven had not screened in ten years or more and one in six were unsure how long it had been since their last cervical screen."*

Alternatively, one study suggested a deep interest and inclination towards health, where family and personal wellness were identified as motivators[38]:

> *"It's about listening to my body. It's about trying to keep it healthy for my children. I want to live as long as I can to bring them up to see them. Not only that, I want to feel good too. And I have had friends who have been diagnosed and that motivates me also to do it. But all in all I do it for myself and my family."*

Interpersonal Processes

Four of the included studies described factors at the interpersonal level[17,18,20,38]. Family relations were discussed as a way to facilitate a generational shift from an environment of sexual taboo to one of love and support[17,18,20,38]. Role-modelling by older family members, specifically mothers, was associated with preventive health care practices[20]. One mother explained the importance of talking openly about sex with her children[17]:

> *"Like my mother, you know, was the kind of woman that… when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your personal, private, so when it happened, I had absolutely no knowledge what was going on in my body… And with my daughter I didn't want her to have that feeling, so I mean, of course I changed and I explained everything."*

Intergenerational communication was also identified as important for educating children and as a critical aspect of cultural traditions[18]:

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> *"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children and out of respect… he's getting that message across [in his sweat lodge ceremonies with men]."*

Institutional/Organisational Factors

All of the included studies discussed factors at the institutional/organisational level that impacted cervical screening. Some items such as cultural awareness among health care workers, integrated services, female health care workers for screening and positive health care experiences were discussed as facilitators for screening. However, other institutional aspects such as negative health care experiences, language barriers, male health care workers for screening, lack of results, privacy concerns and screening availability were identified as barriers to screening. Both participants and health care workers suggested integrated services (Pap smears combined with other services) as a way to eliminate multiple clinic visits[20,34,37]. Offering services in local languages, eliminating medical jargon[20,36] and incorporating holistic healthcare approaches were identified as potential facilitators. Geographically remote communities described difficulties associated with building trusting relationships at health care centres, due to a high turnover of health care workers[17]. Discussions of the gender of the healthcare worker performing screening services (such as a Pap) elicited strong responses, with male health care workers identified as a barrier[18,35] and female health care workers as a facilitator[20,37,38]. Especially in smaller communities, lack of privacy was a substantial concern that impacted decisions to access healthcare services[17,37], as evident in the following quote[17]:

> *"Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk."*

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Lack of communication about results and lack of screening availability[33,37,38] were highlighted as obstacles to accessing services; many participants expressed a desire to receive results regardless of negative or positive outcomes. Positive health care experiences enticed regular utilisation of services[18,20], while negative health care experiences impeded the use of services[17,18,34]. Specifically, participants felt that often procedures were not fully explained[17,18], which competed with a desire for body autonomy. The frustration of this experience was felt by many participants, as shown in the following quotation[18]:

> *"I still don't know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don't know! Nobody explained anything to me because, maybe because I didn't ask. Maybe they thought that I understood everything that was going on, but I didn't."*

Socio-cultural/Community

This theme was separated into five categories, with key quotes arising from six of the ten included studies. Gender imbalances highlighted male dominance in communities, and the consequential inequalities based on gender. Male tendencies towards promiscuity were considered to be a contributing factor for bringing HPV into the home, which affected screening practices due to the associated stigma of an unfaithful partner[37]. Religious and cultural beliefs prevented many women from attending cervical screening and/or treatment. Women shared their experiences of violence and a general disregard for their health by their male partners[33]:

> *"The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn't even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn't believe that."*

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One healthcare worker indicated an urgent need for education to alleviate the negative concerns and sense of humiliation associated with cervical screening[37]. Health care workers in one study described patient requests to not document diagnoses and feelings of nervousness around potential breaches of confidentiality in the community[17]:

> "*Actually in the last 6 months that I've been there, I've had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the community … they're afraid of the stigma surrounding that so and a lot of times too I wasn't able to do any [cervical] screening"*

Participants frequently described a fear of disappointing community and family; that sense of shame and humiliation was a prime concern and barrier to regular screening[36]. Discussing the isolating effects of a cervical cancer diagnosis, one study reflected on how the diagnosis of a single person is the suffering of not one individual but an entire community[17].

Public Policy

While participants in the included studies tended to focus on discussions of institutional/organisational and intrapersonal factors, two of the Canadian studies[17,18] highlighted the impact of traumatic experiences on accessing screening services. Specifically, the sexual abuse experienced at government-mandated Residential Schools in Canada has created an environment of sexual taboo where health concerns are not discussed, and services are not sought. This sentiment is illustrated by the following quotation[17]:

> *"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past."*

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A lack of community connection and relationships was discussed as a consequence of the sexual abuse experienced in residential schools, and substance-dependence was identified as a coping mechanism[17]. Intergenerational trauma and child removal were also mentioned as barriers to accessing health services, both of which are a result of government-mandated public policies[39].

Solutions

Many participants identified solutions to the aforementioned obstacles which were categorised at each level of the social ecological model. Self-sampling[17,20,33,34] and community-level initiatives[20] were discussed as potential ways forward to address some of the concerns and common experiences shared across all included studies. The idea of self-sampling was associated with increased convenience[17,20,34,37], privacy[37] and autonomy[17,37] as well as reduced embarrassment[17,33] and pain[33]**.** The need for education prior to self-sampling was stressed as a means to ensure a sound understanding and completion of tests[34,37]. The potential increase in perceived bodily autonomy increased the acceptance of self-sampling[37]:

> *"I really strongly believe that… because I was the one that was doing it [self-sampling], I was the one that was in control… and this way it gave me the ability to do it myself and I got all the results, they were fine;… it was also self-empowering, great, I like that."*

Maar et al. (2016), discussed a variety of community-level tactics to increase engagement and a desire to seek out and use health care services[23]. Informal community communication at events as well as by word of mouth was seen as valuable. Integrating HPV screening with female-oriented celebrations and workshops was discussed as a way to support screening behaviours. Use of school education, survivor stories, social media, incentives and humour

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were all identified as potential pathways to increasing screening participation. One health care worker described how she used humour in an educational workshop[23]:

> *"We didn't specifically talk about HPV, but we talked about STDs. Yeah I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: 'Are you bringing bananas next week?' You know, so it was a fun thing."*

Social Ecological Model (Figure 2)

Two studies explored experiences relevant on all five levels of the social ecological model. All included studies addressed issues on at least two levels of the model. Five studies incorporated themes or frameworks that focused exclusively on two ecological levels. All studies discussed factors at the institutional and intrapersonal level, while two studies focused primarily on themes related to public policies. The distribution of themes proposed at each ecological level of the model, according to all included studies is included in table 5.

Discussion

The aim of this systematic review was to collate the experiences of Indigenous women and health care workers with regards to HPV, associated cancers and screening methods. A secondary aim was to generate an understanding to inform future HPV prevention strategies. By mapping findings within the social ecological model levels, this review highlights gaps in current HPV practices and identifies areas to be addressed based on first-hand experiences.

The first level of the social ecological model included factors at an intrapersonal level. Lack of knowledge and awareness of HPV infection and its risks persisted in the context of other priorities taking precedence over routine health checks. Two prominent barriers were expressed in all papers: financial burdens and feelings of embarrassment. These factors affect an individual's personal decision to seek screening or preventive health care services.

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The level of interpersonal factors revolved around two main concepts of family relations and intergenerational communication. The communication within one family unit, from parents to children and even the openness of children with their parents, around discussions involving sexual behaviour and practices was taboo. A greater emphasis on, and openness to, education across the life course could increase the acceptability of screening. A lack of intergenerational communication was identified as a potential predisposing factor to higher risk sexual behaviours.

The third level of the social ecological model, institutional/organizational, had nine synthesized findings that demonstrated how positive and negative healthcare experiences impact current perceptions towards screening and HPV infection. There was an emphasis on lack of empathy by some healthcare workers (doctors, nurses etc.) while screening and dealing with sensitive procedures. A preference of being attended to by female health care workers, especially for cervical smears and examination, was observed. Worry due to lack of results after the tests was common, and often was interpreted as bad news which had an impact on participant's daily life and mental health.

The fourth level, socio-cultural/community, includes factors that affect HPV-associated cancer prevention and awareness at a community level. This level included discrimination on the basis of gender and highlights male dominance that exists in some communities. This level also examined the stigma and shame of being diagnosed with a sexually transmitted infection and a common fear of being seen at the doctor's office. The experience of isolation faced after diagnosis/treatment is also a common barrier that impedes regular screening.

The results of the systematic review found trauma as the single synthesized finding at the social ecological model level of public policy. Personal quotes revealed the impacts of sexual abuse, physical abuse, drug abuse and traumatic ancestral experiences which continue to have an

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intergenerational impact across families and communities. Findings revealed a general mistrust towards the health system and health care workers. These findings highlight the urgent need to begin building trust between Indigenous communities and local health services in the provision of culturally safe care. Increased trust could help increase the acceptability of current prevention programs and treatment solutions.

In addition to the themes of the social ecological model, the theme of self-sampling was prominent in most papers. Self-sampling is a solution which may provide more control, independence, privacy and convenience, and in turn increase screening participation. Community-identified solutions included the use of local survivor stories, which were thought to provide strength and courage to other community members to attend regular cervical cancer screenings and to not fear the diagnosis or to overcome the fear by taking preventative steps. The use of humour and social media were both innovative and socially acceptable solutions for creating awareness and educating community members with contemporary knowledge of HPV infection and its effects.

Strengths and limitations

To the best of our knowledge, this systematic review is the first to address the qualitative aspects of HPV infection and associated cancers among Indigenous women at a global level. The review adhered to all protocols to ensure transparency and legitimacy. Another strength is the summary of community-provided solutions, in the form of personal narratives, to help decrease transmission of HPV infection and to, in turn, prevent HPV-related cancers. An innovative strength of this study is the inclusion of the social ecological model as a theoretical framework. This model enables conceptualisation of a more holistic approach to HPV prevention for Indigenous women at a global level. In accordance with other studies, our study highlighted the continuing impact of trauma at the public policy level, with our findings

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providing important evidence of the work required to address the resulting disparities across all health indicators that have resulted from this trauma[39-41].

Limitations include the decision to not include male participants, as the current statistics show a dramatic increase in the incidence of HPV associated oropharyngeal carcinoma among men[42]. Additionally, the majority of articles are from Canada. This highlights the need for more research focusing on Indigenous perspectives and experiences regarding HPV screening and diagnosis in other countries. The community-identified solutions presented are mainly from one study; this demonstrates the need for community feedback to be embedded within Indigenous health research projects so that future policies can be derived from community suggestions that will have greater resonance and likely acceptance.

Conclusion

Qualitative systematic reviews are increasing in the literature. They have particular utility in informing policy decisions, as the success of any new policy or intervention depends upon its acceptance of, and sustainability in, a given population. While quantitative studies provide clarity with respect to disease spread or burden of a health condition, qualitative studies play a significant role in the management and generation of possible solutions to that health condition/disease. The social ecological model has a structured, multi-level approach with sociological, behavioural and individual aspects. Future research needs to expand the geographical scope of the current work, beyond Canada, and integrate community-identified solutions. Policies around HPV screening and diagnosis for Indigenous communities, globally, need to prioritize a holistic approach to health care, addressing barriers and facilitators at each level of the social ecological model, and prioritising specific community needs.

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Author Contribution statement:

SS, BP and JH conceived the presented idea. SS, BP, XJ, LJ developed the theory and performed the data collection. MS, KC, GG verified the analytical methods. JH encouraged SS and BP to explore the cultural limitations and beliefs leading to inaccessibility of cervical cancer screening. MS, KC, GG and LJ supervised the work. SS and BP primarily wrote the manuscript under the supervision of GG, MS, KC, LJ and XJ. All the authors discussed the results and contributed to the final manuscript.

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Figure Legends:

Figure 1: PRISMA flowchart for included studies

Figure 2: Socio-Ecological Model

Table Legends:

Table 1: ENTREQ Checklist

Table 2: Inter-reviewer reliability table

Table 3: Characteristics of Included Studies

Table 4: Illustrations from all the included studies arranged according to all the synthesized categories under each level of the Socio-Ecological Model

Table 5: Social ecological model distribution of studies

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Table 1: ENTREQ Checklist

Item	Description	Reported on Page #
Aim	Awareness of- and barriers to- human papillomavirus infection screening amongst Indigenous women globally	5
Synthesis methodology	Thematic content analysis guided initial data extraction for synthesis, and the Social Ecological Model provided a theoretical framework to understand synthesised findings	5-8
Approach to searching	Pre-established search strategy which involved using terms describing the population of interest, the phenomenon we are researching as well as study designs to be included	6
Inclusion criteria	*Inclusion*: Studies from inception till June 2020 were included in the study. Research based on the experiences of women of Indigenous identity or health care worker of Indigenous identity, personal illustrations of HPV knowledge and cervical cancer, first-person accounts; topics focused on cancers associated with HPV infection; qualitative and/or narrative, or mixed methodology; available in English; accessible via hardcopy or download; peer-reviewed	7

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	Exclusion: research based only on vaccination for HPV; non-Indigenous participants; only Indigenous male narratives; quantitative methodology	
Data sources	MEDLINE, PubMed, SCOPUS, and Web of Science databases; each search tailored per the design of individual database. In our search for published studies, we made use of facilities when given to run 'related' searches and the bibliography of each article was manually scanned for possible additions to the study	7
Electronic Search Strategy	Terms utilised for literature search included: 'HPV' 'Cervical cancer' 'Indigenous' 'female' 'narrative' 'story' 'qualitative' 'mixed methods'	6
Study Screening methods	Two independent researchers screened studies for inclusion in the qualitative systematic review. Titles were first reviewed, then abstracts and those considered relevant by either investigator advanced to full text review.	7
Study characteristic	See Table 3	Table 3
Study selection results	1377 records were returned from initial search, 963 were excluded due to duplication, 24 shortlisted, 11 studies fully satisfied inclusion criteria. After appraisal, one article was removed due to lack of illustrations. 10 papers were included	Figure 1
Rationale for appraisal	Utilizing JBI SUMARI software, articles were appraised according to the CASP (2013) method of quality appraisal.	S2 and S3
Appraisal Items	See S2 and S3	S2 and S3

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Appraisal Process	Appraisal was conducted independently by both reviewers and then findings were discussed, and consensus was required before moving forward.	8
Appraisal Results	One article was excluded after the appraisal because it did not satisfy inclusion criteria of personal illustrations	8
Data extraction	All text under headings “Results” and “Conclusions,” as well as all findings under the heading “Discussion” were analysed. Data was manually extracted with highlighters from printed versions of appraised articles and then imputed into the JBI SUMARI software.	Table 4, Table 3
Software	JBI SUMARI	7
Number of Reviewers	Two reviewers independently reviewed articles and extracted data. Findings were then compared, discussed and compiled.	7
Coding	Data was coded from selected articles, going line by line to search for concepts and considering the author-prescribed themes.	8
Study Comparison	All findings were individually highlighted and written on a white board and then connections were made between findings and categories were created based on similarities within and across extracted data.	8
Derivation of themes	The process of deriving themes was abductive.	-
Quotations	Table 4	Page 9-13 Table 4
Synthesis output	Discussion and Figure 2	Pages 13-15 Table 4, figure 2

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Table 2: Inter-reviewer reliability table

Study	Number of questions in agreement	Number of questions in disagreement	Score
Adcock A, 2019	10	0	10
Allen-Leigh B, 2017	9	1	9
Butler TL, 2020	10	0	10
Cerigo H, 2012	10	0	10
Henderson R, 2018	10	0	10
Maar N, 2016	10	0	10
O'Brien B, 2009	10	0	10
Tratt E, 2020	9	1	9
Wakewich P, 2015	9	1	9
Zehbe, 2017	9	1	9
Mean			**9.60**

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680 **Table 3: Characteristics of Included Studies**

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Study	Methods for data collection and analysis	Country	Phenomena of interest	Setting/context/ culture	Participant characteristics and sample size
Cerigo H et al. 2012.	Mixed methods and focused group interviews Thematic content analysis using NVivo software	Canada	Attitudes and experiences about cervical cancer Pap smear and HPV vaccine	Inuit women in Nunavik, Quebec	Participant characteristics: women, 31-55years Sample size: 6
Henderson R et al. 2018.	Sharing circles (6) Analysis: NVivo	Canada	Identifying and validating known barriers and supports to HPV vaccination among First Nations people in Alberta	First Nations community leaders from Alberta	Participant Characteristics: 21 females, 3 males, all First Nations Elders, Health service directors and Community leaders Sample Size: 24
O'Brien B et al. 2009.	Ethnography, Interviews Analysis: Themes and relationships among transcriptions and researcher perspectives were identified	Canada	Exploration of attitudes of beliefs of First Nations Cree women living on reserve to gain insights into how cervical screening can be better used	Community based health centres and private locations , culture : Cree First Nations	Participant Characteristics: All women, Cree First Nation, Sample size: 8
Tratt E et al. 2020.	Fuzzy Cognitive Mapping (Visual Mapping) mapped and weighted participant knowledge to illustrate the impact of one on the other	Canada	Explore the possible implementation of HPV self sampling	Inuit women in Nunavik Quebec	Participant characteristics: All women over age of 25 years Sample size: 27
Adcock A. 2019.	: Mixed methods approach, Face to face Hui, recruitment by communtiy based	New Zealand	Potential acceptability of HPV self-sampling among Maori women who were never or under screened and also a survey of the HPV self	Maori women and health care providers (both Maori and Non-Maori)	Participant characteristics: Sample size: 106 (Hui), 397 (surveys)

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	researchers Surveys via Qualtrics Hui (focus groups, interviews)was recorded, transcribed then analyzed (Thematic analysis) by NVivo software.		testing among Health care providers.		
Maar N et al. 2016.	Participatory action Research, mixed methods approach, Semi-structured Interviews, Analysis: Thematic analysis	Canada	Understand the effective ways to reach First Nations women with screening information, Potential or promising strategies to help motivate women to participate in cervical cancer screening and address the special issues for First Nations women with respect to cervical cancer screening.	First Nations people in Northwest Ontario (Lake Superior or around Lake Nipigon)	Participant characteristics: First Nations people in Northwest Ontario (Lake Superior or around Lake Nipigon) Sample size: 70-840 members
Allen-Leigh B et al. 2017.	9 focus groups(5-11 per group), 29 interviews, Audio recordings, transcriptions and field notes taken Analysis: two stage coding process (first level - descriptive coding; second stage - pattern codes)	Mexico (South America)	Study the barriers to use of self-sampled HPV testing and cytology among low-income Indigenous women residing in rural areas of Mexico	Mam women in Chiapas, Nahuatl women in Puebla and Huichol Women in Jalisco	Participant Characteristics: 20 years and above, 3 ethnic groups (Mam women in Chiapas, Nahuatl women in Puebla and Huichol Women in Jalisco) Sample size:
Wakewich P et al. 2015.	Participatory action research framework, purpose sampling, Traditional 'talking' circles, Interviews Analysis: Open coding , NVivo software	Canada	Develop culturally sensitive approaches to cervical cancer screening and to explore the feasibility of self-sampling for HPV as an alternative screening modality to Pap cytology	First Nations women in Northwestern Ontario	Participant Characteristics: Health care workers and Elders Sample Size: 16 interviews of healthcare workers and 8 community focus groups (total 69 women)

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Zehbe et al. 2017.	Data Collection: Field Trips, Teleconferences, in depth interviews, 8 focus groups, Semi-structured grounded theory approach Analysis: Nvivo 9 (QSR International)	Canada	Experiences with cancer in their community, followed by community concerns about cancer, knowledge and awareness of cervical cancer	First Nations Women, Northweat Ontario	Participant Characteristics: Health care workers - 20's to 60's age groups, Fifteen HCPs were female and one was male. Community participants - up to 70 years age, Women from First Nations Identity Of these, 10 women self-identified as First Nations. Sample size: 16 (interviews), 8 focus groups = total 69 women
Butler TL AK. 2020.	Convenience sampling Yarning (followed a question guide), interpreting, listening, sharing, health questionnaire Audio recordings, transcriptions (verbatim) and field notes Thematic analysis (NVivo)	Australia	Barriers and enablers to cervical cancer screening	Aboriginal Community Controlled Health Services or Government-run clinics that serve a large proportion of Indigenous clients - Queensland, New South Wales, Northern Territory	50 women (44 Aboriginal, 6 Torres Strait Islander), ages 25-60 6-14 women from 5 different Public Health Care Centres

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690 Table 4: Illustrations from all the included studies arranged according to all the synthesized
691 categories under each level of the Socio-Ecological Model

Social Ecological Model level	Synthesized Finding	Illustration
Public Policy	Trauma	"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past …. I will be here for a long time and whenever you need to see me, to come see me, so that even just that little thing and then when they do come I do see them, hopefully that trust builds up and I think that's a big piece with the First Nations" (p. 374). (*Wakewich et al., 2015*)
		"I think one of the biggest issues um, [short pause] the barriers that prevents people from going to and maybe it's cause it's taboo is because they've been sexually abused" (p. 374). (*Wakewich et al., 2015*)
		"I guess they were told about some kind of vaccine or something, like years ago, and it was something just, to try to get rid of the Native people" (p. 374). (*Wakewich et al., 2015*)
		[Residential schools] "Opened a whole can of worms; when all the illness and substance abuse went on a skyrocket" (p. 96). (*Henderson et al., 2018*)
		Speaking of a sister in the end-stage of cervical cancer, one middle-aged participant attributed the illness to chronic sexual abuse: "There were a couple times she had STIs [sexually transmitted infections] and didn't really know, may have been HPV or may have been another" (p. 96). (*Henderson et al., 2018*)
		In both sharing circles, participants argued that residential school litigation broke students' capacity for intimacy throughout their lives, playing out in self-destructive coping strategies (e.g., substance depen- dence, early sexualization) and increased risk of victimization (p. 96). (*Henderson et al., 2018*)
		"It must be the pesticides that are being sprayed on our cultivated land...You can see our area and there is a layer of dust in spring and harvest" (p. 96). (*Henderson et al., 2018*)
		"Our people are starving for affection, support, respect and love" (p. 96). (*Henderson et al., 2018*)
		"The nuns make sure to remind you no matter what you do, you're going to hell" (p. 96). (*Henderson et al., 2018*)
		"We were taught to be quiet about our private parts...a lot of sexual abuse went on, and spiritual,

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		physical, you name it" (p. 96). (*Henderson et al., 2018*)
		One speaker brought these lingering effects of residential schools together by describing a shift in her society from monogamous relation- ships to relaxed sexual boundaries, noting that with her own daughters she "didn't lay down the law like our parents did" (p. 96). (*Henderson et al., 2018*)
		"She got an STI because she was taken advantage of" (SC2) by someone other than her partner. The mother connected her daughter's exposure to loss of identity and heavy drinking at parties in her community. Lowered inhibitions brought on by alcohol, and illicit drug use were seen by several other speakers to have become normalized among younger generations, who "see the suffering of their parents and grandparents, and they are running to the substance abuse" (p. 96). (*Henderson et al., 2018*)
		In both circles, speakers expressed compassion for younger generations today: "It was a lot easier in my time because we would go to ceremonies...but today, our grandchildren are exposed to drugs, alcohol, everything" (p. 96). (*Henderson et al., 2018*)
		He observed that such statistics reflect disrupted connections between partners, between Elders and youth, as well as with nature and spirituality: "We are not islands, we need to be connected to people and that is what is missing" (p. 96). (*Henderson et al., 2018*)
		Therefore, the burden of HPV in FN people is rooted, at least in part, in efforts by people across the lifespan to cope with the violent disruption of family and community connectedness, as well as connection to land and spirituality. This burden is manifested not only in risk-taking behaviour, but also in avoidance of wider health systems: "I think First Nations don't get checked when they are supposed to, to be honest. They just wait until it is too late to help them" (p. 97). (*Henderson et al., 2018*)
		Coming from a communal society, another participant observed that today many struggle "to fit into an individualistic society, and we don't fit" (p. 98). (*Henderson et al., 2018*)
		[When discussing child welfare] "A lot of those kids when they come back to the reserve they are very angry, there are a lot of things that happened [to them] that we don't know about" (p. 97). (*Henderson et al., 2018*)
		One health director who was herself FN had not

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		approved the HPV vaccine for her own daughter, believing at the time that vaccines are perhaps "not natural, that they are more chemicals given by the government to hurt us" (p. 98). (*Henderson et al., 2018*)
		"If I'm drinking and drugging, then I'm going to feel guilty about taking my children to the clinic to help them live a healthier life" (p. 98). (*Henderson et al., 2018*)
Socio-Cultural/Community	Gender Imbalances	"The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn't even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn't believe that" (p. 4). (*Allen-Leigh et al., 2017*)
		"I decided it [to perform the HPV test] myself, alone. I don't ask anyone's permission. ... How am I going to ask him if he [her husband] wants it or not? It's not for him, it's for me" (p. 4). (*Allen-Leigh et al., 2017*)
		"Because it's a relationship thing" that implied that someone had 'cheated' (p. 7). (*Zehbe et al., 2017*)
		"… I know only women get cervical cancer but … they've helping it along in the same sense where they've passing it back and forth" (p. 7). (*Zehbe et al., 2017*)
	STI Stigma	Focus group participants were concerned that any publication of information about HPV rates on reserve could have negative effects, contributing to the stereotype that 'all native people have HPV' (FG6). As one HCP 38ummarized: 'A lot more education has to be done about HPV and cervical cancer together' to alleviate these concerns (p. 7). (*Zehbe et al., 2017*)
		"Definitely there's always going to be a stigma about any kind of thing that's an STD [STI], because people think it's dirty or whatever. Like I've had people come in, like, even the girl I just saw with genital warts and she just couldn't believe it and couldn't like fathom who she would have got it from because everyone she's been with has only been with her. Right?" (p. 373). (*Wakewich et al., 2015*)
		Actually in the last 6 months that I've been there, I've had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the

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		community … they're afraid of the stigma surrounding that so and a lot of times too I wasn't able to do any [cervical] screening" (p. 373). (*Wakewich et al., 2015*)
		"Health centers on reserves, don't want to spread it out too much how many people have HPV. I know how the stereotype works if we pass on the information in non-native communities they will say all native people have HPV" (p. 373). (*Wakewich et al., 2015*)
	Isolating Effects	It took time for one family to realize that it was "not just happening to them, it's the whole community" (p. 96). (*Henderson et al., 2018*)
	Community Support	"Yeah, just yarning with all the women, you know, getting together, having a cuppa. I think someone just bringing it up, approaching it in a way where it's just a thing where you're sitting there having a yarn about–and not feeling shame." (p.9) (*Butler et al., 2020*)
Institutional/Organisational	Health care worker Cultural Awareness	Cultural awareness of healthcare workers was a major facilitator, reflected in positive patient experiences and more use of adequate communication means (p. 3). (*Tratt et al., 2020*)
		An additional path through which more cultural awareness could have positive impacts on the use of health services was mediated by increased confidence in patients' capabilities; thus, allowing a more active role of patient in decision-making (p. 4). (*Tratt et al., 2020*)
		Previous positive experiences of health personnel in their interaction with Inuit was told to be a factor that highly contributes to increase cultural awareness, countering the negative effect of a healthcare system based on non-Inuit cultural values (p. 4). (*Tratt et al., 2020*)
		"Well, word it a way to have to do with childbearing, [which] is a sacred gift that's only given to women, let's keep our bodies healthy by getting annual check-ups, and you use some culturally appropriate like pictures, or something" (p. 8). (*Maar et al., 2016*)
		"We need to talk to our people in a way they understand, no disrespect to any organization, but people don't look at that material [brochures], it's obvious that the way we deliver that message has to be different" (p. 98). (*Henderson et al., 2018*)
	Integrated Services	"You need to make sure that you train the trainers, like teach the CHRs [community health representatives], and the women in the community

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		that are providing service" (p. 5). (*Maar et al., 2016*)
		"So if you're having something that they're able to come and access, [like a] sewing circle, or community kitchen, something that they're gonna get, then they'll come" (p. 6). (*Maar et al., 2016*)
		"Well, since you're here, you know, we haven't screened for this, and when was your last Pap and what about colon cancer screening?" (p. 7). (*Maar et al., 2016*)
		"If it [other STI testing] can be done on the same swab [as self-sampling], I would say, the value of that would be huge" (p. 8). (*Maar et al., 2016*)
		HCPs felt that education around self-testing might better be done in the context of a well woman healthcare visit, as "part of a physical, that might get best [results]." Women would not necessarily understand that they would still need to go to their HCP for other reproductive healthcare: "They would, they'd probably think, well, I've had that done, so I don't need this done" (p. 7). (*Zehbe et al., 2017*)
		Eliminating multiple clinic visits was stressed (p. 5). (*Adcock et al., 2019*)
	Negative HC Experience	"I still don't know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don't know! Nobody explained anything to me because, maybe because I didn't ask. Maybe they thought that I understood everything that was going on, but I didn't" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"When you're in (western Canadian city A) and (western Canadian city B) sometimes you just get pushed along as if you're just a file. 'Here you go, this is what's wrong, this is what we're going to do, see you later.' I don't understand where I go from there, how's my health going to be affected by it now" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		". . . I think it was [hospital name], and they were the roughest they could be in there. They weren't gentle at all, especially having something like that done and they say, "Lay down," and wham, you know, they're in, and that, and it was awful. I said, no, that's it." (p. 9) (*Butler et al., 2020*)
		Desire for bodily autonomy was often related to negative health experiences (i.e., painful pelvic examinations, inappropriate actions/comments by HCPs) (p. 3). (*Adcock et al., 2019*)
		"We never see a doctor. The health department has to get a hold of us, the ones that are never home. There are a lot of us in the community... We hear

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		suddenly of these workshops and information centres in the community. We wanted to go but we had other commitments so needed others to go. My niece, she died about 3 years ago because she had cancer. They brought her to Edmonton, they started her on chemo...left her kids at home." (p. 97). (*Henderson et al., 2018*)
		"One of the girls I used to visit got cervical cancer. She is very angry, she will not talk to me even though I give her support. She was told 3 years ago that she should get the tests and she didn't because she had problems for a long time" (p. 97). (*Henderson et al., 2018*)
		"She [HCP] thinks it's her [that they don't like], 'like, why is no one coming [for screening]'… 'it's not you, it took me 20 years to get to where I'm at now, and I still get people that don't trust me'" (p. 374). (*Wakewich et al., 2015*)
		"My doctor who I had since my oldest was born, she just up and quit, so when my next appointment, I had this guy looking at me, it's like [pause] right, so it's not like they stick around here" (p. 375). (*Wakewich et al., 2015*)
	Language	Participants reported the major importance of using visual language, ranking higher than using Inuktitut, and they stressed the communication problems behind using French or medical jargon (p. 4). (*Tratt et al., 2020*)
		"I have a sweet little old lady, she's probably close to 80; she won't speak it, she won't speak English, she understands you, she can speak English, but she would rather speak in her own language" (p. 6). (*Maar et al., 2016*)
	Male Health Care worker	"I had it done by a male doctor and I was uncomfortable . . . because of the way I was raised. My [traditional] grandmother always told me that the only man that should know you like that is your husband" (p. 87). (*O'Brien, Mill & Wilson, 2009*)
		"My first Pap smear was done by a male and I was very nervous and I just didn't want it done. Maybe that's why I didn't have another one [Pap smear] . . . because of that [male gender]" (p. 87). (*O'Brien, Mill & Wilson, 2009*)
		"Some people don't go [to get Pap smears], maybe they're shy … all of my friends don't go probably because they are shy or sometimes they don't want to [be] checked by a man, if it is a man that is a nurse" (p. 3). (*Cerigo et al., 2012*)
	Positive Health	"My appointment was scheduled when the offices

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	Care Experience	are closed so I knew he was going to be giving me some fairly bad news. The office was empty, and the lights were all turned down. It was a real comfortable surroundings. He gave me a lot of information, pamphlets, and people to phone for support" (p. 85). (*O'Brien, Mill & Wilson, 2009*)
		"Oh, when I have my testing done here they just make you feel so welcome and so warm, you don't feel so invaded, like a hospital. . .Yeah, you feel like a person, you don't seem–feel like a number." (p.11) (*Butler et al., 2020)*
		"The women are, you know, fairly open once they're comfortable" (p. 6). (*Maar et al., 2016*)
		"It's all about trust and relationships, like if you can't build a relationship you're not gonna get anywhere, you might not even see them again" (p. 7). (*Maar et al., 2016*)
	Female Health Care Worker	"The girls are always looking for female doctors" (p. 8). (*Maar et al., 2016*)
		"Feel more comfortable [with a female physician or nurse practitioner] because maybe she's going through the same thing that you're going through" (p. 6). (*Zehbe et al., 2017*)
		"I only got a Pap test once and I got it from a woman … and she was trying to make me feel comfortable, but it's still really awkward" (p. 6). (*Zehbe et al., 2017*)
		". . . they have got the body parts as me, you know what I mean? And if they've gone through something similar to this, they'll understand what it's like. . . . [We have] strong cultural val- ues on Women's Business too. And it's an invasive procedure, that's how I see it." (p.10) (*Butler et al., 2020)*
	Lack of Results	"A lot of us say, and we've talked about this before, when we get a Papanicolaou, the results don't arrive, and we don't know what it is that's going on. There we are with the doctor, asking why the results don't arrive. ... Whatever it [the result] is, they should give it to me" (p. 4). (*Allen-Leigh et al., 2017*)
		"It's crazy here, how long you have to wait for anything and half the time, you don't even get a call … you go to your next doctor's appointment for something else and [have to ask] … by the way, how about my Pap two months ago?" (p. 5) (*Zehbe et al., 2017*)
	Privacy	Health centers where staff 'phoning people, telling them, 'we've got your results … you need to make another appointment' was upsetting (p. 5). (*Zehbe et*

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		al., 2017)
		"Especially after, you know, you finish your pap smear and then you're having lunch with them in the tea room, no thanks." (p.12) (*Butler et al., 2020)*
		"You know what, just sitting in the waiting room, the receptionists are phoning people telling them, we've got your results, you need an, you know, make another appointment, they say the name right out loud, they say what the test is" (p. 375). (*Wakewich et al., 2015*)
		"Privacy, a small town, I mean, you can hear through the walls, you know, walls talk because there's, it's all in one building and coming in here and everybody sees the first person coming in here, they come to see me, they come to see welfare, they come to see housing, they're going and everybody knows everything" (p. 375). (*Wakewich et al., 2015*)
		"Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk" (p. 375). (*Wakewich et al., 2015*)
	Screening convenience	"I don't know how many times they've cancelled Pap tests and [when I asked] okay well can I get it on Friday? …. [heard back] 'No, we don't do them on Friday'" (p. 5). (*Zehbe et al., 2017*)
		"Most of us do not have doctors here … you have to drive up to [nearby town]" (p. 4). (*Zehbe et al., 2017*)
		". . . when I tell them it was five years, it's a five-year one, they're like, "Five years, even better," knowing that they were right for five years and it picks up early, detects things earlier, they were all for it." (p.10) (*Butler et al., 2020*)
		"The availability of screening services outside of work hours and the potential to reduce waiting times in the clinic by having a nurse available to complete screening rather than a doctor were raised as potential solutions." p.11 (*Butler et al., 2020*)
		"The size of the examination table was a concern for some women, who worried that they may not fit comfort- ably on the table due to having a larger body size. One woman with a physical disability stated the importance of an accessible, height-adjustable examination table that made screening more comfortable and easy for her." (p.11) (*Butler et al., 2020*)
		"Women were time-poor and struggled to fit screening appointments in amongst commitments to

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		work and family, which took higher priority in women's lives." p.11 (Butler et al., 2020)
Interpersonal Factors	Family Relations	"Well, my mother, you know, always went for her physicals; thus I learned that I was to go for my physicals... the families that do have it, you can see that progression of preventative health care" (p. 5). (*Maar et al., 2016*)
		". . . I didn't get it explained too much about that growing up into a woman off my mum. Whereas I want to do the opposite with my daughter and sort of be one step ahead of her, pre- pared and ready. . .. having that information would be good to have it there for her if she ever does want to know about it" *(p. 8) (Butler et al., 2020)*
		"I am learning a lot in these workshops. My mother died of stomach cancer, my sister of stomach cancer. I had 5 girls, and 4 of them went through breast cancer. My oldest daughter, her cancer spread. When they were younger I made sure they all got their needles. But, you know, I have never had a workshop like this. If I get a cold, I can fight it off. When I got those needles, I was told I was able to fight the sicknesses; it won't kill you—that is what I was told. This is really good for my grandchildren; I will take this message home to my family. I have two nurses in my family, they probably know about it, but this is a really good thing I am still learning...[*speaking in Cree] *I was worried the white people would not take care of us, but they have so far [group laughs]. We need to talk to young ladies about how to take care of themselves" (p. 97). (*Henderson et al., 2018*)
		Many of the Elders emphasized that health education within Indige- nous contexts is anchored in the love and care for children. For one speaker, this affection involves "hugging your children and telling them 'I love you', you don't say goodbye to anyone" (p. 98). (*Henderson et al., 2018*)
		"There are anti-bullying programs...and there are cultural programs; some kids are brave and some are afraid of getting immunized, but all the children support each other" (p. 98). (*Henderson et al., 2018*)
		"The best teachers are your parents" (p. 98). (*Henderson et al., 2018*)
		"I just taught them that sex isn't, it's not what you call dirty, you know, but there's a certain way you got to go about having, like for sex with a partner, you know, you got to explain that part to them, but to talk openly about sex, it's not what you call a dirty

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		subject, yeah. If you want to know something just feel free to ask" (p. 376). (*Wakewich et al., 2015*)
		"Like my mother, you know, was the kind of woman that … when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your own personal, private, so when it happened, I had absolutely no knowledge what was going on in my body … And with my daughter, I didn't want her to have that feeling, so I mean, of course I changed and I explained everything but, and then that's the way things are now, women are more informed than they have been in the past" (p. 376). (*Wakewich et al., 2015*)
	Intergenerational Communication	"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children, and out of respect...he's getting that message across [in his sweat lodge ceremonies with men]" (p. 98). (*Henderson et al., 2018*)
Intrapersonal Factors	Lack of Knowledge	The vast majority of participating women expressed the need to get screened after understanding medical facts, including but not limited to the absence of symptoms (p. 4). (*Tratt et al., 2020*)
		"Between these two sicknesses [HPV and cervical cancer] we're in danger, we should go to the clinic or a doctor. If we feel pain in the womb, go to a doctor" (p. 4). (*Allen-Leigh et al., 2017*)
		"Well, as far as I know, the virus is transmitted through sexual contact. Then this human papillomavirus, it begins without any warning. Then later it progresses, then the discomfort in our parts [genitals] begins and that's when the discharge starts and it progresses to the cervix and when it gets to the cervix it goes into the uterus and that is when the doctor sends us for an operation" (p. 4). (*Allen-Leigh et al., 2017*)
		"Moderator: What is cervical cancer, what do you know about cervical cancer? Huichol woman 1: They say you can die of cancer, if you don't detect it early. Moderator: And how do you detect it? Huichol woman 1: With the Papanicolaou, doing it periodically. Moderator: And what's periodically? Huichol woman 1: Every three months. Moderator: Everyone, how often do you think you need to get a Papanicolaou? Huichol woman 2: Once a year. Huichol woman 3: Depends on how you feel, once a year or every two years, I get it every two years. Moderator: What do you mean, how you feel?

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		Huichol woman 3: If you feel burning" (p. 4). (*Allen-Leigh et al., 2017*)
		Further, we found that some women did not fully understand the purpose of the Pap smear as a method of cervical cancer screening (p. 5). (*Cerigo et al., 2012)*
		A sizable proportion of the women were unable to identify a cervical cancer risk factor and were unsure if detecting cervical cancer early would affect the chance for a cure (p. 5). (*Cerigo et al., 2012*)
		Although unintended, participants of both focus groups directed the conversation towards an educational focus due to the limited knowledge about HPV, cervical cancer and Pap smears among participants (p. 6). (*Cerigo et al., 2012*)
		A lack of health literacy about HPV and cervical cancer and a lack of appropriate/empathetic services were also raised as barriers (p. 3). (*Adcock et al., 2019*)
		"The health centre sent out a notification and a consent form, and they listed the benefits and risks...and I paid more attention to the risks, and I decided not to allow her to be vaccinated, because as a parent I needed to do what was best for my children" (p. 98). (*Henderson et al., 2018*)
		"These young people are lucky to get these different resources...its scary when you think about it, you didn't think about [the health risk] before, because you didn't know anything about it" (p. 98). (*Henderson et al., 2018*)
	Awareness of HPV	"Explain [what the Pap test is] then they accept the value of early diagnosis: I find that once I explain to them the importance of early detection and [that] it can be treated, they're more agreeable to [a Pap test]" (p. 4). (*Maar et al., 2016*)
		"The biggest motivation is education, I think. You know, just teach them, let them know that the service is there, that they need to take it, it's important that they have it" (p. 4). (*Maar et al., 2016*)
		"We know cervical cancer is 100% preventable, and I know because I read stats and see statistics that Aboriginal women are the number one on the list for dying from this" (p. 5). (*Maar et al., 2016*)
		"If you detect it in time, you can take that sickness out so it won't progress any more and with a treatment you are fine" (p. 4). (*Allen-Leigh et al., 2017*)
		"There is a vaccine for human papillomavirus. I heard it on the radio, that there are vaccines" (p. 4).

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		(*Allen-Leigh et al., 2017*)
	Embarrassment	"For me, I am used to getting my Pap, I'm fine with it. But for the younger girls, [or] the shy ones, [or] those who were sexually abused, they might prefer [the HPV self-sampling method]" (p. 4). (*Tratt et al., 2020*)
		". . .Whatever the doctor says I just put my hand up, because after having children you don't care really. Just do what you have to do." (p. 8) (*Butler et al., 2020*)
		Among our population, older women reported more feelings of embarrassment than the young women (p. 5). (*Cerigo et al., 2012*)
		"Embarrassed anyway, no matter who did it" (p. 6). (*Zehbe et al., 2017*)
		Desire for bodily autonomy (retaining privacy, control over one's body) as a reason for not attending regular cervical screening - encompasses concepts of whakamā (embarrassment/shyness/reticence), tapu (sacred/taboo/forbidden) (p. 3). (*Adcock et al., 2019)*
		"I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this …. There's got to be a better, kinder, gentler, more humane way to do it" (p. 373). (*Wakewich et al., 2015)*
	Fear of Prognosis	" People are scared. I'm thinking of my sister. She hasn't gone back to get retested. She did have an abnormal result the first time. I think she's afraid. Some people are afraid that if they do find cancer, it's downhill from there" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"My fear is of having [cervical cancer] . . . motivates me. I want to have good health. I don't want that kind of a disease" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"I would try and keep encouraging her . . . to tell her, "If they find it early they can do something about it." I try and set examples for my clients. I tell people, "Did you know this lady had this kind of cancer because she never got tested, and they could have done something about it right away and she would still be here" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"HPV, like, whoa, I don't have that, like, I don't even want to know if I have that" (p. 7). (*Zehbe et al., 2017*)
		"Once they corrected the cervical cancer you would

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		end up with cancer somewhere else" (p. 96). (*Henderson et al., 2018*)
	Financial Barriers	"If the van was full, [you would] have to get a ride … some of us don't have cars, you know" (p. 5). (*Zehbe et al., 2017*)
		Women with children and work commitments also had difficulty arranging childcare and time off work to keep their appointments. As a HCP commented: 'It certainly does pose challenges with regards to babysitting care and the mother being away for an entire day' (p. 5). (*Zehbe et al., 2017*)
		Older women were more likely to mention a previous bad experience and were less likely than younger women to mention cost or other financial barriers. Opinions varied about cost being a barrier (as many clinics offer free cervical screening). Some highlighted hidden costs (transport, parking, childcare) (p. 3). (*Adcock et al., 2019*)
	Health Care Priority	One in seven had not screened in ten years or more and one in six were unsure how long it had been since their last cervical screen (p. 2). (*Adcock et al., 2019*)
		"It's about listening to my body. It's about trying to keep it healthy for my children. I want to live as long as I can to bring them up to see them. Not only that, I want to feel good too. And I have had friends who have been diagnosed and that motivates me also to do it. But all in all I do it for myself and my family." *(p. 6) (Butler et al., 2020)*
		". . . it's no shame, you have to go and do your tests and all that. . . Every women [sic] in Aus- tralia have to do it every time, black or white. . .I don't get shame, because I want to look after myself and for my health too." (p.8) *(Butler et al., 2020)*
		"It's just what you do. I brush my teeth. It keeps coming down to that because it really for me is just about general body maintenance, you brush your teeth, you eat your food, I take my tablets. . . I do what I have to do." (p. 6) *(Butler et al., 2020)*
		"I'll always remember that lady, and she was so nice too. And she's going, your job now is mummy, and you've got to be here, have this test every two years. And because that was at my six week check-up and I didn't want to have the pap smear, and she's going, you don't want to leave him. And I'm thinking, oh my God, because I was a single mother, oh my God, I've got to look after him. So I did; never missed it." (p. 9) (*Butler et al., 2020)*
	Pain	"[When they did the Papanicolaou test], maybe they

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		did it wrong. I don't know, but I felt something like a scrape. Then I thought maybe the equipment wasn't disinfected. ... I thought about it a lot before doing it again, because I was afraid" (p. 4). (*Allen-Leigh et al., 2017*)
		"The Pap is uncomfortable but it has to be done" (p. 5). (*Zehbe et al., 2017*)
		"It doesn't get any easier, like the first time and then the next year, it didn't get any easier for me" (p. 5). (*Zehbe et al., 2017*)
		"I didn't like the way it felt, that's why I didn't want to go back there" (p. 5). (*Zehbe et al., 2017*)
		"They don't want their Paps [be]cause it hurt" (p. 5) (*Zehbe et al., 2017*)
		The test also could be exceptionally painful if providers were 'in a rush' or 'rough' and did the procedure 'real quick' (p. 5). (*Zehbe et al., 2017*)
		"Some women, when they come to the appointment, they decide they don't want to get a Pap, because it's uncomfortable, they're just afraid" (p. 5). (*Zehbe et al., 2017*)
Solutions – Self Sampling	Convenience	"Informant #14: I think that's the best idea that has come out of anything, like it's not invasive, you do it in your own home, at your own time... I.Z.: Whenever you, you know, are ready for it and... Informant #14 I think that with education along with it that this is what it is, I think that the success rate will be phenomenal with it because it's in their own home" (p. 8). (*Maar et al., 2016*)
		"It is more comfortable to do it at home … it's simple" (p. 6). (*Zehbe et al., 2017*)
		"I think the self-test is beneficial for them all because sometimes people don't have time for appointments to take off work, it [the Pap] is kind of an inconvenience" (p. 6). (*Zehbe et al., 2017*)
		"Even hearing [about self-testing], people are just, What? Oh, I'd do that, for sure, instead of me going to the doctor" (p. 6). (*Zehbe et al., 2017*)
		"Doing a self-test doesn't take very long … it's something that can be dealt with, done, gone" (p. 6). *Zehbe et al., 2017*)
		"I think more people would monitor it that way and test it themselves, like 'well maybe not today, but … eventually I'm going to try it" (p. 6) (*Zehbe et al., 2017*)
		"…I think the prevalence [participation in self-testing] rate would go up" (p. 5). (*Zehbe et al., 2017*)
		"Women wouldn't be so agitated and nervous about having the [self-] test" (p. 5). (*Zehbe et al., 2017*)

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		With participants using terms such as ‘easier’, ‘more comfortable’, ‘less intrusive’ and ‘brilliant’ [to describe self-testing] (p. 3). (*Adcock et al., 2019*)
		Participants discussed the value of providing multiple options and flexibility to cater to diverse populations, such as through community outreach services, was suggested for optimum engagement (p. 5). (*Adcock et al., 2019*)
		"It’s simple, it’s not like you are having forceps in you" (p. 376). (*Wakewich et al., 2015*)
	Decreased Pain	“Well, this one [the self-sampled HPV test] is better, because it is more comfortable to do it” (p. 5). (*Allen-Leigh et al., 2017*)
	Education	"If you hand me a kit, I won't touch it. Because I wouldn’t know what to do, what if I did it wrong, or whatever, right, so? And I think a lot of people would be that way" (p. 7). (*Zehbe et al., 2017*)
		“You just give a test kit to them at home, the majority of them are just going to throw them out” (p. 7). (*Zehbe et al., 2017*)
		HCPs agreed that with appropriate support and education, HPV self-testing will benefit never/under-screened Māori (p. 4). (*Adcock et al., 2019*)
		Most frequently [participants] said they would be happy to do the HPV self-test in a clinic or their own home, and emphasised having good support and education to increase their confidence about properly doing it (p. 5). (*Adcock et al., 2019*)
	Privacy	“Would feel more comfortable with that [self-test] than a male doctor” (p. 6). (*Zehbe et al., 2017*)
		Self-testing would also address the ‘trust issues’ that discouraged women from seeking care from non-indigenous HCPs (p. 6). (*Zehbe et al., 2017*)
		"It’s a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference” (p. 5). (*Zehbe et al., 2017*)
		“A lot more private, at home, if you do it by yourself” (p. 5). (*Zehbe et al., 2017*)
		“Women would prefer to be discreet, do it themselves and get their own results and not have their results shared with others” (p. 5). (*Zehbe et al., 2017*)
	Autonomy	“I really strongly believe that … because I was the one that was doing it [self-sampling], I was the one that was in control … and this way it gave me the

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		ability to do it myself and I got all the results, they were fine; … it was also self-empowering, great, I like that" (p. 6) (*Zehbe et al., 2017*)
		Option of taking it home might make women "feel more empowered" and contribute to a better relationship with HCPs (p. 6). (*Zehbe et al., 2017*)
		"Think it [self-sampling] would be, I think it would be great, because it provides them with some autonomy and allows them to take control of the situation. That would be really great actually …. I think it would increase [education opportunities] actually, yup, because it's a lot less clinical right, cause like you say, it's a lot less you know, stripping down and you know, allowing someone else to do the scraping of the cervix, you know, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference" (p. 376). (*Wakewich et al., 2015*)
	Decreased Embarrassment	"This one [the HPV test] is good because it [cytology or the Papanicolaou test] really does embarrass you, not because your husband doesn't want it or doesn't let us, but because it's embarrassing and because of embarrassment we don't do it, and so this [self-sampled HPV test] is good for us" (p. 5). (*Allen-Leigh et al., 2017*)
		"Because you do it yourself, since always, even if there is trust, you feel a little embarrassed to undress in front of someone else" (p. 5). (*Allen-Leigh et al., 2017*)
		"I think it [self-sampling] would definitely be more private … They wouldn't be so embarrassed … all kinds of people are easily embarrassed" (p. 376). (*Wakewich et al. 2015*)
Solutions – Community Level	Informal Community Communication	"Workshops, information sessions: I have lunch and learns. That's a start. There's all different venues bringing that into the community" (p. 5). (*Maar et al., 2016*)
		"I would concentrate on a day that's important to the women, like Mother's Day. The mothers would all come and the grandmas would come and the aunts would come. I would have guest speakers come in as well and there would be a dinner, a luncheon, or a feast of some sort. I would bring somebody in who had a little bit of charisma like a Tai Chi instructor who cooks meals" (p. 6). (*Maar et al., 2016*)
		"For our little group of work girls, [the] service team, we always tend to remind each other as well, like, 'Hey, is it time for your check-up yet, like?'" (p. 7).

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		(*Maar et al., 2016*)
		"I think it's just sort of word of mouth, spreading things around, and just kind of bringing it up, you know, all the time" (p. 7). (*Maar et al., 2016*)
		"I call four people and then each one of the people I call, they call four people and it's something that we're trying to work on" (p. 7). (*Maar et al., 2016*)
	Humor	"We didn't specifically talk about HPV, but we talked about STDs [during the education session]. Yeah, I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: "Are you bringing bananas next week?" You know, so, it was a fun thing" (p. 6). (*Maar et al., 2016*)
	Incentives	"It's hard to get people coming unless we have food and incentives" (p. 6). (*Maar et al., 2016*)
	School Education	"Start early, in school already: we should get our younger girls out there, and have that part as [education], at [grade] 8, and I know they're doing great in the school, getting it out there, about STDs" (p. 5). (*Maar et al., 2016*)
	Social Media	"You could even [do] something as simple as Facebook. Everybody's on Facebook" (p. 5). (*Maar et al., 2016*)
	Survivor Stories	"I think if there was someone just sharing stories. If a woman was willing to share her story that she's had this and got screened early. Girls like to listen to things like that, and the women like to listen to things like that. I beat this and I did that. Something like that would motivate them" (p. 5). (*Maar et al., 2016*)
	Positive Inspiration	"... and being a health worker too because then I can say, "No, I do mine," ... it feels a bit shame but you can talk to your patient and encourage them to do it because you've done it and if I hadn't done it, well, I shouldn't be saying those things to the patient." (p.12) (*Butler et al., 2020*)

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694 **Table 5: Social ecological model distribution of studies**

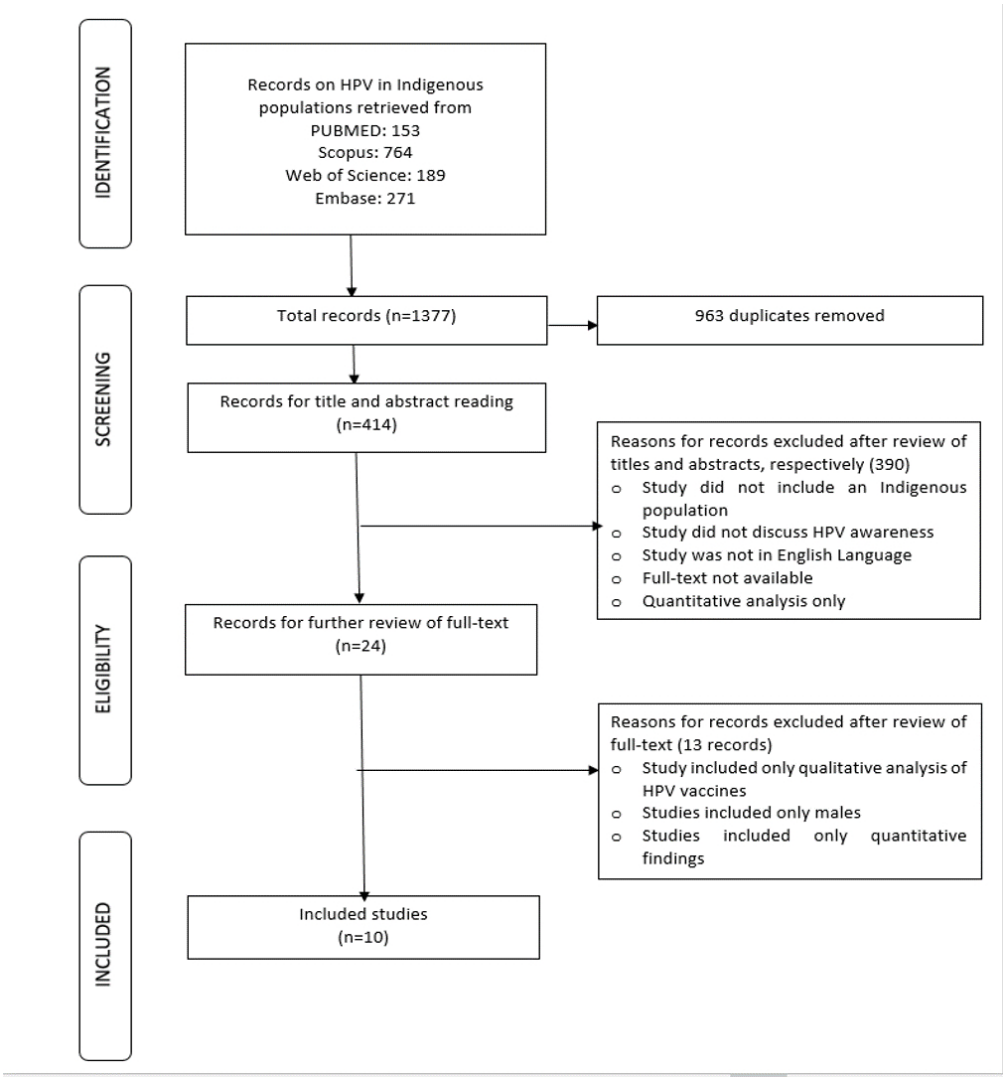
695

Studies	Public Policy	Socio-cultural/ community Factors	Institutional/ Organizational factors	Interpersonal factors	Intrapersonal Factors
Adcock A. 2019					
Allen-Leigh B et al. 2017.					
Cerigo H et al. 2012					
Henderson R et al. 2018					
Maar N et al. 2016					
O'Brien B et al. 2009					
Tratt E et al. 2020					
Wakewich P et al. 2015					
Zehbe et al. 2017					
Butler TL et al, 2020					

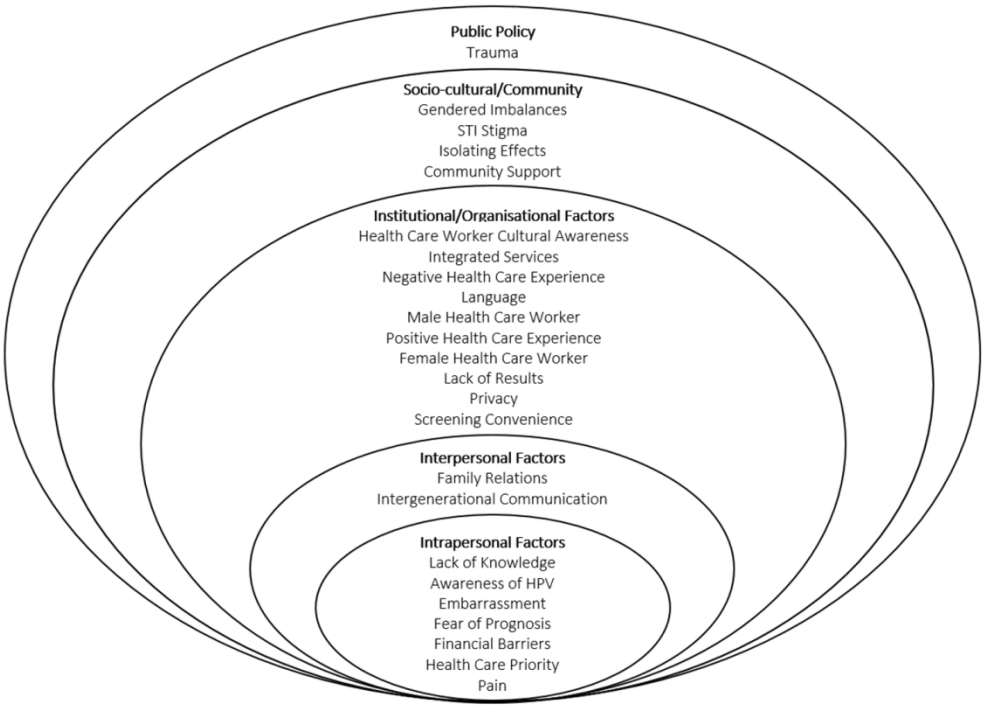
696 *Highlighted boxes show the contribution of a study to a particular ecological level.

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PRISMA flowchart for included studies



Socio-Ecological Model

Supplementary File 1 (S1)

Logic Grid: **Pubmed: 153 : 29/06/2020**

HPV AND INDIGENOUS POPULATIONS

HPV	Indigenous Populations
HPV[Text Word] OR "papillomaviridae"[MeSH Terms] OR "human papillomavirus"[MeSH] OR "human papillomavirus"[Text Word] OR "human papillomavirus 16"[MeSH Terms] OR human papillomavirus 16 [Text Word] OR "human papillomavirus 18" [MeSH Terms] OR human papillomavirus 18 [Text Word] OR papillomavirus[Text Word]	("first nation"[Text Word]) OR "first nations"[Text Word]) OR "pacific islander"[Text Word]) OR "pacific islanders"[Text Word]) OR "torres strait islander"[Text Word]) OR "torres strait islanders"[Text Word]) OR aborigin*[Text Word]) OR africa*[Text Word]) OR alaska*[Text Word]) OR aleut*[Text Word]) OR amerind*[Text Word]) OR arctic[Text Word]) OR aymara[Text Word]) OR bushmen[Text Word]) OR chukchi[Text Word]) OR chukotka*[Text Word]) OR circumpolar[Text Word]) OR eskimo*[Text Word]) OR greenland*[Text Word]) OR hmong[Text Word]) OR indian*[Text Word]) OR indigen*[Text Word]) OR inuit*[Text Word]) OR inupiaq[Text Word]) OR inupiat[Text Word]) OR khanty[Text Word]) OR maori*[Text Word]) OR mapuche[Text Word]) OR metis[Text Word]) OR native*[Text Word]) OR navaho*[Text Word]) OR navajo*[Text Word]) OR nenets[Text Word]) OR quechua[Text Word]) OR saami[Text Word]) OR sami[Text Word]) OR samoan*[Text Word]) OR siberia*[Text Word]) OR skolt[Text Word]) OR tribal[Text Word]) OR tribe*[Text Word]) OR xingu*[Text Word]) OR yup'ik[Text Word]) OR yupik[Text Word]) OR zuni[Text Word]) OR "African continental ancestry group"[Mesh]) OR "African continental ancestry group"[Mesh]) OR "Asian continental ancestry group"[Mesh]) OR "Health Services, Indigenous"[Mesh]) OR "Oceanic ancestry group"[Mesh]) OR "arctic regions"[Mesh]) OR "ethnic groups"[mesh]

Logic Grid: **EMBASE: 271 : 29/06/2020**

HPV	Indigenous Populations

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('human papillomavirus type 16':ti,ab,kw OR 'human papillomavirus type 18':ti,ab,kw OR 'wart virus':ti,ab,kw OR 'oral human papillomavirus infection':ti,ab,kw OR 'hpv':ti,ab,kw)	'indigenous population':ti,ab,kw OR 'indigenous people':ti,ab,kw OR 'Samoan (people)':ti,ab,kw OR 'First Nation':ti,ab,kw OR 'Pacific Islander':ti,ab,kw OR 'Torres Strait Islander':ti,ab,kw OR 'Black person':ti,ab,kw OR 'Alaska Native':ti,ab,kw OR 'Aleut (people)':ti,ab,kw OR 'Amerind people':ti,ab,kw OR 'Aymara (people)':ti,ab,kw OR 'Chukchi (people)':ti,ab,kw OR 'Chukotka':ti,ab,kw OR 'Eskimo-Aleut people':ti,kw,ab OR 'Greenland':ti,ab,kw OR 'Hmong (people)':ti,kw,ab OR 'Indian':ti,ab,kw OR 'indigenous people':ti,kw,ab OR 'Inuit':ti,kw,ab OR 'Inupiat (people)':ti,kw,ab OR 'Khanty (people)':ti,ab,kw OR 'Maori (people)':ti,kw,ab OR 'Mapuche (people)':ti,kw,ab OR 'indigenous people':ti,kw,ab OR 'Navajo (people)':ti,kw,ab OR 'Nenets (people)':ti,kw,ab OR 'Quechua (people)':ti,kw,ab OR 'Sami (people)':ti,kw,ab OR 'Yupik (people)':ti,ab,kw OR 'Zuni (people)':ti,ab,kw OR 'Asian continental ancestry group':ti,ab,kw OR 'Oceanic ancestry group':ti,ab,kw OR 'Arctic':ti,ab,kw OR 'ethnic group':ti,kw,ab

Logic Grid **SCOPUS: 764 : 29/06/2020**

HPV	Indigenous Populations
TITLE-ABS-KEY (HPV) OR TITLE-ABS-KEY (papillomaviridae) OR TITLE-ABS-KEY (human papillomavirus) OR TITLE-ABS-KEY (human papillomavirus type 16) OR TITLE-ABS-KEY (human papillomavirus type 18)	TITLE-ABS-KEY (indigenous AND population) OR TITLE-ABS-KEY (ethnic AND groups) OR TITLE-ABS-KEY (aborigin) OR TITLE-ABS-KEY (torres AND strait AND islander) OR TITLE-ABS-KEY (black AND person)

Web of Science: 189 :29/06/2020

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HPV	Indigenous Populations
(HPV OR human papillomavirus OR human papillomavirus type 16 OR human papillomavirus type 18 OR Oral HPV OR wart virus)	**TOPIC:** ((first nation OR first nations OR pacific islander OR pacific islanders OR torres strait islander OR torres strait islanders OR aborigin* OR africa* OR alaska* OR aleut* OR amerind* OR arctic OR aymara OR bushmen OR chukchi OR chukotka* OR circumpolar OR eskimo* OR greenland* OR hmong OR indian* OR indigen* OR inuit* OR inupiaq OR inupiat OR khanty OR maori* OR mapuche OR metis OR native* OR navaho* OR navajo* OR nenets OR quechua OR saami OR sami OR samoan* OR siberia* OR skolt OR tribal OR tribe* OR xingu* OR yup'ik OR yupik OR zuni OR African continental ancestry group OR African continental ancestry group OR Asian continental ancestry group OR Health Services, Indigenous OR Oceanic ancestry group OR arctic regions OR ethnic groups))

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S2: Supplementary File 2: JBI Summary appraisal tool questionnaire for qualitative systematic reviews

1. Is there congruity between the stated philosophical perspective and research methodology?
2. Is there congruity between the research methodology and the research question or objectives?
3. Is there congruity between the research methodology and the methods used to collect the data?
4. Is there congruity between the research methodology and the representation and analysis of data?
5. Is there congruity between the research methodology and the interpretation of results?
6. Is there a statement placing the researcher culturally or theoretically?
7. Is the influence of the research on the researcher and vice versa, addressed?
8. Are the participants and their voices adequately represented?
9. Is there research ethical according to current criteria or, for recent studies, and there evidence for ethical approval by an appropriate body?
10. Do the conclusions drawn in the research report flow from the analysis, or interpretation of the data?

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S3: Supplementary file 3: Appraisal of included studies

Table: Qualitative Research

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Adcock A. 2019.	Y	Y	Y	Y	Y	N	N	Y	Y	Y
Allen-Leigh B et al. 2017.	Y	Y	Y	Y	Y	N	U	Y	Y	Y
Butler et al., 2020	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Cerigo H et al. 2012.	Y	Y	Y	Y	Y	N	N	Y	Y	Y
Cui Y et al. 2010.	Y	Y	Y	Y	Y	N	N	N	Y	Y
Henderson R et al. 2018.	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Maar N et al. 2016.	Y	Y	Y	Y	U	Y	U	Y	Y	Y
O'Brien B et al. 2009.	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Tratt E et al. 2020.	Y	Y	Y	Y	Y	Y	U	Y	Y	Y
Wakewich P et al. 2015.	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
Zehbe et al. 2017.	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
%	100.0	100.0	100.0	100.0	90.9	63.63	18.18	90.9	100.0	100.0

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S4: Supplementary file 4: List of Study Findings with Illustrations

Study: Tratt E et al 2020	
Finding	Cultural Awareness (C)
Illustration	Cultural awareness of healthcare workers was a major facilitator, reflected in positive patient experiences and more use of adequate communication means. (pg 3)
Finding	Previous Experiences (C)
Illustration	Previous positive experiences of health personnel in their interaction with Inuit was told to be a factor that highly contributes to increase cultural awareness, countering the negative effect of a healthcare system based on non-Inuit cultural values. (pg 4)
Finding	Increased confidence in patients' capabilities (C)
Illustration	Thus, allowing a more active role of patient in decision-making. (pg 4)
Finding	Access to medical information (C)
Illustration	The vast majority of participating women expressed the need to get screened after understanding medical facts, including but not limited to the absence of symptoms (pg 4)
Finding	Personal Preference (U)
Illustration	“For me, I am used to getting my Pap, I’m fine with it. But for the younger girls, [or] the shy ones, [or] those who were sexually abused, they might prefer [the HPV self-sampling method].” pg 4
Finding	Means of communication (C)
Illustration	Participants reported the major importance of using visual language, ranking higher than using Inuktitut, and they stressed the communication problems behind using French or medical jargon
Study: O’Brien B et al 2009	
Finding	Vivid Recollection of healthcare encounters (U)
Illustration	My appointment was scheduled when the offices are closed so I knew he was going to be giving me some fairly bad news. The office was empty, and the lights were all turned down. It was a real comfortable surrounding. He gave me a lot of information, pamphlets, and people to phone for support. (pg 85)
Finding	Not Knowing what was happening (inadequate explanations) (U)

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Illustration	I still don't know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don't know! Nobody explained anything to me because, maybe because I didn't ask. Maybe they thought that I understood everything that was going on, but I didn't. (pg 86)
Finding	Not knowing what was happening (Confusion) (U)
Illustration	When you're in (western Canadian city A) and (western Canadian city B) sometimes you just get pushed along as if you're just a file. "Here you go, this is what's wrong, this is what we're going to do, see you later." I don't understand where I go from there, how's my health going to be affected by it now. (pg 86)
Finding	Encouragement and Support (Fear of process) (U)
Illustration	"My mom has gone through the process. She told me that she had cancer in her cervix and that she had to go for surgery. So it was a big fear for me because as soon as you hear cancer you think of the ultimate. There's nothing more then; but for this person to die. "(pg 86) "My dad had a really difficult time with it, because he was already in his early 60s at the time, and as soon as he heard cancer [he thought] I was going to die. He started cutting off the emotional ties with me then." (pg 86) "People are scared. I'm thinking of my sister. She hasn't gone back to get retested. She did have an abnormal result the first time. I think she's afraid. Some people are afraid that if they do find cancer, it's downhill from there." (pg 86) "To find out that I have some kind of illness . . . like to know you're sick you just want to block it out. . . . If I knew I had an illness . . . I would probably just shut down from everything and totally change, and not for the better either." (pg 86)
Finding	Encouragement and Support (Motivation) (U)
Illustration	"My fear is of having [cervical cancer] . . . motivates me. I want to have good health. I don't want that kind of a disease." (pg 86)
Finding	Encouragement and Support (Support) (U)
Illustration	"I would try and keep encouraging her . . . to tell her, "If they find it early they can do something about it." I try and set examples for my clients. I tell people, "Did you know this lady had this kind of cancer because she never got tested, and they could have done something about it right away and she would still be here."" (pg 86)
Finding	Encountering a male provider (Taditional Beliefs) (U)
Illustration	"I had it done by a male doctor and I was uncomfortable . . . because of the way I was raised. My [traditional] grandmother always told

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	me that the only man that should know you like that is your husband." (pg 87)
Finding	Encountering a male provider (Nervousness) (U)
Illustration	"My first Pap smear was done by a male and I was very nervous and I just didn't want it done. Maybe that's why I didn't have another one [Pap smear] . . . because of that [male gender]." (pg 87)
Study: Maar N et al 2016	
Finding	(Explain the value of cervical screening) Importance & Benefits (U)
Illustration	"Explain [what the Pap test is] then they accept the value of early diagnosis: I find that once I explain to them the importance of early detection and [that] it can be treated, they're more agreeable to [a Pap test]." (p. 4)
Finding	(Explain the value of cervical cancer screening) Awareness of screening services (U)
Illustration	"The biggest motivation is education, I think. You know, just teach them, let them know that the service is there, that they need to take it, it's important that they have it." (p. 4)
Finding	(Explain the value of cervical cancer screening) Screening can prevent cancer (U)
Illustration	"We know cervical cancer is 100% preventable, and I know because I read stats and see statistics that Aboriginal women are the number one on the list for dying from this." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Build local HCP capacity (U)
Illustration	"You need to make sure that you train the trainers, like teach the CHRs [community health representatives], and the women in the community that are providing service." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Education by community cancer survivors (U)
Illustration	"I think if there was someone just sharing stories. If a woman was willing to share her story that she's had this and got screened early. Girls like to listen to things like that, and the women like to listen to things like that. I beat this and I did that. Something like that would motivate them." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Education by extended female relatives (U)

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Illustration	"Well, my mother, you know, always went for her physicals; thus I learned that I was to go for my physicals... the families that do have it, you can see that progression of preventative health care." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Early education in schools (U)
Illustration	"Start early, in school already: we should get our younger girls out there, and have that part as [education], at [grade] 8, and I know they're doing great in the school, getting it out there, about STDs." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Regular workshops (U)
Illustration	"Workshops, information sessions: I have lunch and learns. That's a start. There's all different venues bringing that into the community." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Passive intervention for whole community (U)
Illustration	"You could even [do] something as simple as Facebook. Everybody's on Facebook." (p. 5)
Finding	(Motivate through specific educational strategies and capacity building) Include humour in health education (U)
Illustration	"We didn't specifically talk about HPV, but we talked about STDs [during the education session]. Yeah, I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: "Are you bringing bananas next week?" You know, so, it was a fun thing." (p. 6)
Finding	(Motivate through specific educational strategies and capacity building) Use local language (U)
Illustration	"I have a sweet little old lady, she's probably close to 80; she won't speak it, she won't speak English, she understands you, she can speak English, but she would rather speak in her own language." (p. 6)
Finding	(Embed cervical cancer screening into community events) Integrate screening into wellness events (U)
Illustration	"So if you're having something that they're able to come and access, [like a] sewing circle, or community kitchen, something that they're gonna get, then they'll come." (p. 6)

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Finding	(Embed cervical cancer screening into community events) Screening on days that honour women (U)
Illustration	"I would concentrate on a day that's important to the women, like Mother's Day. The mothers would all come and the grandmas would come and the aunts would come. I would have guest speakers come in as well and there would be a dinner, a luncheon, or a feast of some sort. I would bring somebody in who had a little bit of charisma like a Tai Chi instructor who cooks meals" (p. 6)
Finding	(Embed cervical cancer screening into community events) Small incentives (U)
Illustration	"It's hard to get people coming unless we have food and incentives." (p. 6)
Finding	(Health care providers need a trusting rapport with women) Friendly Relationship (U)
Illustration	"The women are, you know, fairly open once they're comfortable." (p. 6)
Finding	(Health care providers need a trusting rapport with women) Trusting Relationship (U)
Illustration	"It's all about trust and relationships, like if you can't build a relationship you're not gonna get anywhere, you might not even see them again." (p. 7)
Finding	Integrate cervical cancer screening recruitment into existing health care services (U)
Illustration	"Well, since you're here, you know, we haven't screened for this, and when was your last Pap and what about colon cancer screening?" (p. 7)
Finding	(Create a dialogue on cervical cancer screening in day-to-day activities) Sustain screening culture with HCP (U)
Illustration	"For our little group of work girls, [the] service team, we always tend to remind each other as well, like, "Hey, is it time for your check-up yet, like?"" (p. 7)
Finding	(Create a dialogue on cervical cancer screening in day-to-day activities) Word of mouth reminders (U)
Illustration	"I think it's just sort of word of mouth, spreading things around, and just kind of bringing it up, you know, all the time." (p. 7) " I call four people and then each one of the people I call, they call four people and it's something that we're trying to work on." (p. 7)

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Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Screening as a way for women to re-gain teaching roles (C)
Illustration	"The regular program for girls meets every second Monday, and it's called the Moon Time Girls."
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Holistic approach for cultural integrity (U)
Illustration	"Well, word it a way to have to do with childbearing, [which] is a sacred gift that's only given to women, let's keep our bodies healthy by getting annual check-ups, and you use some culturally appropriate like pictures, or something." (p. 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Using non-invasive screening (U)
Illustration	"Informant #14: I think that's the best idea that has come out of anything, like it's not invasive, you do it in your own home, at your own time... I.Z.: Whenever you, you know, are ready for it and... Informant #14: I think that with education along with it that this is what it is, I think that the success rate will be phenomenal with it because it's in their own home." (p. 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Multi-purpose screening (U)
Illustration	"If it [other STI testing] can be done on the same swab [as self-sampling], I would say, the value of that would be huge." (p. 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Preference of female HCP (U)
Illustration	"The girls are always looking for female doctors" (p. 8)
Study: Allen-Leigh B et al 2017	
Finding	Beliefs and knowledge about cervical cancer (Correct knowledge) (U)
Illustration	"If you detect it in time, you can take that sickness out so it won't progress any more and with a treatment you are fine." (p. 4)
Finding	Beliefs and knowledge about cervical cancer (Incorrect knowledge) (U)

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Illustration	"Between these two sicknesses [HPV and cervical cancer] we're in danger, we should go to the clinic or a doctor. If we feel pain in the womb, go to a doctor." (p. 4)
Finding	Perceptions of and knowledge about HPV (Correct knowledge) (U)
Illustration	[Cervical cancer is caused] "by infections, because later they become like a tumor in the womb." (p. 4) "There is a vaccine for human papillomavirus. I heard it on the radio, that there are vaccines." (p. 4)
Finding	Perceptions of and knowledge about HPV (Combined correct and incorrect knowledge) (U)
Illustration	"Well, as far as I know, the virus is transmitted through sexual contact. Then this human papillomavirus, it begins without any warning. Then later it progresses, then the discomfort in our parts [genitals] begins and that's when the discharge starts and it progresses to the cervix and when it gets to the cervix it goes into the uterus and that is when the doctor sends us for an operation." (p. 4)
Finding	Perceptions, knowledge and experiences related to cytology (Combined correct and incorrect knowledge) (U)
Illustration	"Moderator: What is cervical cancer, what do you know about cervical cancer? Huichol woman 1: They say you can die of cancer, if you don't detect it early. Moderator: And how do you detect it? Huichol woman 1: With the Papanicolaou, doing it periodically. Moderator: And what's periodically? Huichol woman 1: Every three months. Moderator: Everyone, how often do you think you need to get a Papanicolaou? Huichol woman 2: Once a year. Huichol woman 3: Depends on how you feel, once a year or every two years, I get it every two years. Moderator: What do you mean, how you feel? Huichol woman 3: If you feel burning." (p. 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Pain and fear of unsterile equipment as a barrier) (U)
Illustration	"[When they did the Papanicolaou test], maybe they did it wrong. I don't know, but I felt something like a scrape. Then I thought maybe the equipment wasn't disinfected. ... I thought about it a lot before doing it again, because I was afraid." (p. 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Not receiving test results as a barrier) (U)
Illustration	"A lot of us say, and we've talked about this before, when we get a Papanicolaou, the results don't arrive, and we don't know what it is that's going on. There we are with the doctor, asking why the results don't arrive. ... Whatever it [the result] is, they should give it to me." (p. 4)

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Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Unequal gender relations as a barrier) (U)
Illustration	“The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn’t even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn’t believe that.” (p. 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Lack of gender-related barriers) (U)
Illustration	“Because with all these [health education] talks they give us, women are more secure in themselves. So, we don’t really ask men for permission anymore, because it’s something that’s good for us.” (p. 4) “I decided it [to perform the HPV test] myself, alone. I don’t ask anyone’s permission. ... How am I going to ask him if he [her husband] wants it or not? It’s not for him, it’s for me.” (p. 4)
Finding	Perceived advantages of the self-sampled HPV test (Less embarrassment) (U)
Illustration	“This one [the HPV test] is good because it [cytology or the Papanicolaou test] really does embarrass you, not because your husband doesn’t want it or doesn’t let us, but because it’s embarrassing and because of embarrassment we don’t do it, and so this [self-sampled HPV test] is good for us.” (p. 5) “Because you do it yourself, since always, even if there is trust, you feel a little embarrassed to undress in front of someone else” (p. 5)
Finding	Perceived advantages of the self-sampled HPV test (More comfortable) (U)
Illustration	“Well, this one [the self-sampled HPV test] is better, because it is more comfortable to do it.” (p. 5)
Study: Cerigo H et al 2012	
Finding	Experiences, Attitudes and beliefs about cervical cancer (C)
Illustration	A sizable proportion of the women were unable to identify a cervical cancer risk factor and were unsure if detecting cervical cancer early would affect the chance for a cure. (pg 5)
Finding	Pap smear history, understanding and experiences (female preference) (U)
Illustration	""“Some people don't go [to get Pap smears], maybe they're shy … all of my friends don't go probably because they are shy or sometimes they don't want to [be] checked by a man, if it is a man that is a nurse”." (pg 3)

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Finding	Pap smear history, understanding and experiences (embarrasment) (C)
Illustration	Feelings of embarrassment and pain during Pap smears were common among our study population and others 23. Among our population, older women reported more feelings of embarrassment than the young women. (pg 5)
Finding	Pap smear history, understanding and experiences (lack of knowledge) (C)
Illustration	Further, we found that some women did not fully understand the purpose of the Pap smear as a method of cervical cancer screening, as 30% reported that the purpose of the Pap smear test is to screen for STIs. (pg 5)
Study: Zehbe et al 2017	
Finding	Educational implications (contribution to stereotype) (U)
Illustration	Focus group participants were concerned that any publication of information about HPV rates on reserve could have negative effects, contributing to the stereotype that 'all native people have HPV' (FG6). As one HCP summarised: 'A lot more education has to be done about HPV and cervical cancer together' to alleviate these concerns (pg 7)
Finding	Educational implications (women blamed for STI's) (U)
Illustration	"'HPV, like, whoa, I don't have that, like, I don't even want to know if I have that" (pg 7) 'because it's a relationship thing' that implied that someone had 'cheated' (pg 7) '…. do they [the men] not need to know some of this stuff too?' (pg 7) '… I know only women get cervical cancer but … they've helping it along in the same sense where they've passing it back and forth' (pg 7)
Finding	Educational implications (need to explain self-sampling) (U)
Illustration	HCPs felt that education around self-testing might better be done in the context of a well woman healthcare visit, as 'part of a physical, that might get best [results]' (Pg 7) women would not necessarily understand that they would still need to go to their HCP for other reproductive healthcare: 'They would, they'd probably think, well, I've had that done, so I don't need this done'" (pg 7) "If you hand me a kit, I won't touch it. Because I wouldn't know what to do, what if I did it wrong, or whatever, right, so? And I think a lot of people would be that way" (pg 7) 'you just give a test kit to them at home, the majority of them are just going to throw them out'. (pg 7)
Finding	Convenience factor self sampling (no appointment needed) (U)

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Illustration	'It is more comfortable to do it at home … it's simple'. (pg 6) I think the self-test is beneficial for them all because sometimes people don't have time for appointments to take off work, it [the Pap] is kind of an inconvenience' (pg 6) 'would go for it' (pg 6) ;older woman who had fewer constraints'were used to getting Paps' (pg 6) Even hearing [about self-testing], people are just, What? Oh, I'd do that, for sure, instead of me going to the doctor' (pg 6) Doing a self-test doesn't take very long … it's something that can be dealt with, done, gone' (pg 6) " I really strongly believe that … because I was the one that was doing it [self-sampling], I was the one that was in control … and this way it gave me the ability to do it myself and I got all the results, they were fine; … it was also self-empowering, great, I like that. (pg 6)
Finding	Convenience factor self-sampling (more privacy) (U)
Illustration	'I think more people would monitor it that way and test it themselves, like 'well maybe not today, but … eventually I'm going to try it' (pg 6) option of taking it home might make women 'feel more empowered' and contribute to a better relationship with HCPs. (pg 6) An option, they can either do it themselves, or you can offer to do it for them' (pg 6)
Finding	Psychological comfort (high comfort with self-sampling) (U)
Illustration	'doing it yourself would be better' 'would feel more comfortable with that [self-test] than a male doctor' (pg 6) 'shy' of doctors or nurses who visited the communities. (pg 6) Self-testing would also address the 'trust issues' that discouraged women from seeking care from non-indigenous HCPs. (pg 6)
Finding	Psychological comfort (male providers) (U)
Illustration	'feel more comfortable [with a female physician or nurse practitioner] because maybe she's going through the same thing that you're going through." (pg 6) ''embarrassed anyway, no matter who did it (pg 6) ''I only got a Pap test once and I got it from a woman … and she was trying to make me feel comfortable, but it's still really awkward' (pg 6)
Finding	Physical comfort level (slef sampling is less painful) (U)
Illustration	"Women wouldn't be so agitated and nervous about having the [self-] test'"(pg 5) "It's a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference.' '…I think the prevalence [participation] rate would go up' (Pg 5) "It's a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of

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	going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference.' '…I think the prevalence [participation] rate would go up'" (pg 5)
Finding	Physical comfort level (paps hurt) (U)
Illustration	'it's kind of a routine after a while … they open you up and they swab and then you're done' (pg 5) 'the Pap is uncomfortable but it has to be done' (pg 5) 'it doesn't get any easier, like the first time and then the next year, it didn't get any easier for me' (pg 5) 'I didn't like the way it felt, that's why I didn't want to go back there' They don't want their Paps [be]cause it hurt' ". (pg 5) The test also could be exceptionally painful if providers were 'in a rush' or 'rough' and did the procedure 'real quick' (pg 5) 'Some women, when they come to the appointment, they decide they don't want to get a Pap, because it's uncomfortable, they're just afraid' (pg 5)
Finding	Cervical Screening practices in partner communities (privacy concerns) (U)
Illustration	"health centres where staff 'phoning people, telling them, 'we've got your results … you need to make another appointment' (FG2) was upsetting. (pg 5) 'I don't want nobody else to know' 'a lot more private, at home, if you do it by yourself'(pg 5) 'Women would prefer to be discreet, do it themselves and get their own results and not have their results shared with others' (pg 5)
Finding	Cervical Screening practices in partner communities (no efficient follow up) (U)
Illustration	" 'It's crazy here, how long you have to wait for anything and half the time, you don't even get a call … you go to your next doctor's appointment for something else and [have to ask] … by the way, how about my Pap two months ago?'" (pg 5)
Finding	Cervical Screening practices in partner communities (transportation and child care issues) (U)
Illustration	'If the van was full, [you would] have to get a ride … some of us don't have cars, you know' (pg 5) Women with children and work commitments also had difficulty arranging childcare and time off work to keep their appointments. As a HCP commented: 'It certainly does pose challenges with regards to babysitting care and the mother being away for an entire day' (pg 5)
Finding	Cervical Screening practices in partner communities (lack of provider flexibility) (U)

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Illustration	'I don't know how many times they've cancelled Pap tests and [when I asked] "okay well can I get it on Friday? …. [heard back] "No, we don't do them on Friday"' (pg 5)
Finding	Cervical Screening practices in partner communities (no regular screening service) (U)
Illustration	"'Most of us do not have doctors here … you have to drive up to [nearby town]'" (pg 4)
Study: Adcock A 2019	
Finding	Engagement with health services (C)
Illustration	One in seven (14.36%) had not screened in ten years or more (including those never screened), and one in six (16.37%) were unsure how long it had been since their last cervical screen. (pg 2)
Finding	Barriers to screening (shyness, embarrasment) (C)
Illustration	desire for bodily autonomy (retaining privacy, control over one's body) as a reason for not attending regular cervical screening. (pg 3) encompasses concepts of whakamā (embarrassment/shyness/reticence), (Pg 3) tapu (sacred/taboo/forbidden) (pg 3) .desire for bodily autonomy was often related to negative health experiences (eg, painful pelvic examinations, inappropriate actions/comments by HCPs) (pg 3)
Finding	Barriers to screening (cost) (C)
Illustration	Older women were more likely to mention a previous bad experience and were less likely than younger women to mention cost or other financial barriers (pg 3) Opinions varied about cost being a barrier (as many clinics offer free cervical screening). (pg 3) Some highlighted hidden costs (transport, parking, childcare). (pg 3)
Finding	Barriers to screening (education) (C)
Illustration	A lack of health literacy about HPV and cervical cancer and a lack of appropriate/empathetic services were also raised as barriers. (pg 3) relationship-building between communities and health promoters; including whānau/family in HPV education; and ensuring clear information about HPV vaccination (pg 3) HCPs agreed that with appropriate support and education, HPV self-testing will benefit never/under-screened Māori. (pg 4)
Finding	Acceptability (U)
Illustration	positive, with participants using terms such as 'easier', 'more comfortable', 'less intrusive' and 'brilliant'. (pg 3) Nearly two-thirds (61.21%) said they would prefer an HPV self-test to a clinician-

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	collected vaginal swab or a cervical sample collected with a speculum, or to other options. (pg 4)
Finding	Implementation (C)
Illustration	picking up the HPV self-test from a clinic, a pharmacy or other community venue, or receiving it by post/mail (pg 4) concerns were raised about the reliability of post. Participants discussed the value of providing multiple options (pg 4) Flexibility to cater to diverse populations, such as through community outreach services, was suggested for optimum engagement. (pg 5)
Finding	Doing the self-test? (U)
Illustration	most frequently said they would be happy to do the HPV self-test in a clinic or their own home, and emphasised having good support and education to increase their confidence about properly doing it. (pg 5) need for a flexible program, with different options. (pg 5) Urban women were more likely (than rural) to say they would self-test at a community centre. (pg 5) One in five said they would like whānau/family or friends to be around in case they needed help, while some (<15%) said they would like either a community health worker or health practitioner to be nearby, available to help. (pg 5) importance of delivering empathetic cervical screening services, ie 'the process' – taking the time to put mind and body at ease. They emphasised that this process will be important for any new HPV-based cervical screening program. (pg 5)
Finding	HPV self test results (C)
Illustration	if negative a text is fine, but if positive it is better to find out in person or by phone. (Pg 5) ome women emphasised the importance of results being delivered in an empathetic way.(pg 5) Older women were less likely than younger women to want to receive a text asking them to call their health practitioner.(pg 5) Non-PHO-enrolled/unsure women were less likely to choose to receive results by email and more likely to choose a visit with a health practitioner need for clear communication (pg 5) appropriate/empathetic support for women. (pg 5) Concerns were raised about who would be responsible for HPV self-test results and making sure that women are added to the screening register when doing a self-test. (pg 5)
Finding	Postive test results (C)
Illustration	mostly positive response to the idea of seeking further screening (cytology) or diagnosis (colposcopy) if a HPV test was positive – because knowing that something needs follow up would be a good motivator.(pg 5) HCPs wanted assurance that women would be supported appropriately and empathetically to have cytology or colposcopy if they had a positive HPV self-test result, and many

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	supported going straight to colposcopy.(pg 5) Eliminating multiple clinic visits was stressed. (pg 5)
Study: Henderson R et al 2018	
Finding	(Cancer & HPV Experiences) Disease Progression (U)
Illustration	"Once they corrected the cervical cancer you would end up with cancer somewhere else" (p. 96)
Finding	(Cancer & HPV Experiences) Isolating Effects (U)
Illustration	It took time for one family to realize that it was "not just happening to them, it's the whole community" (p. 96)
Finding	(Cancer & HPV Experiences) Mistrust of external influences (U)
Illustration	"It must be the pesticides that are being sprayed on our cultivated land...You can see our area and there is a layer of dust in spring and harvest" (p. 96)
Finding	(Cancer & HPV Experiences) Residential school disruption of community relationships (U)
Illustration	"Opened a whole can of worms; when all the illness and substance abuse went on a skyrocket" (p. 96) Speaking of a sister in the end-stage of cervical cancer, one middle-aged partici- pant attributed the illness to chronic sexual abuse: "There were a couple times she had STIs [sexually transmitted infections] and didn't really know, may have been HPV or may have been another" (p. 96) In both sharing circles, participants argued that residential school litigation broke students' capacity for intimacy throughout their lives, playing out in self-destructive coping strategies (e.g., substance depen- dence, early sexualization) and increased risk of victimization
Finding	(Trauma-informed lens) Contributions of higher Indigenous HPV burden (U)
Illustration	"Our people are starving for affection, support, respect and love." (p. 96)
Finding	(Trauma-informed lens) Residential schools (U)
Illustration	"The nuns make sure to remind you no matter what you do, you're going to hell" (p. 96) "We were taught to be quiet about our private parts...a lot of sexual abuse went on, and spiritual, physical, you name it" (p. 96) One speaker brought these lingering effects of residential schools to- gether by describing a shift in her society from monogamous relation- ships to relaxed sexual boundaries, noting that with her own daughters she "didn't lay down the law like our parents did" (p. 96)

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Finding	(Trauma-informed lens) Normalization of substance abuse among youth (U)
Illustration	"She got an STI because she was taken advantage of" (SC2) by someone other than her partner. The mother connected her daughter's exposure to loss of identity and heavy drinking at parties in her community. Lowered inhibitions brought on by alcohol, and illicit drug use were seen by several other speakers to have become normalized among younger generations, who "see the suffering of their parents and grandparents, and they are running to the substance abuse" (p. 96) In both circles, speakers expressed compassion for younger generations today: "It was a lot easier in my time because we would go to ceremonies...but today, our grandchildren are exposed to drugs, alcohol, everything" (p. 96)
Finding	(Trauma-informed lens) Isolation (U)
Illustration	He observed that such statis- tics reflect disrupted connections between partners, between Elders and youth, as well as with nature and spirituality: "We are not islands, we need to be connected to people and that is what is missing" (p. 96) Therefore, the burden of HPV in FN people is rooted, at least in part, in efforts by people across the lifespan to cope with the vi- olent disruption of family and community connectedness, as well as connection to land and spirituality. This burden is manifested not only in risk-taking behaviour, but also in avoidance of wider health systems: "I think First Nations don't get checked when they are supposed to, to be honest. They just wait until it is too late to help them" (p. 97) Coming from a com- munal society, another participant observed that today many struggle "to fit into an individualistic society, and we don't fit" (p. 98)
Finding	(Trauma-Informed lens) Child welfare (U)
Illustration	"A lot of those kids when they come back to the reserve they are very angry, there are a lot of things that happened [to them] that we don't know about" (p. 97) "It's sad to say that lots of babies have been taken because their mothers were on opium, and the mothers are in treatment and are so sorry" (p. 97)
Finding	(Family & Community) Communication (U)
Illustration	"I am learning a lot in these workshops. My mother died of stomach cancer, my sister of stomach cancer. I had 5 girls, and 4 of them went through breast cancer. My oldest daughter, her cancer spread. When they were younger I made sure they all got their needles. But, you know, I have never had a workshop like this. If I get a cold, I can fight it off. When I got those needles, I was told I was able to fight the sicknesses; it won't kill you—that is what I was told. This is

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	really good for my grandchildren; I will take this message home to my family. I have two nurses in my family, they probably know about it, but this is a really good thing I am still learning...[*speaking in Cree] *I was worried the white people would not take care of us, but they have so far [group laughs]. We need to talk to young ladies about how to take care of themselves." (p. 97)
Finding	(Family & Community) Family (U)
Illustration	Many of the Elders emphasized that health education within Indigenous contexts is anchored in the love and care for children. For one speaker, this affection involves "hugging your children and telling them 'I love you', you don't say goodbye to anyone" (p. 98) "If I'm drinking and drugging, then I'm going to feel guilty about taking my children to the clinic to help them live a healthier life" (p. 98)
Finding	(Family & Community) Role of elders (U)
Illustration	"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children, and out of respect...he's getting that message across [in his sweat lodge ceremonies with men]" (p. 98)
Finding	(Family & Community) Mistrust in HCP (U)
Illustration	"We need to talk to our people in a way they understand, no disrespect to any organization, but people don't look at that material [bro- chures], it's obvious that the way we deliver that message has to be differ- ent" (p. 98) "The health centre sent out a notification and a consent form, and they listed the benefits and risks...and I paid more attention to the risks, and I decided not to allow her to be vaccinated, because as a parent I needed to do what was best for my children" (p. 98)
Finding	(Family & Community) Youth (U)
Illustration	"There are anti-bullying programs...and there are cultural programs; some kids are brave and some are afraid of getting immunized, but all the children support each other" (p. 98)
Finding	(Family & community) Parents & grandparents (U)
Illustration	"The best teachers are your parents" (p. 98)
Finding	(Changing information landscapes) Lack of knowledge (U)
Illustration	"We never see a doctor. The health department has to get a hold of us, the ones that are never home. There are a lot of us in the community... We hear suddenly of these workshops and information centres in the community. We wanted to go but we had other commitments so needed others to go. My niece, she died about 3

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	years ago because she had cancer. They brought her to Edmonton, they started her on chemo...left her kids at home." (p. 97) “One of the girls I used to visit got cervical cancer. She is very angry, she will not talk to me even though I give her support. She was told 3 years ago that she should get the tests and she didn't because she had problems for a long time.” (p. 97) One health director who was herself FN had not approved the HPV vaccine for her own daughter, believing at the time that vaccines are perhaps “not natural, that they are more chemicals given by the government to hurt us” (p. 98)
Finding	(Changing information landscapes) new knowledge (U)
Illustration	“These young people are lucky to get these dif- ferent resources...its scary when you think about it, you didn't think about [the health risk] before, because you didn't know anything about it” (p. 98)
Study: Wakewich P et al 2015	
Finding	Negative body perceptions (stigma of sexually tranmitted infections) (U)
Illustration	"Definitely there’s always going to be a stigma about any kind of thing that’s an STD [STI], because people think it’s dirty or whatever. Like I’ve had people come in, like, even the girl I just saw with genital warts and she just couldn’t believe it, and couldn’t like fathom who she would have got it from because everyone she’s been with has only been with her. Right? " (pg 373) "Actually in the last 6 months that I’ve been there, I’ve had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the community … they’re afraid of the stigma surrounding that so and a lot of times too I wasn’t able to do any [cervical] screening." (pg 373) "Health centers on reserves, don’t want to spread it out too much how many people have HPV. I know how the stereotype works if we pass on the information in non-native communities they will say all native people have HPV." (pg 373)
Finding	Negative body perceptions (body shyness) (U)
Illustration	"I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this …. There’s got to be a better, kinder, gentler, more humane way to do it." (pg 373)
Finding	Negative body perceptions (sexual abuse) (U)
Illustration	"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past …. I will be here

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	for a long time and whenever you need to see me, to come see me, so that even just that little thing and then when they do come I do see them, hopefully that trust builds up and I think that's a big piece with the First Nations." (pg 374) "I think one of the biggest issues um, [short pause] the barriers that prevents people from going to and maybe it's cause it's taboo is because they've been sexually abused." (pg 374)
Finding	Negative experience with governmental helath service (distrust of health authorities) (U)
Illustration	"She [HCP] thinks it's her [that they don't like], 'like, why is no one coming [for screening]'… 'it's not you, it took me 20 years to get to where I'm at now, and I still get people that don't trust me.'" (pg 374) "I guess they were told about some kind of vaccine or something, like years ago, and it was something just, to try to get rid of the Native people." (pg 374) "My doctor who I had since my oldest was born, she just up and quit, so when my next appointment, I had this guy looking at me, it's like [pause] right, so it's not like they stick around here." (pg 375)
Finding	Negative experience with government health service (lack of confidentiality and privacy) (U)
Illustration	"You know what, just sitting in the waiting room, the receptionists are phoning people telling them, we've got your results, you need an, you know, make another appointment, they say the name right out loud, they say what the test is." (pg 375) "Privacy, a small town, I mean, you can hear through the walls, you know, walls talk because there's, it's all in one building and coming in here and everybody sees the first person coming in here, they come to see me, they come to see welfare, they come to see housing, they're going and everybody knows everything."(pg 375) " Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk." (pg 375)
Finding	Role of family (mother - daughter relations) (U)
Illustration	I just taught them that sex isn't, it's not what you call dirty, you know, but there's a certain way you got to go about having, like for sex with a partner, you know, you got to explain that part to them, but to talk openly about sex, it's not what you call a dirty subject, yeah. If you want to know something just feel free to ask." (pg 376) " Like my mother, you know, was the kind of woman that … when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your own personal, private, so when it happened, I had absolutely no knowledge what was going on in my body … And

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	with my daughter, I didn't want her to have that feeling, so I mean, of course I changed and I explained everything but, and then that's the way things are now, women are more informed than they have been in the past." (pg 376)
Finding	Self-collected sampling as a way of increasing women's control of cervical screening (U)
Illustration	"think it [self-sampling] would be, I think it would be great, because it provides them with some autonomy and allows them to take control of the situation. That would be really great actually …. I think it would increase [education opportunities] actually, yup, because it's a lot less clinical right, cause like you say, it's a lot less you know, stripping down and you know, allowing someone else to do the scraping of the cervix, you know, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference." (pg 376) "I think it would definitely be more private … They wouldn't be so embarrassed … all kinds of people are easily embarrassed." (pg 376) "It's simple, it's not like you are having forceps in you." (pg 376)
Study: Butler TL 2020	
Finding	Desire to educate daughter (U)
Illustration	". . . I didn't get it explained too much about that growing up into a woman off my mum. Whereas I want to do the opposite with my daughter and sort of be one step ahead of her, pre- pared and ready. . .. having that information would be good to have it there for her if she ever does want to know about it" (p. 8)
Finding	Health care is a priority (U)
Illustration	"It's about listening to my body. It's about trying to keep it healthy for my children. I want to live as long as I can to bring them up to see them. Not only that, I want to feel good too. And I have had friends who have been diagnosed and that motivates me also to do it. But all in all I do it for myself and my family" (p. 6)
Finding	Health care routine (U)
Illustration	"It's just what you do. I brush my teeth. It keeps coming down to that because it really for me is just about general body maintenance, you brush your teeth, you eat your food, I take my tablets. . . I do what I have to do." (p.6)
Finding	Health more important than shame (U)
Illustration	". . . it's no shame, you have to go and do your tests and all that. . . Every women [sic] in Australia have to do it every time, black or

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	white. . .I don't get shame, because I want to look after myself and for my health too." (p.8)
Finding	Reduced embarrassment after birth (U)
Illustration	". . .Whatever the doctor says I just put my hand up, because after having children you don't care really. Just do what you have to do." (p. 8)
Finding	Children as motivation for prioritising mother health (U)
Illustration	"I'll always remember that lady, and she was so nice too. And she's going, your job now is mummy, and you've got to be here, have this test every two years. And because that was at my six week check-up and I didn't want to have the pap smear, and she's going, you don't want to leave him. And I'm thinking, oh my God, because I was a single mother, oh my God, I've got to look after him. So I did; never missed it." (p. 9)
Finding	Rough examination (U)
Illustration	". . . I think it was [hospital name], and they were the roughest they could be in there. They wer- en't gentle at all, especially having something like that done and they say, "Lay down," and wham, you know, they're in, and that, and it was awful. I said, no, that's it." (p. 9)
Finding	Community as support for screening (U)
Illustration	"Yeah, just yarning with all the women, you know, getting together, having a cuppa. I think someone just bringing it up, approaching it in a way where it's just a thing where you're sitting there having a yarn about–and not feeling shame." (p.9)
Finding	Convenience of 5 year screening (U)
Illustration	". . . when I tell them it was five years, it's a five-year one, they're like, "Five years, even better," knowing that they were right for five years and it picks up early, detects things earlier, they were all for it." (p.10)
Finding	Preference for female HCW (U)
Illustration	". . . they have got the body parts as me, you know what I mean? And if they've gone through something similar to this, they'll understand what it's like. . . . [We have] strong cultural val- ues on Women's Business too. And it's an invasive procedure, that's how I see it." (p.10)
Finding	Comfortable and secure health care visit (U)

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Illustration	"Oh, when I have my testing done here they just make you feel so welcome and so warm, you don't feel so invaded, like a hospital. . .Yeah, you feel like a person, you don't seem–feel like a number." (p.11)
Finding	Availability of screening (C)
Illustration	The availability of screening services outside of work hours and the potential to reduce waiting times in the clinic by having a nurse available to complete screening rather than a doctor were raised as potential solutions. p.11
Finding	Inaccessibility of screening facilities (C)
Illustration	The size of the examination table was a concern for some women, who worried that they may not fit comfort- ably on the table due to having a larger body size. One woman with a physical disability stated the importance of an accessible, height-adjustable examination table that made screening more comfortable and easy for her. p.11
Finding	Time restrictions (C)
Illustration	Women were time-poor and struggled to fit screening appointments in amongst commitments to work and family, which took higher priority in women's lives. p.11
Finding	Lack of privacy for HCW (U)
Illustration	"Especially after, you know, you finish your pap smear and then you're having lunch with them in the tea room, no thanks." (p.12)
Finding	Leading by example (U)
Illustration	"... and being a health worker too because then I can say, "No, I do mine," ... it feels a bit shame but you can talk to your patient and encourage them to do it because you've done it and if I hadn't done it, well, I shouldn't be saying those things to the patient." (p.12)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page – Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract – Page 2,3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Background - Page 3,4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Background – Page 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Materials and methods - Page 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Methods – Page 5,6,7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Data sources and Searches in Methods section - page 5 and 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Data sources and

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PRISMA 2009 Checklist

			Searches in Methods section - page 5 and 6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Inclusion and Exclusion criteria – Page 7 Critical Appraisal – Page 8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Critical Appraisal – Page 8 Data extraction and quality assessment in methods section page 7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and synthesis in methods section page 8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	NA
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Themes and

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			illustrations – table 4 Pages 9-16
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Pages 9-16 and SEM model figure 2

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA for qualitative reviews
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Meta regression and socioecological model on page 16
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	NA in qualitative systematic review
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	All forest plots and study data summarised in tables and supplementary files provided
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results pages 9-16, socioecological model figure 2
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA in qualitative systematic review

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Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Socioecological model figure 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion page 16, 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion ending page 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Declaration page 20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Declaration page 13

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Working towards a comprehensive understanding of HPV and cervical cancer amongst Indigenous women: A qualitative systematic review

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Title: Working towards a comprehensive understanding of HPV and cervical cancer amongst Indigenous women: A qualitative systematic review

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Abstract:

Rationale: Indigenous peoples carry a disproportionate burden of infectious diseases and cancers and are over-represented among the socially disadvantaged of most countries. Human papillomavirus (HPV) is a risk factor and causative agent of cervical, oropharyngeal and other cancers. Recent literature shows evidence of Indigenous populations being at increased risk of HPV infections and its associated cancers.

Objective: This is a qualitative systematic review. The objective of this study was to explore the experiences and barriers Indigenous women face in relation to HPV awareness, knowledge and cervical screening, in order to better understand factors that may mitigate against or facilitate prevention efforts for HPV infection and associated cancers.

Methods: Two investigators independently searched MEDLINE, PubMed, SCOPUS and Web of Science databases (for articles published from inception until 30th June, 2020) using a pre-specified search strategy to identify qualitative studies on narratives of Indigenous women regarding HPV infection awareness, knowledge and cervical screening, across all geographic and income-level settings. Using a 'meta-study' approach, a social ecological model of cervical screening, infection and associated cancer prevention among Indigenous populations was formulated.

Results: Five core themes were identified and formulated within the Social Ecological Model; Intrapersonal factors, Interpersonal factors, Institutional/Organizational factors, Socio-cultural/Community factors and Public Policy. These collectively formed the proposed social ecological model of HPV infection awareness and cervical cancer prevention among Indigenous women. This model has been synthesized by taking into account personal stories of Indigenous women and health care workers, thus offering a more nuanced, organised, structured and culturally sensitive approach to policy translation.

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Conclusion: The social ecological model of HPV infection awareness and cervical cancer prevention among Indigenous women offers a holistic and practical approach for Indigenous health policy makers. It clearly addresses the high risk of Indigenous populations at a global level in experience of both HPV infection and HPV-related cancers.

Keywords: Indigenous women health, Human papillomavirus infections, cervical cancer, Social ecological model, qualitative systematic review, self-sampling strategies

Article Summary:

Strengths and Limitations:

- This systematic review is the first to address the qualitative aspects of HPV infection and associated cancers among Indigenous women at a global level. The review adhered to all protocols to ensure transparency and legitimacy.
- Another strength is the summary of community-provided solutions, in the form of personal narratives, to help decrease transmission of HPV infection and to, in turn, prevent HPV-related cancers.
- Our study highlights the continuing impact of trauma at the public policy level, with our findings providing important evidence of the work required to address the resulting disparities across all health indicators that have resulted from this trauma
- Limitations include the decision to not include male participants, as the current statistics show a dramatic increase in the incidence of HPV associated oropharyngeal carcinoma among men.
- The community-identified solutions presented are mainly from one study; this demonstrates the need for community feedback to be embedded within Indigenous health research projects so that future policies can be derived from community suggestions that will have greater resonance and likely acceptance.

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Introduction

Indigenous peoples, as defined by the United Nations (2004), includes all "people with a historical continuity with pre-invasion and pre-colonial societies that developed on their territories, and who consider themselves distinct from other sectors of the societies now prevailing on those territories." Globally, Indigenous peoples experience disproportionate health inequalities, in comparison to non-Indigenous populations, due to a unique history of colonial settlement. This has resulted in environmental dispossession of traditional lands and resources, which has negatively impacted spiritual connections with the land and with each other[1-4]. Indigenous peoples are over-represented among the socially disadvantaged in almost every country, particularly in developed countries[5]. Public policy created by colonial settlers in many countries has been aimed at the assimilation and cultural annihilation of Indigenous peoples, resulting in mass loss of culture, language and community. Examples of assimilation policies include the Residential School System in Canada[6], the Stolen Generations of Australia[7] and racial amalgamation in New Zealand[8]. Adelson's (2005) notion of "the embodiment of inequity," and similar works[9,10] have demanded attention be paid to the ways in which the impacts of colonial legacies, marginalisation and discrimination manifest in Indigenous peoples' health outcomes[10]. Previous works[11,12] have explored the relationship between colonization and the intersectional nature of social determinants of health, such as race, demonstrating that health care patterns are rooted in historic economic, social and political circumstances and power relations[11]. This literature provides insight into the commonality of high-risk health profiles experienced by Indigenous peoples around the world and has created awareness of the need to take special measures to protect the rights of Indigenous peoples[13,14].

Human papillomavirus (HPV) infection and associated cancers have gained significant public and research attention in recent decades. WHO has launched a three step program for elimination of cervical cancer by 2050[15], and the International Society for Papillomavirus

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Research has demonstrated support[16] for the program with a call for equity in elimination strategies, particularly reinforcing best practices for Indigenous populations. While screening tools, such as the Papanicolaou test (Pap test), have been successful in reducing cervical cancer overall, and HPV vaccination programs are expected to reduce rates of HPV infection and all HPV-related cancers, Indigenous communities continue to experience significantly higher rates of cervical cancer and oncogenic HPV infection compared with non-Indigenous populations[17-21]. This discrepancy in cancer is related to Indigenous women being less likely to have been screened[22-24] and compounded by implications of colonial legacies[1-4].

It has been reported[25] that the prevalence of high risk HPV infection related to invasive cervical cancer was 32.7% in American Indian/Alaskan Native populations in USA compared to 24.9% in the non-Indigenous populations. A crude incidence of HPV related cervical cancer in 9 Indigenous populations of the Brazillian Amazon was 46/100,000; which indicated a public health concern, also suggesting a weakness in the current secondary prevention program.[26] Additionally, a study of 3 specific tribes (Aymara, Mestizo, Quechua) of Bolivian Andean women has revealed an unexpected number of cases infected with HPV.[27] The number of sexual partners was an important predictor of HPV infection in the Metit population of Canada.[28] The younger age of onset of sexual practices and less likely use of condoms has been observed in Maori tribes also.[29] A Pilaga community in Northern Argentina showed very high prevalence of HPV infection (46.7%).[30] Several reasons for not attending cervical cancer screening were described by Canadian Inuit women, including previous painful experiences and embarrassment to be examined by a male nurse.[28]

HPV-related cancers are mediated by risk factors such as sexual behaviour, tobacco use and screening participation, but these relevant health behaviours of affected Indigenous Peoples co-exist within multidimensional contexts[31,32]. Bronfenbrenner's (1974) Social Ecological Model provides a theoretical framework in which to understand the micro and macro

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dimensions of a phenomenon, by simultaneously considering related political, social, physical and economic environments[33,34]. Many health promotion efforts focus on individual choice, ignoring the social context of health-related behaviours. The social ecological model acknowledges the importance of the social environmental impacts on human health, beyond the ideology of individual responsibility[34]. It may be that current HPV prevention programs are not successfully recognising and/or incorporating strategies targeted at factors beyond the individual. This is a particular shortcoming among initiatives targeting Indigenous communities who, at a global level, are much more collectivistic and group-oriented than their non-Indigenous peers[17].

This review therefore seeks to better understand facilitators and barriers to HPV and associated cancer prevention, as experienced by Indigenous peoples on an international scale at each level of the social ecological model[34]. Combining existing literature within the framework of the model acknowledges the significance of Indigenous culture and its interaction with socioeconomic conditions[17,34]. The objective of this systematic review was to synthesise qualitative perspectives and experiences of Indigenous women and health care workers in the context of HPV infection, associated cervical cancer and screening methods to establish a foundation that might better facilitate targeted, holistic, and culturally relevant prevention strategies.

Materials and methods

This systematic review has been registered in PROSPERO (CRD42020207643) and the Joanna Briggs Systematic Reviews register. A prior search of the PROSPERO register revealed no similar studies. Both the PRISMA guidelines[35] and the Enhancing Transparency in Reporting the Synthesis of Qualitative Research (ENTREQ) statement (Table 1) were followed in the conduct of this systematic review.

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Patient and Public involvement

No patient or public involvement was undertaken in this study.

Table 1: ENTREQ Checklist

Item	Description	Reported on Page #
Aim	Awareness of- and barriers to- human papillomavirus infection screening amongst Indigenous women globally	5
Synthesis methodology	Thematic content analysis guided initial data extraction for synthesis, and the Social Ecological Model provided a theoretical framework to understand synthesised findings	5-8
Approach to searching	Pre-established search strategy which involved using terms describing the population of interest, the phenomenon we are researching as well as study designs to be included	6
Inclusion criteria	*Inclusion*: Studies from inception till June 2020 were included in the study. Research based on the experiences of women of Indigenous identity or health care worker of Indigenous identity, personal illustrations of HPV knowledge and cervical cancer, first-person accounts; topics focused on cancers associated with HPV infection; qualitative and/or narrative, or mixed methodology; available in English; accessible via hardcopy or download; peer-reviewed *Exclusion*: research based only on vaccination for HPV; non-Indigenous participants; only Indigenous male narratives; quantitative methodology	7

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Data sources	MEDLINE, PubMed, SCOPUS, and Web of Science databases; each search tailored per the design of individual database. In our search for published studies, we made use of facilities when given to run 'related' searches and the bibliography of each article was manually scanned for possible additions to the study	7
Electronic Search Strategy	Terms utilised for literature search included: 'HPV' 'Cervical cancer' 'Indigenous' 'female' 'narrative' 'story' 'qualitative' 'mixed methods'	6
Study Screening methods	Two independent researchers screened studies for inclusion in the qualitative systematic review. Titles were first reviewed, then abstracts and those considered relevant by either investigator advanced to full text review.	7
Study characteristic	See Table 2	Table 2
Study selection results	1377 records were returned from initial search, 963 were excluded due to duplication, 24 shortlisted, 11 studies fully satisfied inclusion criteria. After appraisal, one article was removed due to lack of illustrations. 10 papers were included	Figure 1
Rationale for appraisal	Utilizing JBI SUMARI software, articles were appraised according to the CASP (2013) method of quality appraisal.	S2 and S3
Appraisal Items	See S2 and S3	S2 and S3
Appraisal Process	Appraisal was conducted independently by both reviewers and then findings were discussed, and consensus was required before moving forward.	8

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Appraisal Results	One article was excluded after the appraisal because it did not satisfy inclusion criteria of personal illustrations	8
Data extraction	All text under headings “Results” and “Conclusions,” as well as all findings under the heading “Discussion” were analysed. Data was manually extracted with highlighters from printed versions of appraised articles and then imputed into the JBI SUMARI software.	Table 3, Table 2
Software	JBI SUMARI	7
Number of Reviewers	Two reviewers independently reviewed articles and extracted data. Findings were then compared, discussed and compiled.	7
Coding	Data was coded from selected articles, going line by line to search for concepts and considering the author-prescribed themes.	8
Study Comparison	All findings were individually highlighted and written on a white board and then connections were made between findings and categories were created based on similarities within and across extracted data.	8
Derivation of themes	The process of deriving themes was abductive.	-
Quotations	Table 4	Page 9-13 Table 3
Synthesis output	Discussion and Figure 2	Pages 13-15 Table 3, figure 2

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Table 2: Characteristics of Included Studies

Study	Methods for data collection and analysis	Country	Phenomena of interest	Setting/context/ culture	Participant characteristics and sample size
Cerigo H et al. 2012.	Mixed methods and focused group interviews Thematic content analysis using NVivo software	Canada	Attitudes and experiences about cervical cancer Pap smear and HPV vaccine	Inuit women in Nunavik, Quebec	Participant characteristics: women, 31-55years Sample size: 6
Henderson R et al. 2018.	Sharing circles (6) Analysis: NVivo	Canada	Identifying and validating known barriers and supports to HPV vaccination among First Nations people in Alberta	First Nations community leaders from Alberta	Participant Characteristics: 21 females, 3 males, all First Nations Elders, Health service directors and Community leaders Sample Size: 24
O'Brien B et al. 2009.	Ethnography, Interviews Analysis: Themes and relationships among transcriptions and researcher perspectives were identified	Canada	Exploration of attitudes of beliefs of First Nations Cree women living on reserve to gain insights into how cervical screening can be better used	Community based health centres and private locations , culture : Cree First Nations	Participant Characteristics: All women, Cree First Nation, Sample size: 8
Tratt E et al. 2020.	Fuzzy Cognitive Mapping (Visual Mapping) mapped and weighted participant knowledge to illustrate the impact of one on the other	Canada	Explore the possible implementation of HPV self sampling	Inuit women in Nunavik Quebec	Participant characteristics: All women over age of 25 years Sample size: 27
Adcock A. 2019.	: Mixed methods approach, Face to face Hui, recruitment by communtiy based	New Zealand	Potential acceptability of HPV self-sampling among Maori women who were never or under screened and also a survey of the HPV self	Maori women and health care providers (both Maori and Non-Maori)	Participant characteristics: Sample size: 106 (Hui), 397 (surveys)

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	researchers Surveys via Qualtrics Hui (focus groups, interviews)was recorded, transcribed then analyzed (Thematic analysis) by NVivo software.		testing among Health care providers.		
Maar N et al. 2016.	Participatory action Research, mixed methods approach, Semi-structured Interviews, Analysis: Thematic analysis	Canada	Understand the effective ways to reach First Nations women with screening information, Potential or promising strategies to help motivate women to participate in cervical cancer screening and address the special issues for First Nations women with respect to cervical cancer screening.	First Nations people in Northwest Ontario (Lake Superior or around Lake Nipigon)	Participant characteristics: First Nations people in Northwest Ontario (Lake Superior or around Lake Nipigon) Sample size: 70-840 members
Allen-Leigh B et al. 2017.	9 focus groups(5-11 per group), 29 interviews, Audio recordings, transcriptions and field notes taken Analysis: two stage coding process (first level - descriptive coding; second stage - pattern codes)	Mexico (South America)	Study the barriers to use of self-sampled HPV testing and cytology among low-income Indigenous women residing in rural areas of Mexico	Mam women in Chiapas, Nahuatl women in Puebla and Huichol Women in Jalisco	Participant Characteristics: 20 years and above, 3 ethnic groups (Mam women in Chiapas, Nahuatl women in Puebla and Huichol Women in Jalisco) Sample size:
Wakewich P et al. 2015.	Participatory action research framework, purpose sampling, Traditional 'talking' circles, Interviews Analysis: Open coding , NVivo software	Canada	Develop culturally sensitive approaches to cervical cancer screening and to explore the feasibility of self-sampling for HPV as an alternative screening modality to Pap cytology	First Nations women in Northwestern Ontario	Participant Characteristics: Health care workers and Elders Sample Size: 16 interviews of healthcare workers and 8 community focus groups (total 69 women)

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Zehbe et al. 2017.	Data Collection: Field Trips, Teleconferences, in depth interviews, 8 focus groups, Semi-structured grounded theory approach Analysis: Nvivo 9 (QSR International)	Canada	Experiences with cancer in their community, followed by community concerns about cancer, knowledge and awareness of cervical cancer	First Nations Women, Northweat Ontario	Participant Characteristics: Health care workers - 20's to 60's age groups, Fifteen HCPs were female and one was male. Community participants - up to 70 years age, Women from First Nations Identity Of these, 10 women self-identified as First Nations. Sample size: 16 (interviews), 8 focus groups = total 69 women
Butler TL AK. 2020.	Convenience sampling Yarning (followed a question guide), interpreting, listening, sharing, health questionnaire Audio recordings, transcriptions (verbatim) and field notes Thematic analysis (NVivo)	Australia	Barriers and enablers to cervical cancer screening	Aboriginal Community Controlled Health Services or Government-run clinics that serve a large proportion of Indigenous clients - Queensland, New South Wales, Northern Territory	50 women (44 Aboriginal, 6 Torres Strait Islander), ages 25-60 6-14 women from 5 different Public Health Care Centres

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Table 3: Illustrations from all the included studies arranged according to all the synthesized categories under each level of the Socio-Ecological Model

Social Ecological Model level	Synthesized Finding	Illustration
Public Policy	Trauma	"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past …. I will be here for a long time and whenever you need to see me, to come see me, so that even just that little thing and then when they do come I do see them, hopefully that trust builds up and I think that's a big piece with the First Nations" (p. 374). (*Wakewich et al., 2015*)
		"I think one of the biggest issues um, [short pause] the barriers that prevents people from going to and maybe it's cause it's taboo is because they've been sexually abused" (p. 374). (*Wakewich et al., 2015*)
		"I guess they were told about some kind of vaccine or something, like years ago, and it was something just, to try to get rid of the Native people" (p. 374). (*Wakewich et al., 2015*)
		[Residential schools] "Opened a whole can of worms; when all the illness and substance abuse went on a skyrocket" (p. 96). (*Henderson et al., 2018*)
		Speaking of a sister in the end-stage of cervical cancer, one middle-aged participant attributed the illness to chronic sexual abuse: "There were a couple times she had STIs [sexually transmitted infections] and didn't really know, may have been HPV or may have been another" (p. 96). (*Henderson et al., 2018*)
		In both sharing circles, participants argued that residential school litigation broke students' capacity for intimacy throughout their lives, playing out in self-destructive coping strategies (e.g., substance depen-dence, early sexualization) and increased risk of victimization (p. 96). (*Henderson et al., 2018*)
		"It must be the pesticides that are being sprayed on our cultivated land...You can see our area and there is a layer of dust in spring and harvest" (p. 96). (*Henderson et al., 2018*)
		"Our people are starving for affection, support, respect and love" (p. 96). (*Henderson et al., 2018*)
		"The nuns make sure to remind you no matter what you do, you're going to hell" (p. 96). (*Henderson et al., 2018*)
		"We were taught to be quiet about our private parts...a lot of sexual abuse went on, and spiritual, physical, you name it" (p. 96). (*Henderson et al., 2018*)

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		One speaker brought these lingering effects of residential schools together by describing a shift in her society from monogamous relation- ships to relaxed sexual boundaries, noting that with her own daughters she "didn't lay down the law like our parents did" (p. 96). (*Henderson et al., 2018*)
		"She got an STI because she was taken advantage of" (SC2) by someone other than her partner. The mother connected her daughter's exposure to loss of identity and heavy drinking at parties in her community. Lowered inhibitions brought on by alcohol, and illicit drug use were seen by several other speakers to have become normalized among younger generations, who "see the suffering of their parents and grandparents, and they are running to the substance abuse" (p. 96). (*Henderson et al., 2018*)
		In both circles, speakers expressed compassion for younger generations today: "It was a lot easier in my time because we would go to ceremonies...but today, our grandchildren are exposed to drugs, alcohol, everything" (p. 96). (*Henderson et al., 2018*)
		He observed that such statistics reflect disrupted connections between partners, between Elders and youth, as well as with nature and spirituality: "We are not islands, we need to be connected to people and that is what is missing" (p. 96). (*Henderson et al., 2018*)
		Therefore, the burden of HPV in FN people is rooted, at least in part, in efforts by people across the lifespan to cope with the violent disruption of family and community connectedness, as well as connection to land and spirituality. This burden is manifested not only in risk-taking behaviour, but also in avoidance of wider health systems: "I think First Nations don't get checked when they are supposed to, to be honest. They just wait until it is too late to help them" (p. 97). (*Henderson et al., 2018*)
		Coming from a communal society, another participant observed that today many struggle "to fit into an individualistic society, and we don't fit" (p. 98). (*Henderson et al., 2018*)
		[When discussing child welfare] "A lot of those kids when they come back to the reserve they are very angry, there are a lot of things that happened [to them] that we don't know about" (p. 97). (*Henderson et al., 2018*)
		One health director who was herself FN had not approved the HPV vaccine for her own daughter, believing at the time that vaccines are perhaps "not natural, that they are more chemicals given by the government to hurt us" (p. 98). (*Henderson et al., 2018*)
		"If I'm drinking and drugging, then I'm going to feel

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		guilty about taking my children to the clinic to help them live a healthier life" (p. 98). (*Henderson et al., 2018*)
Socio-Cultural/Community	Gender Imbalances	"The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn't even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn't believe that" (p. 4). (*Allen-Leigh et al., 2017*)
		"I decided it [to perform the HPV test] myself, alone. I don't ask anyone's permission. ... How am I going to ask him if he [her husband] wants it or not? It's not for him, it's for me" (p. 4). (*Allen-Leigh et al., 2017*)
		"Because it's a relationship thing" that implied that someone had 'cheated' (p. 7). (*Zehbe et al., 2017*)
		"… I know only women get cervical cancer but … they've helping it along in the same sense where they've passing it back and forth" (p. 7). (*Zehbe et al., 2017*)
	STI Stigma	Focus group participants were concerned that any publication of information about HPV rates on reserve could have negative effects, contributing to the stereotype that 'all native people have HPV' (FG6). As one HCP 15ummarized: 'A lot more education has to be done about HPV and cervical cancer together' to alleviate these concerns (p. 7). (*Zehbe et al., 2017*)
		"Definitely there's always going to be a stigma about any kind of thing that's an STD [STI], because people think it's dirty or whatever. Like I've had people come in, like, even the girl I just saw with genital warts and she just couldn't believe it and couldn't like fathom who she would have got it from because everyone she's been with has only been with her. Right?" (p. 373). (*Wakewich et al., 2015*)
		Actually in the last 6 months that I've been there, I've had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the community … they're afraid of the stigma surrounding that so and a lot of times too I wasn't able to do any [cervical] screening" (p. 373). (*Wakewich et al., 2015*)
		"Health centers on reserves, don't want to spread it out too much how many people have HPV. I know how the stereotype works if we pass on the information in non-native communities they will say all native people have HPV" (p. 373). (*Wakewich et al., 2015*)
	Isolating Effects	It took time for one family to realize that it was "not just happening to them, it's the whole community" (p. 96). (*Henderson et al., 2018*)

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	Community Support	"Yeah, just yarning with all the women, you know, getting together, having a cuppa. I think someone just bringing it up, approaching it in a way where it's just a thing where you're sitting there having a yarn about–and not feeling shame." (p.9) (*Butler et al., 2020*)
Institutional/Organisational	Health care worker Cultural Awareness	Cultural awareness of healthcare workers was a major facilitator, reflected in positive patient experiences and more use of adequate communication means (p. 3). (*Tratt et al., 2020*)
		An additional path through which more cultural awareness could have positive impacts on the use of health services was mediated by increased confidence in patients' capabilities; thus, allowing a more active role of patient in decision-making (p. 4). (*Tratt et al., 2020*)
		Previous positive experiences of health personnel in their interaction with Inuit was told to be a factor that highly contributes to increase cultural awareness, countering the negative effect of a healthcare system based on non-Inuit cultural values (p. 4). (*Tratt et al., 2020*)
		"Well, word it a way to have to do with childbearing, [which] is a sacred gift that's only given to women, let's keep our bodies healthy by getting annual check-ups, and you use some culturally appropriate like pictures, or something" (p. 8). (*Maar et al., 2016*)
		"We need to talk to our people in a way they understand, no disrespect to any organization, but people don't look at that material [brochures], it's obvious that the way we deliver that message has to be different" (p. 98). (*Henderson et al., 2018*)
	Integrated Services	"You need to make sure that you train the trainers, like teach the CHRs [community health representatives], and the women in the community that are providing service" (p. 5). (*Maar et al., 2016*)
		"So if you're having something that they're able to come and access, [like a] sewing circle, or community kitchen, something that they're gonna get, then they'll come" (p. 6). (*Maar et al., 2016*)
		"Well, since you're here, you know, we haven't screened for this, and when was your last Pap and what about colon cancer screening?" (p. 7). (*Maar et al., 2016*)
		"If it [other STI testing] can be done on the same swab [as self-sampling], I would say, the value of that would be huge" (p. 8). (*Maar et al., 2016*)
		HCPs felt that education around self-testing might better be done in the context of a well woman healthcare visit, as "part of a physical, that might get best [results]." Women would not necessarily understand that they would still need to go to their HCP for other

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		reproductive healthcare: "They would, they'd probably think, well, I've had that done, so I don't need this done" (p. 7). (*Zehbe et al., 2017*)
		Eliminating multiple clinic visits was stressed (p. 5). (*Adcock et al., 2019*)
	Negative HC Experience	"I still don't know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don't know! Nobody explained anything to me because, maybe because I didn't ask. Maybe they thought that I understood everything that was going on, but I didn't" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"When you're in (western Canadian city A) and (western Canadian city B) sometimes you just get pushed along as if you're just a file. 'Here you go, this is what's wrong, this is what we're going to do, see you later.' I don't understand where I go from there, how's my health going to be affected by it now" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		". . . I think it was [hospital name], and they were the roughest they could be in there. They weren't gentle at all, especially having something like that done and they say, "Lay down," and wham, you know, they're in, and that, and it was awful. I said, no, that's it." (p. 9) (*Butler et al., 2020*)
		Desire for bodily autonomy was often related to negative health experiences (i.e., painful pelvic examinations, inappropriate actions/comments by HCPs) (p. 3). (*Adcock et al., 2019*)
		"We never see a doctor. The health department has to get a hold of us, the ones that are never home. There are a lot of us in the community... We hear suddenly of these workshops and information centres in the community. We wanted to go but we had other commitments so needed others to go. My niece, she died about 3 years ago because she had cancer. They brought her to Edmonton, they started her on chemo...left her kids at home." (p. 97). (*Henderson et al., 2018*)
		"One of the girls I used to visit got cervical cancer. She is very angry, she will not talk to me even though I give her support. She was told 3 years ago that she should get the tests and she didn't because she had problems for a long time" (p. 97). (*Henderson et al., 2018*)
		"She [HCP] thinks it's her [that they don't like], 'like, why is no one coming [for screening]'… 'it's not you, it took me 20 years to get to where I'm at now, and I still get people that don't trust me'" (p. 374). (*Wakewich et al., 2015*)
		"My doctor who I had since my oldest was born, she just up and quit, so when my next appointment, I had this

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		guy looking at me, it's like [pause] right, so it's not like they stick around here" (p. 375). (*Wakewich et al., 2015*)
	Language	Participants reported the major importance of using visual language, ranking higher than using Inuktitut, and they stressed the communication problems behind using French or medical jargon (p. 4). (*Tratt et al., 2020*)
		"I have a sweet little old lady, she's probably close to 80; she won't speak it, she won't speak English, she understands you, she can speak English, but she would rather speak in her own language" (p. 6). (*Maar et al., 2016*)
	Male Health Care worker	"I had it done by a male doctor and I was uncomfortable . . . because of the way I was raised. My [traditional] grandmother always told me that the only man that should know you like that is your husband" (p. 87). (*O'Brien, Mill & Wilson, 2009*)
		"My first Pap smear was done by a male and I was very nervous and I just didn't want it done. Maybe that's why I didn't have another one [Pap smear] . . . because of that [male gender]" (p. 87). (*O'Brien, Mill & Wilson, 2009*)
		"Some people don't go [to get Pap smears], maybe they're shy … all of my friends don't go probably because they are shy or sometimes they don't want to [be] checked by a man, if it is a man that is a nurse" (p. 3). (*Cerigo et al., 2012*)
	Positive Health Care Experience	"My appointment was scheduled when the offices are closed so I knew he was going to be giving me some fairly bad news. The office was empty, and the lights were all turned down. It was a real comfortable surroundings. He gave me a lot of information, pamphlets, and people to phone for support" (p. 85). (*O'Brien, Mill & Wilson, 2009*)
		"Oh, when I have my testing done here they just make you feel so welcome and so warm, you don't feel so invaded, like a hospital. . .Yeah, you feel like a person, you don't seem–feel like a number." (p.11) (*Butler et al., 2020)*
		"The women are, you know, fairly open once they're comfortable" (p. 6). (*Maar et al., 2016*)
		"It's all about trust and relationships, like if you can't build a relationship you're not gonna get anywhere, you might not even see them again" (p. 7). (*Maar et al., 2016*)
	Female Health Care Worker	"The girls are always looking for female doctors" (p. 8). (*Maar et al., 2016*)
		"Feel more comfortable [with a female physician or nurse practitioner] because maybe she's going through the same thing that you're going through" (p. 6). (*Zehbe et al., 2017*)

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		"I only got a Pap test once and I got it from a woman … and she was trying to make me feel comfortable, but it's still really awkward" (p. 6). (*Zehbe et al., 2017*)
		". . . they have got the body parts as me, you know what I mean? And if they've gone through something similar to this, they'll understand what it's like. . . . [We have] strong cultural val- ues on Women's Business too. And it's an invasive procedure, that's how I see it." (p.10) (*Butler et al., 2020)*
	Lack of Results	"A lot of us say, and we've talked about this before, when we get a Papanicolaou, the results don't arrive, and we don't know what it is that's going on. There we are with the doctor, asking why the results don't arrive. ... Whatever it [the result] is, they should give it to me" (p. 4). (*Allen-Leigh et al., 2017*)
		"It's crazy here, how long you have to wait for anything and half the time, you don't even get a call … you go to your next doctor's appointment for something else and [have to ask] … by the way, how about my Pap two months ago?" (p. 5) (*Zehbe et al., 2017*)
	Privacy	Health centers where staff 'phoning people, telling them, 'we've got your results … you need to make another appointment' was upsetting (p. 5). (*Zehbe et al., 2017*)
		"Especially after, you know, you finish your pap smear and then you're having lunch with them in the tea room, no thanks." (p.12) (*Butler et al., 2020)*
		"You know what, just sitting in the waiting room, the receptionists are phoning people telling them, we've got your results, you need an, you know, make another appointment, they say the name right out loud, they say what the test is" (p. 375). (*Wakewich et al., 2015*)
		"Privacy, a small town, I mean, you can hear through the walls, you know, walls talk because there's, it's all in one building and coming in here and everybody sees the first person coming in here, they come to see me, they come to see welfare, they come to see housing, they're going and everybody knows everything" (p. 375). (*Wakewich et al., 2015*)
		"Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk" (p. 375). (*Wakewich et al., 2015*)
	Screening convenience	"I don't know how many times they've cancelled Pap tests and [when I asked] okay well can I get it on Friday? …. [heard back] 'No, we don't do them on Friday'" (p. 5). (*Zehbe et al., 2017*)
		"Most of us do not have doctors here … you have to drive up to [nearby town]" (p. 4). (*Zehbe et al., 2017*)

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		". . . when I tell them it was five years, it's a five-year one, they're like, "Five years, even better," knowing that they were right for five years and it picks up early, detects things earlier, they were all for it." (p.10) (*Butler et al., 2020*)
		"The availability of screening services outside of work hours and the potential to reduce waiting times in the clinic by having a nurse available to complete screening rather than a doctor were raised as potential solutions." p.11 (*Butler et al., 2020*)
		"The size of the examination table was a concern for some women, who worried that they may not fit comfort- ably on the table due to having a larger body size. One woman with a physical disability stated the importance of an accessible, height-adjustable examination table that made screening more comfortable and easy for her." (p.11) (*Butler et al., 2020*)
		"Women were time-poor and struggled to fit screening appointments in amongst commitments to work and family, which took higher priority in women's lives." p.11 (Butler et al., 2020)
Interpersonal Factors	Family Relations	"Well, my mother, you know, always went for her physicals; thus I learned that I was to go for my physicals... the families that do have it, you can see that progression of preventative health care" (p. 5). (*Maar et al., 2016*)
		". . . I didn't get it explained too much about that growing up into a woman off my mum. Whereas I want to do the opposite with my daughter and sort of be one step ahead of her, pre- pared and ready. . .. having that information would be good to have it there for her if she ever does want to know about it" *(p. 8) (Butler et al., 2020)*
		"I am learning a lot in these workshops. My mother died of stomach cancer, my sister of stomach cancer. I had 5 girls, and 4 of them went through breast cancer. My oldest daughter, her cancer spread. When they were younger I made sure they all got their needles. But, you know, I have never had a workshop like this. If I get a cold, I can fight it off. When I got those needles, I was told I was able to fight the sicknesses; it won't kill you—that is what I was told. This is really good for my grandchildren; I will take this message home to my family. I have two nurses in my family, they probably know about it, but this is a really good thing I am still learning...[*speaking in Cree] *I was worried the white people would not take care of us, but they have so far [group laughs]. We need to talk to young ladies about how to take care of themselves" (p. 97). (*Henderson et al., 2018*)

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		Many of the Elders emphasized that health education within Indige- nous contexts is anchored in the love and care for children. For one speaker, this affection involves "hugging your children and telling them 'I love you', you don't say goodbye to anyone" (p. 98). (*Henderson et al., 2018*)
		"There are anti-bullying programs...and there are cultural programs; some kids are brave and some are afraid of getting immunized, but all the children support each other" (p. 98). (*Henderson et al., 2018*)
		"The best teachers are your parents" (p. 98). (*Henderson et al., 2018*)
		"I just taught them that sex isn't, it's not what you call dirty, you know, but there's a certain way you got to go about having, like for sex with a partner, you know, you got to explain that part to them, but to talk openly about sex, it's not what you call a dirty subject, yeah. If you want to know something just feel free to ask" (p. 376). (*Wakewich et al., 2015*)
		"Like my mother, you know, was the kind of woman that … when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your own personal, private, so when it happened, I had absolutely no knowledge what was going on in my body … And with my daughter, I didn't want her to have that feeling, so I mean, of course I changed and I explained everything but, and then that's the way things are now, women are more informed than they have been in the past" (p. 376). (*Wakewich et al., 2015*)
	Intergenerational Communication	"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children, and out of respect...he's getting that message across [in his sweat lodge ceremonies with men]" (p. 98). (*Henderson et al., 2018*)
Intrapersonal Factors	Lack of Knowledge	The vast majority of participating women expressed the need to get screened after understanding medical facts, including but not limited to the absence of symptoms (p. 4). (*Tratt et al., 2020*)
		"Between these two sicknesses [HPV and cervical cancer] we're in danger, we should go to the clinic or a doctor. If we feel pain in the womb, go to a doctor" (p. 4). (*Allen-Leigh et al., 2017*)
		"Well, as far as I know, the virus is transmitted through sexual contact. Then this human papillomavirus, it begins without any warning. Then later it progresses, then the discomfort in our parts [genitals] begins and that's when the discharge starts and it progresses to the cervix and when it gets to the cervix it goes into the uterus and that is when the doctor sends us for an

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		operation" (p. 4). (*Allen-Leigh et al., 2017*)
		"Moderator: What is cervical cancer, what do you know about cervical cancer? Huichol woman 1: They say you can die of cancer, if you don't detect it early. Moderator: And how do you detect it? Huichol woman 1: With the Papanicolaou, doing it periodically. Moderator: And what's periodically? Huichol woman 1: Every three months. Moderator: Everyone, how often do you think you need to get a Papanicolaou? Huichol woman 2: Once a year. Huichol woman 3: Depends on how you feel, once a year or every two years, I get it every two years. Moderator: What do you mean, how you feel? Huichol woman 3: If you feel burning" (p. 4). (*Allen-Leigh et al., 2017*)
		Further, we found that some women did not fully understand the purpose of the Pap smear as a method of cervical cancer screening (p. 5). (*Cerigo et al., 2012)*
		A sizable proportion of the women were unable to identify a cervical cancer risk factor and were unsure if detecting cervical cancer early would affect the chance for a cure (p. 5). (*Cerigo et al., 2012*)
		Although unintended, participants of both focus groups directed the conversation towards an educational focus due to the limited knowledge about HPV, cervical cancer and Pap smears among participants (p. 6). (*Cerigo et al., 2012*)
		A lack of health literacy about HPV and cervical cancer and a lack of appropriate/empathetic services were also raised as barriers (p. 3). (*Adcock et al., 2019*)
		"The health centre sent out a notification and a consent form, and they listed the benefits and risks...and I paid more attention to the risks, and I decided not to allow her to be vaccinated, because as a parent I needed to do what was best for my children" (p. 98). (*Henderson et al., 2018*)
		"These young people are lucky to get these different resources...its scary when you think about it, you didn't think about [the health risk] before, because you didn't know anything about it" (p. 98). (*Henderson et al., 2018*)
	Awareness of HPV	"Explain [what the Pap test is] then they accept the value of early diagnosis: I find that once I explain to them the importance of early detection and [that] it can be treated, they're more agreeable to [a Pap test]" (p. 4). (*Maar et al., 2016*)
		"The biggest motivation is education, I think. You know, just teach them, let them know that the service is there, that they need to take it, it's important that they have it" (p. 4). (*Maar et al., 2016*)

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		"We know cervical cancer is 100% preventable, and I know because I read stats and see statistics that Aboriginal women are the number one on the list for dying from this" (p. 5). (*Maar et al., 2016*)
		"If you detect it in time, you can take that sickness out so it won't progress any more and with a treatment you are fine" (p. 4). (*Allen-Leigh et al., 2017*)
		"There is a vaccine for human papillomavirus. I heard it on the radio, that there are vaccines" (p. 4). (*Allen-Leigh et al., 2017*)
	Embarrassment	"For me, I am used to getting my Pap, I'm fine with it. But for the younger girls, [or] the shy ones, [or] those who were sexually abused, they might prefer [the HPV self-sampling method]" (p. 4). (*Tratt et al., 2020*)
		". . .Whatever the doctor says I just put my hand up, because after having children you don't care really. Just do what you have to do." (p. 8) (*Butler et al., 2020*)
		Among our population, older women reported more feelings of embarrassment than the young women (p. 5). (*Cerigo et al., 2012*)
		"Embarrassed anyway, no matter who did it" (p. 6). (*Zehbe et al., 2017*)
		Desire for bodily autonomy (retaining privacy, control over one's body) as a reason for not attending regular cervical screening - encompasses concepts of whakamā (embarrassment/shyness/reticence), tapu (sacred/taboo/forbidden) (p. 3). (*Adcock et al., 2019)*
		"I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this …. There's got to be a better, kinder, gentler, more humane way to do it" (p. 373). (*Wakewich et al., 2015)*
	Fear of Prognosis	" People are scared. I'm thinking of my sister. She hasn't gone back to get retested. She did have an abnormal result the first time. I think she's afraid. Some people are afraid that if they do find cancer, it's downhill from there" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"My fear is of having [cervical cancer] . . . motivates me. I want to have good health. I don't want that kind of a disease" (p. 86). (*O'Brien, Mill & Wilson, 2009*)
		"I would try and keep encouraging her . . . to tell her, "If they find it early they can do something about it." I try and set examples for my clients. I tell people, "Did you know this lady had this kind of cancer because she never got tested, and they could have done something about it right away and she would still be here" (p. 86). (*O'Brien, Mill & Wilson, 2009*)

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		"HPV, like, whoa, I don't have that, like, I don't even want to know if I have that" (p. 7). (*Zehbe et al., 2017*)
		"Once they corrected the cervical cancer you would end up with cancer somewhere else" (p. 96). (*Henderson et al., 2018*)
	Financial Barriers	"If the van was full, [you would] have to get a ride … some of us don't have cars, you know" (p. 5). (*Zehbe et al., 2017*)
		Women with children and work commitments also had difficulty arranging childcare and time off work to keep their appointments. As a HCP commented: 'It certainly does pose challenges with regards to babysitting care and the mother being away for an entire day' (p. 5). (*Zehbe et al., 2017*)
		Older women were more likely to mention a previous bad experience and were less likely than younger women to mention cost or other financial barriers. Opinions varied about cost being a barrier (as many clinics offer free cervical screening). Some highlighted hidden costs (transport, parking, childcare) (p. 3). (*Adcock et al., 2019*)
	Health Care Priority	One in seven had not screened in ten years or more and one in six were unsure how long it had been since their last cervical screen (p. 2). (*Adcock et al., 2019*)
		"It's about listening to my body. It's about trying to keep it healthy for my children. I want to live as long as I can to bring them up to see them. Not only that, I want to feel good too. And I have had friends who have been diagnosed and that motivates me also to do it. But all in all I do it for myself and my family." *(p. 6) (Butler et al., 2020)*
		". . . it's no shame, you have to go and do your tests and all that. . . Every women [sic] in Aus- tralia have to do it every time, black or white. . .I don't get shame, because I want to look after myself and for my health too." (p.8) *(Butler et al., 2020)*
		"It's just what you do. I brush my teeth. It keeps coming down to that because it really for me is just about general body maintenance, you brush your teeth, you eat your food, I take my tablets. . . I do what I have to do." (p. 6) *(Butler et al., 2020)*
		"I'll always remember that lady, and she was so nice too. And she's going, your job now is mummy, and you've got to be here, have this test every two years. And because that was at my six week check-up and I didn't want to have the pap smear, and she's going, you don't want to leave him. And I'm thinking, oh my God, because I was a single mother, oh my God, I've got to look after him. So I did; never missed it." (p. 9) (*Butler et al., 2020)*

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	Pain	"[When they did the Papanicolaou test], maybe they did it wrong. I don't know, but I felt something like a scrape. Then I thought maybe the equipment wasn't disinfected. ... I thought about it a lot before doing it again, because I was afraid" (p. 4). (*Allen-Leigh et al., 2017*)
		"The Pap is uncomfortable but it has to be done" (p. 5). (*Zehbe et al., 2017*)
		"It doesn't get any easier, like the first time and then the next year, it didn't get any easier for me" (p. 5). (*Zehbe et al., 2017*)
		"I didn't like the way it felt, that's why I didn't want to go back there" (p. 5). (*Zehbe et al., 2017*)
		"They don't want their Paps [be]cause it hurt" (p. 5) (*Zehbe et al., 2017*)
		The test also could be exceptionally painful if providers were 'in a rush' or 'rough' and did the procedure 'real quick' (p. 5). (*Zehbe et al., 2017*)
		"Some women, when they come to the appointment, they decide they don't want to get a Pap, because it's uncomfortable, they're just afraid" (p. 5). (*Zehbe et al., 2017*)
Solutions – Self Sampling	Convenience	"Informant #14: I think that's the best idea that has come out of anything, like it's not invasive, you do it in your own home, at your own time... I.Z.: Whenever you, you know, are ready for it and... Informant #14 I think that with education along with it that this is what it is, I think that the success rate will be phenomenal with it because it's in their own home" (p. 8). (*Maar et al., 2016*)
		"It is more comfortable to do it at home … it's simple" (p. 6). (*Zehbe et al., 2017*)
		"I think the self-test is beneficial for them all because sometimes people don't have time for appointments to take off work, it [the Pap] is kind of an inconvenience" (p. 6). (*Zehbe et al., 2017*)
		"Even hearing [about self-testing], people are just, What? Oh, I'd do that, for sure, instead of me going to the doctor" (p. 6). (*Zehbe et al., 2017*)
		"Doing a self-test doesn't take very long … it's something that can be dealt with, done, gone" (p. 6). *Zehbe et al., 2017*)
		"I think more people would monitor it that way and test it themselves, like 'well maybe not today, but … eventually I'm going to try it" (p. 6) (*Zehbe et al., 2017*)
		"…I think the prevalence [participation in self-testing] rate would go up" (p. 5). (*Zehbe et al., 2017*)
		"Women wouldn't be so agitated and nervous about having the [self-] test" (p. 5). (*Zehbe et al., 2017*)
		With participants using terms such as 'easier', 'more

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		comfortable', 'less intrusive' and 'brilliant' [to describe self-testing] (p. 3). (*Adcock et al., 2019*)
		Participants discussed the value of providing multiple options and flexibility to cater to diverse populations, such as through community outreach services, was suggested for optimum engagement (p. 5). (*Adcock et al., 2019*)
		"It's simple, it's not like you are having forceps in you" (p. 376). (*Wakewich et al., 2015*)
	Decreased Pain	"Well, this one [the self-sampled HPV test] is better, because it is more comfortable to do it" (p. 5). (*Allen-Leigh et al., 2017*)
	Education	"If you hand me a kit, I won't touch it. Because I wouldn't know what to do, what if I did it wrong, or whatever, right, so? And I think a lot of people would be that way" (p. 7). (*Zehbe et al., 2017*)
		"You just give a test kit to them at home, the majority of them are just going to throw them out" (p. 7). (*Zehbe et al., 2017*)
		HCPs agreed that with appropriate support and education, HPV self-testing will benefit never/under-screened Māori (p. 4). (*Adcock et al., 2019*)
		Most frequently [participants] said they would be happy to do the HPV self-test in a clinic or their own home, and emphasised having good support and education to increase their confidence about properly doing it (p. 5). (*Adcock et al., 2019*)
	Privacy	"Would feel more comfortable with that [self-test] than a male doctor" (p. 6). (*Zehbe et al., 2017*)
		Self-testing would also address the 'trust issues' that discouraged women from seeking care from non-indigenous HCPs (p. 6). (*Zehbe et al., 2017*)
		"It's a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference" (p. 5). (*Zehbe et al., 2017*)
		"A lot more private, at home, if you do it by yourself" (p. 5). (*Zehbe et al., 2017*)
		"Women would prefer to be discreet, do it themselves and get their own results and not have their results shared with others" (p. 5). (*Zehbe et al., 2017*)
	Autonomy	"I really strongly believe that … because I was the one that was doing it [self-sampling], I was the one that was in control … and this way it gave me the ability to do it myself and I got all the results, they were fine; … it was also self-empowering, great, I like that" (p. 6) (*Zehbe et al., 2017*)
		Option of taking it home might make women "feel more

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		empowered" and contribute to a better relationship with HCPs (p. 6). (*Zehbe et al., 2017*)
		"Think it [self-sampling] would be, I think it would be great, because it provides them with some autonomy and allows them to take control of the situation. That would be really great actually …. I think it would increase [education opportunities] actually, yup, because it's a lot less clinical right, cause like you say, it's a lot less you know, stripping down and you know, allowing someone else to do the scraping of the cervix, you know, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference" (p. 376). (*Wakewich et al., 2015*)
	Decreased Embarrassment	"This one [the HPV test] is good because it [cytology or the Papanicolaou test] really does embarrass you, not because your husband doesn't want it or doesn't let us, but because it's embarrassing and because of embarrassment we don't do it, and so this [self-sampled HPV test] is good for us" (p. 5). (*Allen-Leigh et al., 2017*)
		"Because you do it yourself, since always, even if there is trust, you feel a little embarrassed to undress in front of someone else" (p. 5). (*Allen-Leigh et al., 2017*)
		"I think it [self-sampling] would definitely be more private … They wouldn't be so embarrassed … all kinds of people are easily embarrassed" (p. 376). (*Wakewich et al. 2015*)
Solutions – Community Level	Informal Community Communication	"Workshops, information sessions: I have lunch and learns. That's a start. There's all different venues bringing that into the community" (p. 5). (*Maar et al., 2016*)
		"I would concentrate on a day that's important to the women, like Mother's Day. The mothers would all come and the grandmas would come and the aunts would come. I would have guest speakers come in as well and there would be a dinner, a luncheon, or a feast of some sort. I would bring somebody in who had a little bit of charisma like a Tai Chi instructor who cooks meals" (p. 6). (*Maar et al., 2016*)
		"For our little group of work girls, [the] service team, we always tend to remind each other as well, like, 'Hey, is it time for your check-up yet, like?'" (p. 7). (*Maar et al., 2016*)
		"I think it's just sort of word of mouth, spreading things around, and just kind of bringing it up, you know, all the time" (p. 7). (*Maar et al., 2016*)
		"I call four people and then each one of the people I call, they call four people and it's something that we're trying to work on" (p. 7). (*Maar et al., 2016*)

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	Humor	"We didn't specifically talk about HPV, but we talked about STDs [during the education session]. Yeah, I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: "Are you bringing bananas next week?" You know, so, it was a fun thing" (p. 6). (*Maar et al., 2016*)
	Incentives	"It's hard to get people coming unless we have food and incentives" (p. 6). (*Maar et al., 2016*)
	School Education	"Start early, in school already: we should get our younger girls out there, and have that part as [education], at [grade] 8, and I know they're doing great in the school, getting it out there, about STDs" (p. 5). (*Maar et al., 2016*)
	Social Media	"You could even [do] something as simple as Facebook. Everybody's on Facebook" (p. 5). (*Maar et al., 2016*)
	Survivor Stories	"I think if there was someone just sharing stories. If a woman was willing to share her story that she's had this and got screened early. Girls like to listen to things like that, and the women like to listen to things like that. I beat this and I did that. Something like that would motivate them" (p. 5). (*Maar et al., 2016*)
	Positive Inspiration	"... and being a health worker too because then I can say, "No, I do mine," ... it feels a bit shame but you can talk to your patient and encourage them to do it because you've done it and if I hadn't done it, well, I shouldn't be saying those things to the patient." (p.12) (*Butler et al., 2020*)

Positionality

Acknowledging that personal experiences and beliefs heavily influence research perspectives, it is critical for researchers to self-situate. This review is a result of the desire to prioritize individual voices and stories of Indigenous women. After hearing first-hand accounts of various health disparities experienced by Indigenous women in South Australia, while conducting field work for a different HPV project, the primary reviewers (S.S and B.P) discussed the importance of the person behind a statistic and the desire to synthesise existing knowledge in HPV literature was established in order to identify future research steps. While both non-Indigenous researchers, S.S is an oral pathologist with experience working with Indigenous populations in Australia and B.P has qualitative experience with community-

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engaged scholarship in the context of Indigenous health in Canada and Australia. The supporting research team consists of Indigenous and non-Indigenous scholars with vast experience in the realm of Indigenous health research.

Identifying studies for inclusion

The reviewers used a pre-established search strategy[36], which involved using terms (and their edited variants) describing the population of interest, the phenomenon being researched, as well as the included study designs (Supplementary file 1). Two investigators (S.S and B.P) independently screened the literature for eligible articles using MEDLINE, PubMed, SCOPUS and Web of Science databases from inception until June 2020. For example, the search strategy used for PubMed Database was as follows: First Nation/First Nations/Pacific Islander/Pacific Islanders/Torres Strait Islander/Torres Strait* Islanders/ Aborigin*/ Alaska*/ Aleut*/ Amerind*/ American Indian/ Arctic/ Aymara/ Bushmen/ Chukchi/ Chukotka*/ Circumpolar/ Eskimo*/ Greenland*/ Hmong/ Indian*/ Indigen*/ Inuit*/ Inupiaq/ Inupiat/ Khanty/ Maori*/ Mapuche/ Metis/ Native*/ Navaho*/ Navajo*/ Nenets/ Quechua/ Saami/ Sami/ Samoan*/ Siberia*/ Skolt/ Tribal/ Tribe*/ Xingu*/ Yup'ik/ Yupik/ Zuni/"African continental ancestry group"/"African continental ancestry group"/ "Asian continental ancestry group"/"Health Services, Indigenous"/"Oceanic ancestry group"/"arctic regions"/"ethnic groups", "HPV", "Human Papillomavirus", "Papillomavirus", "HPV 18" , "HPV*", "Qualitative", "awareness", "barriers", "screening". The search was tailored as per the design of individual databases.

In the search for published studies, the reviewers made use of facilities where the option was given to run 'related' searches, where similar studies are automatically identified. The bibliography of each article was scanned manually for possible additions to the search. The titles and abstracts were screened by both reviewers independently to assess eligibility, with those considered relevant by either investigator advancing to a full-text review. The initial

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search prioritized first-hand experiences, however, a number of studies included perspectives from health care workers that provided valuable insights into patient experiences. The decision was made to include these studies. The investigator pair fully screened articles to identify studies that fulfilled the following criteria:

- The study focused on the experiences of women and/or health care workers of Indigenous identity across the world
- Findings contained personal illustrations or first-person accounts of HPV knowledge and cervical cancer screening
- The study was qualitative or mixed methods (with clear qualitative examples)
- Where cancer was the phenomenon of interest, studies only focused on cancers associated with HPV infection (mostly cervical cancer, but did not exclude on basis of oropharyngeal cancer)
- The study was available in English
- The study was available in hardcopy or in downloadable form
- The study was published prior to June 30, 2020

Exclusion criteria

- Based only on HPV vaccination
- Quantitative only studies
- Studies that included narratives from males only

While efforts were made to decrease publication bias, the reviewers recognize that limiting the search to the English language could result in loss of data in other native languages. Additionally, the inclusion of all grey literature could have provided additional findings for the study and decreased possible impacts of publication bias.

Critical Appraisal

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There are various validated tools for appraisal of studies; this review employed the JBI (Joanna Briggs Institute) System for the Unified Management, Assessment and Review of Information (SUMARI) critical appraisal tool[37] (Supplementary file 2). This tool includes questions regarding sample sizes, locations, methods used, techniques utilised and themes derived.

Data Extraction and Synthesis

Data were extracted in two phases. For the first phase, the reviewers utilized concepts of thematic analysis and comprehensively identified findings from each of the included studies. These findings were extracted via a piloted extraction form and then uploaded to JBI SUMARI. The second phase utilised the JBI data extraction tool for all studies. Each reviewer independently scored the findings within JBI SUMARI as "Credible (C)", "Not Supported (NS)" and "Unequivocal (U)"; the score for each finding was based on inter-reviewer agreement. The synthesis of findings was done manually by reviewers, which included writing all findings on a white board and identifying common phrases, themes and concepts. Common themes were grouped, with connections between other themes explored in the context of the lived experience of HPV prevention for an Indigenous woman. These categories were then transferred from the white board to the JBI SUMARI tool and each individual finding was placed within the appropriate category.

Social Ecological Model

Upon reviewing the categories and creating synthesized themes, it was apparent that findings were present at varying societal levels that are comprehensively identified in the social ecological model. The overarching themes were therefore generated to fit the social ecological model including: (1) intrapersonal factors; (2) interpersonal factors; (3) institutional and organizational factors; (4) socio-cultural and community factors; (5) public policy[33]. This was

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additional to the solutions provided by the community which included two main themes: self-sampling and community interventions.

Results

The literature search returned 1377 records, of which 963 were duplicates and consequently excluded. After being screened against the pre-specified inclusion and exclusion criteria, 24 studies were shortlisted. Of the 24 potentially eligible studies reviewed in full, 10 fully satisfied the inclusion criteria (Figure 1). The average inter-reviewer score was 9.60, indicating a very high level of agreement between reviewers (Table 4). Although some studies (n = 4) did not perform well according to the criteria set by the Qualitative Assessment and Review Instrument of method (Supplementary file 3) of quality appraisal[37], the reviewers felt that studies should not be excluded on the basis of being 'less rigorous' according to one set of criteria. On the contrary, such studies were included precisely because of the richness of textual narrative provided. Other studies had minor methodological issues pertaining primarily to whether the researchers had adopted a reflexive stance by examining their own role in formulating and conducting the research[38].

Table 4: Inter-reviewer reliability table

Study	Number of questions in agreement	Number of questions in disagreement	Score
Adcock A, 2019	10	0	10
Allen-Leigh B, 2017	9	1	9
Butler TL, 2020	10	0	10
Cerigo H, 2012	10	0	10
Henderson R, 2018	10	0	10
Maar N, 2016	10	0	10
O'Brien B, 2009	10	0	10

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Tratt E, 2020	9	1	9
Wakewich P, 2015	9	1	9
Zehbe, 2017	9	1	9
Mean			**9.60**

After the literature search and article appraisal, ten studies were deemed eligible for inclusion in this review. Studies were conducted in four countries: with Māori communities in New Zealand, Aboriginal and Torres Strait Islander population in Australia, Inuit and First Nations communities in Canada, as well as Mam and Huichol communities in Mexico (Table 2).

Reviewers identified 118 findings (Supplementary file 4) from the included ten studies, a chart with each finding, associated quotes and score was generated (Table 3). This collaborative process resulted in 36 distinct categories, which provided an appropriate template to meta-aggregate the findings. Five main themes were synthesized by the end of this process, with reviewers in complete agreement. After completion of this process, the JBI SUMARI tool generated a meta-aggregation flowchart of all findings, categories and themes.

Intrapersonal Factors

A theme strongly reflected in all included studies was intrapersonal factors. Findings revealed marginal awareness of HPV and cervical cancer, including a lack of knowledge regarding HPV symptoms and disease progression. The lack of awareness of disease progression was highlighted in more than one paper[17,39-41]. A deficiency of empathetic services was identified as a potential barrier to seeking timely health support[40]. Feelings of embarrassment and shyness were the most commonly stated experiences when visiting a doctor for screening. One study

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suggested that younger women feel shy, especially individuals with a history of sexual abuse[42], while another suggested that older women are more hesitant to pursue screening options due to feeling shy[41]. Māori women have referred to this feeling as whakamā and that its tapu (a taboo) to let go of your body autonomy[40]. Feelings of vulnerability were common, and experiences of cervical screening were described as vulnerable[17]:

> "*I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this .... There's got to be a better, kinder, gentler, more humane way to do it."*

Associated with feelings of shyness were previous negative and painful healthcare experiences, which may serve to demotivate women from regular screenings. Specifically, concerns were expressed around sterilization, empathetic procedures and rushed appointments[39,43]. Another common intrapersonal factor was fear of prognosis; for example, some participants identified this as a barrier because they feared a cancer diagnosis[17,18,43]. However, other participants found fear to be a motivator, and the desire to avoid suffering or death from this disease ensured prioritization of prevention[18]. The financial burden associated with the Pap test and other costs including transport or childcare were substantial barriers among low income families[40,43]. Some studies suggested that health care as a whole was not identified as a high priority for many individuals as stated by health care workers[40]:

> "*One in seven had not screened in ten years or more and one in six were unsure how long it had been since their last cervical screen."*

Alternatively, one study suggested a deep interest and inclination towards health, where family and personal wellness were identified as motivators[44]:

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"It's about listening to my body. It's about trying to keep it healthy for my children. I want to live as long as I can to bring them up to see them. Not only that, I want to feel good too. And I have had friends who have been diagnosed and that motivates me also to do it. But all in all I do it for myself and my family."

Interpersonal Processes

Four of the included studies described factors at the interpersonal level[17,18,20,44]. Family relations were discussed as a way to facilitate a generational shift from an environment of sexual taboo to one of love and support[17,18,20,44]. Role-modelling by older family members, specifically mothers, was associated with preventive health care practices[20]. One mother explained the importance of talking openly about sex with her children[17]:

"Like my mother, you know, was the kind of woman that… when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your personal, private, so when it happened, I had absolutely no knowledge what was going on in my body… And with my daughter I didn't want her to have that feeling, so I mean, of course I changed and I explained everything."

Intergenerational communication was also identified as important for educating children and as a critical aspect of cultural traditions[18]

"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children and out of respect… he's getting that message across [in his sweat lodge ceremonies with men]."

Institutional/Organisational Factors

All of the included studies discussed factors at the institutional/organisational level that impacted cervical screening. Some items such as cultural awareness among health care

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workers, integrated services, female health care workers for screening and positive health care experiences were discussed as facilitators for screening. However, other institutional aspects such as negative health care experiences, language barriers, male health care workers for screening, lack of results, privacy concerns and screening availability were identified as barriers to screening. Both participants and health care workers suggested integrated services (Pap smears combined with other services) as a way to eliminate multiple clinic visits[20,40,43]. Offering services in local languages, eliminating medical jargon[20,42] and incorporating holistic healthcare approaches were identified as potential facilitators. Geographically remote communities described difficulties associated with building trusting relationships at health care centres, due to a high turnover of health care workers[17]. Discussions of the gender of the healthcare worker performing screening services (such as a Pap) elicited strong responses, with male health care workers identified as a barrier[18,41] and female health care workers as a facilitator[20,43,44]. Especially in smaller communities, lack of privacy was a substantial concern that impacted decisions to access healthcare services[17,43], as evident in the following quote[17]:

> *"Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk."*

Lack of communication about results and lack of screening availability[39,43,44] were highlighted as obstacles to accessing services; many participants expressed a desire to receive results regardless of negative or positive outcomes. Positive health care experiences enticed regular utilisation of services[18,20], while negative health care experiences impeded the use of services[17,18,40]. Specifically, participants felt that often procedures were not fully explained[17,18], which competed with a desire for body autonomy. The frustration of this experience was felt by many participants, as shown in the following quotation[18]:

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"I still don't know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don't know! Nobody explained anything to me because, maybe because I didn't ask. Maybe they thought that I understood everything that was going on, but I didn't."

Socio-cultural/Community

This theme was separated into five categories, with key quotes arising from six of the ten included studies. Gender imbalances highlighted male dominance in communities, and the consequential inequalities based on gender. Male tendencies towards promiscuity were considered to be a contributing factor for bringing HPV into the home, which affected screening practices due to the associated stigma of an unfaithful partner[43]. Religious and cultural beliefs prevented many women from attending cervical screening and/or treatment. Women shared their experiences of violence and a general disregard for their health by their male partners[39]:

"The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn't even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn't believe that."

One healthcare worker indicated an urgent need for education to alleviate the negative concerns and sense of humiliation associated with cervical screening[43]. Health care workers in one study described patient requests to not document diagnoses and feelings of nervousness around potential breaches of confidentiality in the community[17]:

"*Actually in the last 6 months that I've been there, I've had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the*

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community … they're afraid of the stigma surrounding that so and a lot of times too I wasn't able to do any [cervical] screening"

Participants frequently described a fear of disappointing community and family; that sense of shame and humiliation was a prime concern and barrier to regular screening[42]. Discussing the isolating effects of a cervical cancer diagnosis, one study reflected on how the diagnosis of a single person is the suffering of not one individual but an entire community[17].

Public Policy

While participants in the included studies tended to focus on discussions of institutional/organisational and intrapersonal factors, two of the Canadian studies[17,18] highlighted the impact of traumatic experiences on accessing screening services. Specifically, the sexual abuse experienced at government-mandated Residential Schools in Canada has created an environment of sexual taboo where health concerns are not discussed, and services are not sought. This sentiment is illustrated by the following quotation[17]:

"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past."

A lack of community connection and relationships was discussed as a consequence of the sexual abuse experienced in residential schools, and substance-dependence was identified as a coping mechanism[17]. Intergenerational trauma and child removal were also mentioned as barriers to accessing health services, both of which are a result of government-mandated public policies[45].

Solutions

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Many participants identified solutions to the aforementioned obstacles which were categorised at each level of the social ecological model. Self-sampling[17,20,39,40] and community-level initiatives[20] were discussed as potential ways forward to address some of the concerns and common experiences shared across all included studies. The idea of self-sampling was associated with increased convenience[17,20,40,43], privacy[43] and autonomy[17,43] as well as reduced embarrassment[17,39] and pain[39]**.** The need for education prior to self-sampling was stressed as a means to ensure a sound understanding and completion of tests[40,43]. The potential increase in perceived bodily autonomy increased the acceptance of self-sampling[43]:

> *"I really strongly believe that… because I was the one that was doing it [self-sampling], I was the one that was in control… and this way it gave me the ability to do it myself and I got all the results, they were fine;… it was also self-empowering, great, I like that."*

Maar et al. (2016), discussed a variety of community-level tactics to increase engagement and a desire to seek out and use health care services[23]. Informal community communication at events as well as by word of mouth was seen as valuable. Integrating HPV screening with female-oriented celebrations and workshops was discussed as a way to support screening behaviours. Use of school education, survivor stories, social media, incentives and humour were all identified as potential pathways to increasing screening participation. One health care worker described how she used humour in an educational workshop[23]:

> *"We didn't specifically talk about HPV, but we talked about STDs. Yeah I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: 'Are you bringing bananas next week?' You know, so it was a fun thing."*

Social Ecological Model (Figure 2)

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Two studies explored experiences relevant on all five levels of the social ecological model. All included studies addressed issues on at least two levels of the model. Five studies incorporated themes or frameworks that focused exclusively on two ecological levels. All studies discussed factors at the institutional and intrapersonal level, while two studies focused primarily on themes related to public policies. The distribution of themes proposed at each ecological level of the model, according to all included studies is included in table 5.

Table 5: Social ecological model distribution of studies

Studies	Public Policy	Socio-cultural/ community Factors	Institutional/ Organizational factors	Interpersonal factors	Intrapersonal Factors
Adcock A. 2019			■		■
Allen-Leigh B et al. 2017.		■			■
Cerigo H et al. 2012			■		■
Henderson R et al. 2018	■	■	■	■	■
Maar N et al. 2016			■	■	■
O'Brien B et al. 2009			■		■
Tratt E et al. 2020			■		■
Wakewich P et al. 2015	■	■	■	■	■
Zehbe et al. 2017		■	■		■
Butler TL et al, 2020		■	■	■	■

*Highlighted boxes show the contribution of a study to a particular ecological level.

Discussion

The aim of this systematic review was to collate the experiences of Indigenous women and health care workers with regards to HPV, associated cancers and screening methods. A secondary aim was to generate an understanding to inform future HPV prevention strategies. By mapping findings within the social ecological model levels, this review highlights gaps in current HPV practices and identifies areas to be addressed based on first-hand experiences.

The first level of the social ecological model included factors at an intrapersonal level. Lack of knowledge and awareness of HPV infection and its risks persisted in the context of other priorities taking precedence over routine health checks. Two prominent barriers were expressed in all papers: financial burdens and feelings of embarrassment. These factors affect an individual's personal decision to seek screening or preventive health care services.

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The level of interpersonal factors revolved around two main concepts of family relations and intergenerational communication. The communication within one family unit, from parents to children and even the openness of children with their parents, around discussions involving sexual behaviour and practices was taboo. A greater emphasis on, and openness to, education across the life course could increase the acceptability of screening. A lack of intergenerational communication was identified as a potential predisposing factor to higher risk sexual behaviours.

The third level of the social ecological model, institutional/organizational, had nine synthesized findings that demonstrated how positive and negative healthcare experiences impact current perceptions towards screening and HPV infection. There was an emphasis on lack of empathy by some healthcare workers (doctors, nurses etc.) while screening and dealing with sensitive procedures. A preference of being attended to by female health care workers, especially for cervical smears and examination, was observed. Worry due to lack of results after the tests was common, and often was interpreted as bad news which had an impact on participant's daily life and mental health.

The fourth level, socio-cultural/community, includes factors that affect HPV-associated cancer prevention and awareness at a community level. This level included discrimination on the basis of gender and highlights male dominance that exists in some communities. This level also examined the stigma and shame of being diagnosed with a sexually transmitted infection and a common fear of being seen at the doctor's office. The experience of isolation faced after diagnosis/treatment is also a common barrier that impedes regular screening.

The results of the systematic review found trauma as the single synthesized finding at the social ecological model level of public policy. Personal quotes revealed the impacts of sexual abuse, physical abuse, drug abuse and traumatic ancestral experiences which continue to have an

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intergenerational impact across families and communities. Findings revealed a general mistrust towards the health system and health care workers. These findings highlight the urgent need to begin building trust between Indigenous communities and local health services in the provision of culturally safe care. Increased trust could help increase the acceptability of current prevention programs and treatment solutions.

In addition to the themes of the social ecological model, the theme of self-sampling was prominent in most papers. Self-sampling is a solution which may provide more control, independence, privacy and convenience, and in turn increase screening participation. Community-identified solutions included the use of local survivor stories, which were thought to provide strength and courage to other community members to attend regular cervical cancer screenings and to not fear the diagnosis or to overcome the fear by taking preventative steps. The use of humour and social media were both innovative and socially acceptable solutions for creating awareness and educating community members with contemporary knowledge of HPV infection and its effects.

Strengths and limitations

To the best of our knowledge, this systematic review is the first to address the qualitative aspects of HPV infection and associated cancers among Indigenous women at a global level. The review adhered to all protocols to ensure transparency and legitimacy. Another strength is the summary of community-provided solutions, in the form of personal narratives, to help decrease transmission of HPV infection and to, in turn, prevent HPV-related cancers. An innovative strength of this study is the inclusion of the social ecological model as a theoretical framework. This model enables conceptualisation of a more holistic approach to HPV prevention for Indigenous women at a global level. In accordance with other studies, our study highlighted the continuing impact of trauma at the public policy level, with our findings

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providing important evidence of the work required to address the resulting disparities across all health indicators that have resulted from this trauma[45-47].

Limitations include the decision to not include male participants, as the current statistics show a dramatic increase in the incidence of HPV associated oropharyngeal carcinoma among men[48]. Additionally, the majority of articles are from Canada. This highlights the need for more research focusing on Indigenous perspectives and experiences regarding HPV screening and diagnosis in other countries. The community-identified solutions presented are mainly from one study. There is a lack of embedded community feedback in the studies included in this review despite the well-documented importance of community suggestions in improving the acceptability and success of Indigenous health programs. Future policies and initiatives should prioritise Indigenous voices through incorporation of community suggestions.

Conclusion

Qualitative systematic reviews are increasing in the literature. They have particular utility in informing policy decisions, as the success of any new policy or intervention depends upon its acceptance of, and sustainability in, a given population. While quantitative studies provide clarity with respect to disease spread or burden of a health condition, qualitative studies play a significant role in the management and generation of possible solutions to that health condition/disease. The social ecological model has a structured, multi-level approach with sociological, behavioural and individual aspects. Future research needs to expand the geographical scope of the current work, beyond Canada, and integrate community-identified solutions. Policies around HPV screening and diagnosis for Indigenous communities, globally, need to prioritize a holistic approach to health care, addressing barriers and facilitators at each level of the social ecological model, and prioritising specific community needs.

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Author Contribution statement:

SS, BP and JH conceived the presented idea. SS, BP, XJ, LJ developed the theory and performed the data collection. MS, KC, GG verified the analytical methods. JH encouraged SS and BP to explore the cultural limitations and beliefs leading to inaccessibility of cervical cancer screening. MS, KC, GG and LJ supervised the work. SS and BP primarily wrote the manuscript under the supervision of GG, MS, KC, LJ and XJ. All the authors discussed the results and contributed to the final manuscript.

Ethics Statement: No human participants, so not applicable.

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Figure Legends:

Figure 1: PRISMA flowchart for included studies

Figure 2: Socio-Ecological Model

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Table Legends:

Table 1: ENTREQ Checklist

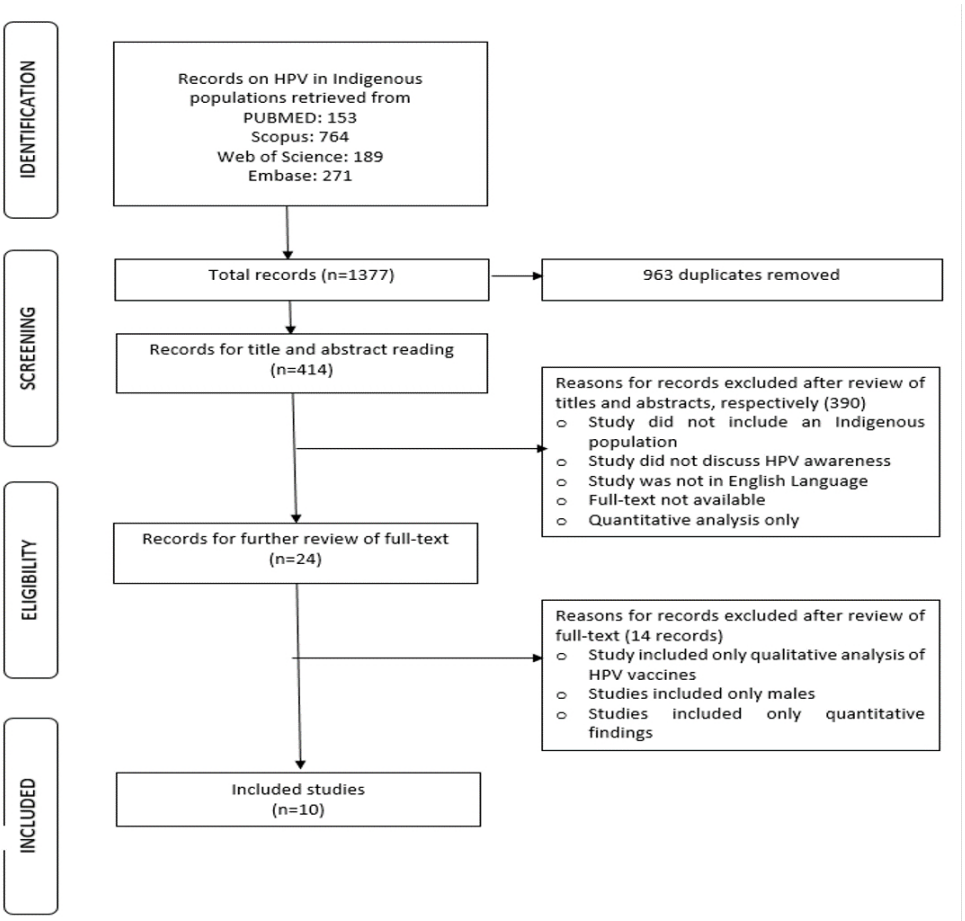
Table 2: Characteristics of Included Studies

Table 3: Illustrations from all the included studies arranged according to all the synthesized categories under each level of the Socio-Ecological Model

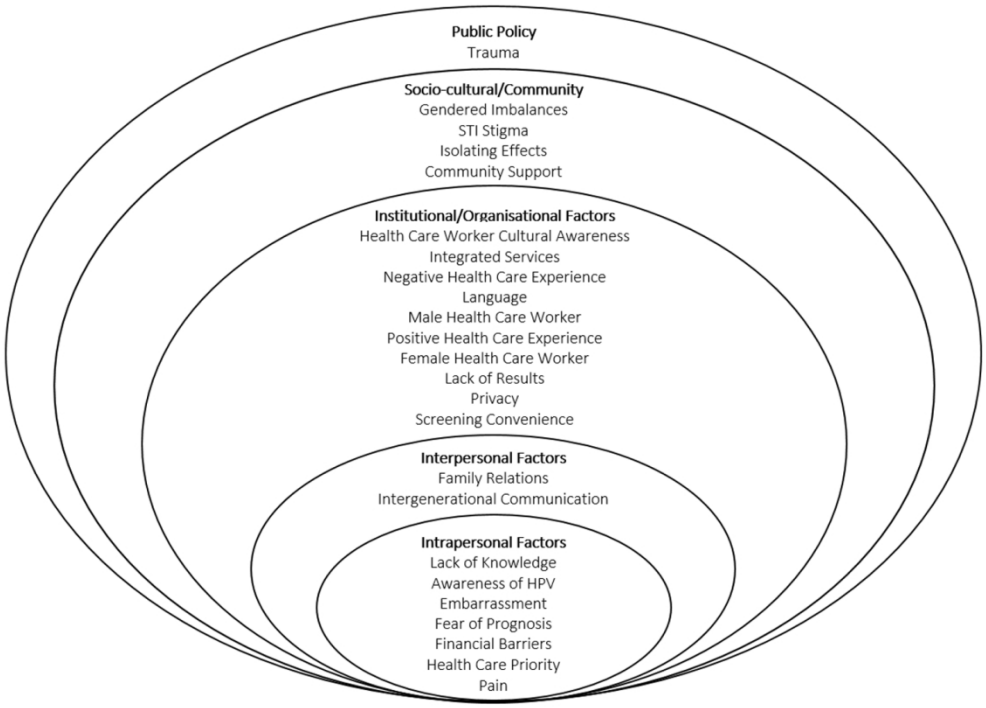
Table 4: Inter-reviewer reliability table

Table 5: Social ecological model distribution of studies

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PRISMA flowchart for included studies



Socio-Ecological Model

Supplementary File 1(S1)

Logic Grid: **Pubmed: 153 : 29/06/2020**

HPV AND INDIGENOUS POPULATIONS

HPV	Indigenous Populations
HPV[Text Word] OR "papillomaviridae"[MeSH Terms] OR “human papillomavirus”[MeSH] OR “human papillomavirus”[Text Word] OR "human papillomavirus 16"[MeSH Terms] OR human papillomavirus 16 [Text Word] OR "human papillomavirus 18" [MeSH Terms] OR human papillomavirus 18 [Text Word] OR papillomavirus[Text Word]	("first nation"[Text Word]) OR "first nations"[Text Word]) OR "pacific islander"[Text Word]) OR "pacific islanders"[Text Word]) OR "torres strait islander"[Text Word]) OR "torres strait islanders"[Text Word]) OR aborigin*[Text Word]) OR africa*[Text Word]) OR alaska*[Text Word]) OR aleut*[Text Word]) OR amerind*[Text Word]) OR arctic[Text Word]) OR aymara[Text Word]) OR bushmen[Text Word]) OR chukchi[Text Word]) OR chukotka*[Text Word]) OR circumpolar[Text Word]) OR eskimo*[Text Word]) OR greenland*[Text Word]) OR hmong[Text Word]) OR indian*[Text Word]) OR indigen*[Text Word]) OR inuit*[Text Word]) OR inupiaq[Text Word]) OR inupiat[Text Word]) OR khanty[Text Word]) OR maori*[Text Word]) OR mapuche[Text Word]) OR metis[Text Word]) OR native*[Text Word]) OR navaho*[Text Word]) OR navajo*[Text Word]) OR nenets[Text Word]) OR quechua[Text Word]) OR saami[Text Word]) OR sami[Text Word]) OR samoan*[Text Word]) OR siberia*[Text Word]) OR skolt[Text Word]) OR tribal[Text Word]) OR tribe*[Text Word]) OR xingu*[Text Word]) OR yup’ik[Text Word]) OR yupik[Text Word]) OR zuni[Text Word]) OR "African continental ancestry group"[Mesh]) OR "African continental ancestry group"[Mesh]) OR "Asian continental ancestry group"[Mesh]) OR "Health Services, Indigenous"[Mesh]) OR "Oceanic ancestry group"[Mesh]) OR "arctic regions"[Mesh]) OR "ethnic groups"[mesh]

Logic Grid: **EMBASE: 271 : 29/06/2020**

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HPV	Indigenous Populations
('human papillomavirus type 16':ti,ab,kw OR 'human papillomavirus type 18':ti,ab,kw OR 'wart virus':ti,ab,kw OR 'oral human papillomavirus infection':ti,ab,kw OR 'hpv':ti,ab,kw)	'indigenous population':ti,ab,kw OR 'indigenous people':ti,ab,kw OR 'Samoan (people)':ti,ab,kw OR 'First Nation':ti,ab,kw OR 'Pacific Islander':ti,ab,kw OR 'Torres Strait Islander':ti,ab,kw OR 'Black person':ti,ab,kw OR 'Alaska Native':ti,ab,kw OR 'Aleut (people)':ti,ab,kw OR 'Amerind people':ti,ab,kw OR 'Aymara (people)':ti,ab,kw OR 'Chukchi (people)':ti,ab,kw OR 'Chukotka':ti,ab,kw OR 'Eskimo-Aleut people':ti,kw,ab OR 'Greenland':ti,ab,kw OR 'Hmong (people)':ti,kw,ab OR 'Indian':ti,ab,kw OR 'indigenous people':ti,kw,ab OR 'Inuit':ti,kw,ab OR 'Inupiat (people)':ti,kw,ab OR 'Khanty (people)':ti,ab,kw OR 'Maori (people)':ti,kw,ab OR 'Mapuche (people)':ti,kw,ab OR 'indigenous people':ti,kw,ab OR 'Navajo (people)':ti,kw,ab OR 'Nenets (people)':ti,kw,ab OR 'Quechua (people)':ti,kw,ab OR 'Sami (people)':ti,kw,ab OR 'Yupik (people)':ti,ab,kw OR 'Zuni (people)':ti,ab,kw OR 'Asian continental ancestry group':ti,ab,kw OR 'Oceanic ancestry group':ti,ab,kw OR 'Arctic':ti,ab,kw OR 'ethnic group':ti,kw,ab

Logic Grid **SCOPUS: 764 : 29/06/2020**

HPV	Indigenous Populations
TITLE-ABS-KEY (HPV) OR TITLE-ABS-KEY (papillomaviridae) OR TITLE-ABS-KEY (human papillomavirus) OR TITLE-ABS-KEY (human papillomavirus type 16) OR TITLE-ABS-KEY (human papillomavirus type 18)	TITLE-ABS-KEY (indigenous AND population) OR TITLE-ABS-KEY (ethnic AND groups) OR TITLE-ABS-KEY (aborigin) OR TITLE-ABS-KEY (torres AND strait AND islander) OR TITLE-ABS-KEY (black AND person)

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Web of Science: 189 :29/06/2020

HPV	Indigenous Populations
(HPV OR human papillomavirus OR human papillomavirus type 16 OR human papillomavirus type 18 OR Oral HPV OR wart virus)	**TOPIC:** ((first nation OR first nations OR pacific islander OR pacific islanders OR torres strait islander OR torres strait islanders OR aborigin* OR africa* OR alaska* OR aleut* OR amerind* OR arctic OR aymara OR bushmen OR chukchi OR chukotka* OR circumpolar OR eskimo* OR greenland* OR hmong OR indian* OR indigen* OR inuit* OR inupiaq OR inupiat OR khanty OR maori* OR mapuche OR metis OR native* OR navaho* OR navajo* OR nenets OR quechua OR saami OR sami OR samoan* OR siberia* OR skolt OR tribal OR tribe* OR xingu* OR yup'ik OR yupik OR zuni OR African continental ancestry group OR African continental ancestry group OR Asian continental ancestry group OR Health Services, Indigenous OR Oceanic ancestry group OR arctic regions OR ethnic groups))

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S2: Supplementary File 2: JBI Summary appraisal tool questionnaire for qualitative systematic reviews

1. Is there congruity between the stated philosophical perspective and research methodology?
2. Is there congruity between the research methodology and the research question or objectives?
3. Is there congruity between the research methodology and the methods used to collect the data?
4. Is there congruity between the research methodology and the representation and analysis of data?
5. Is there congruity between the research methodology and the interpretation of results?
6. Is there a statement placing the researcher culturally or theoretically?
7. Is the influence of the research on the researcher and vice versa, addressed?
8. Are the participants and their voices adequately represented?
9. Is there research ethical according to current criteria or, for recent studies, and there evidence for ethical approval by an appropriate body?
10. Do the conclusions drawn in the research report flow from the analysis, or interpretation of the data?

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S3: Supplementary file 3: Appraisal of included studies

Table: Qualitative Research

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Adcock A. 2019.	Y	Y	Y	Y	Y	N	N	Y	Y	Y
Allen-Leigh B et al. 2017.	Y	Y	Y	Y	Y	N	U	Y	Y	Y
Cerigo H et al. 2012.	Y	Y	Y	Y	Y	N	N	Y	Y	Y
Cui Y et al. 2010.	Y	Y	Y	Y	Y	N	N	N	Y	Y
Henderson R et al. 2018.	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Maar N et al. 2016.	Y	Y	Y	Y	U	Y	U	Y	Y	Y
O'Brien B et al. 2009.	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Tratt E et al. 2020.	Y	Y	Y	Y	Y	Y	U	Y	Y	Y
Wakewich P et al. 2015.	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
Zehbe et al. 2017.	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
%	100.0	100.0	100.0	100.0	90.9	63.63	18.18	90.9	100.0	100.0

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S4: Supplementary file: List of study findings with illustrations

Study: Tratt E et al 2020	
Finding	Cultural Awareness (C)
Illustration	Cultural awareness of healthcare workers was a major facilitator, reflected in positive patient experiences and more use of adequate communication means. (Pg 3)
Finding	Previous Experiences (C)
Illustration	Previous positive experiences of health personnel in their interaction with Inuit was told to be a factor that highly contributes to increase cultural awareness, countering the negative effect of a healthcare system based on non-Inuit cultural values. (Pg 4)
Finding	Increased confidence in patients' capabilities (C)
Illustration	Thus, allowing a more active role of patient in decision-making. (Pg 4)
Finding	Access to medical information (C)
Illustration	The vast majority of participating women expressed the need to get screened after understanding medical facts, including but not limited to the absence of symptoms (Pg 4)
Finding	Personal Preference (U)
Illustration	"For me, I am used to getting my Pap, I'm fine with it. But for the younger girls, [or] the shy ones, [or] those who were sexually abused, they might prefer [the HPV self-sampling method]." (Pg 4)
Finding	Means of communication (C)
Illustration	Participants reported the major importance of using visual language, ranking higher than using Inuktitut, and they stressed the communication problems behind using French or medical jargon. (Pg 4)
Study: O'Brien B et al 2009	
Finding	Vivid Recollection of healthcare encounters (U)
Illustration	"My appointment was scheduled when the offices are closed so I knew he was going to be giving me some fairly bad news. The office was empty, and the lights were all turned down. It was a real comfortable surrounding. He gave me a lot of information, pamphlets, and people to phone for support." (Pg 85)
Finding	Not Knowing what was happening (inadequate explanations) (U)

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Illustration	“I still don’t know, to this day, whether they found anything or what portion they removed, if they even removed anything. I don’t know! Nobody explained anything to me because, maybe because I didn’t ask. Maybe they thought that I understood everything that was going on, but I didn’t.” (Pg 86)
Finding	Not knowing what was happening (Confusion) (U)
Illustration	“When you’re in (western Canadian city A) and (western Canadian city B) sometimes you just get pushed along as if you’re just a file. “Here you go, this is what’s wrong, this is what we’re going to do, see you later.” I don’t understand where I go from there, how’s my health going to be affected by it now.” (Pg 86)
Finding	Encouragement and Support (Fear of process) (U)
Illustration	"My mom has gone through the process. She told me that she had cancer in her cervix and that she had to go for surgery. So it was a big fear for me because as soon as you hear cancer you think of the ultimate. There’s nothing more then; but for this person to die. "(Pg 86) "My dad had a really difficult time with it, because he was already in his early 60s at the time, and as soon as he heard cancer [he thought] I was going to die. He started cutting off the emotional ties with me then." (Pg 86) "People are scared. I’m thinking of my sister. She hasn’t gone back to get retested. She did have an abnormal result the first time. I think she’s afraid. Some people are afraid that if they do find cancer, it’s downhill from there." (Pg 86) "To find out that I have some kind of illness . . . like to know you’re sick you just want to block it out. . . . If I knew I had an illness . . . I would probably just shut down from everything and totally change, and not for the better either." (Pg 86)
Finding	Encouragement and Support (Motivation) (U)
Illustration	"My fear is of having [cervical cancer] . . . motivates me. I want to have good health. I don’t want that kind of a disease." (Pg 86)
Finding	Encouragement and Support (Support) (U)
Illustration	"I would try and keep encouraging her . . . to tell her, “If they find it early they can do something about it.” I try and set examples for my clients. I tell people, “Did you know this lady had this kind of cancer because she never got tested, and they could have done something about it right away and she would still be here.”" (Pg 86)
Finding	Encountering a male provider (Traditional Beliefs) (U)

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Illustration	"I had it done by a male doctor and I was uncomfortable . . . because of the way I was raised. My [traditional] grandmother always told me that the only man that should know you like that is your husband." (Pg 87)
Finding	Encountering a male provider (Nervousness) (U)
Illustration	"My first Pap smear was done by a male and I was very nervous and I just didn't want it done. Maybe that's why I didn't have another one [Pap smear] . . . because of that [male gender]." (Pg 87)
Study: Maar N et al 2016	
Finding	(Explain the value of cervical screening) Importance & Benefits (U)
Illustration	"Explain [what the Pap test is] then they accept the value of early diagnosis: I find that once I explain to them the importance of early detection and [that] it can be treated, they're more agreeable to [a Pap test]." (Pg 4)
Finding	(Explain the value of cervical cancer screening) Awareness of screening services (U)
Illustration	"The biggest motivation is education, I think. You know, just teach them, let them know that the service is there, that they need to take it, it's important that they have it." (Pg 4)
Finding	(Explain the value of cervical cancer screening) Screening can prevent cancer (U)
Illustration	"We know cervical cancer is 100% preventable, and I know because I read stats and see statistics that Aboriginal women are the number one on the list for dying from this." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Build local HCP capacity (U)
Illustration	"You need to make sure that you train the trainers, like teach the CHRs [community health representatives], and the women in the community that are providing service." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Education by community cancer survivors (U)
Illustration	"I think if there was someone just sharing stories. If a woman was willing to share her story that she's had this and got screened early. Girls like to listen to things like that, and the women like to listen to things like that. I beat this and I did that. Something like that would motivate them." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Education by extended female relatives (U)

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Illustration	"Well, my mother, you know, always went for her physicals; thus I learned that I was to go for my physicals... the families that do have it, you can see that progression of preventative health care." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Early education in schools (U)
Illustration	"Start early, in school already: we should get our younger girls out there, and have that part as [education], at [grade] 8, and I know they're doing great in the school, getting it out there, about STDs." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Regular workshops (U)
Illustration	"Workshops, information sessions: I have lunch and learns. That's a start. There's all different venues bringing that into the community." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Passive intervention for whole community (U)
Illustration	"You could even [do] something as simple as Facebook. Everybody's on Facebook." (Pg 5)
Finding	(Motivate through specific educational strategies and capacity building) Include humour in health education (U)
Illustration	"We didn't specifically talk about HPV, but we talked about STDs [during the education session]. Yeah, I think they got it because they still, when they see me on the streets here, we kind of giggle about it because we used bananas [for the sex education], you know, and they'll ask me: "Are you bringing bananas next week?" You know, so, it was a fun thing." (Pg 6)
Finding	(Motivate through specific educational strategies and capacity building) Use local language (U)
Illustration	"I have a sweet little old lady, she's probably close to 80; she won't speak it, she won't speak English, she understands you, she can speak English, but she would rather speak in her own language." (Pg 6)
Finding	(Embed cervical cancer screening into community events) Integrate screening into wellness events (U)
Illustration	"So if you're having something that they're able to come and access, [like a] sewing circle, or community kitchen, something that they're gonna get, then they'll come." (Pg 6)
Finding	(Embed cervical cancer screening into community events) Screening on days that honour women (U)

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Illustration	"I would concentrate on a day that's important to the women, like Mother's Day. The mothers would all come and the grandmas would come and the aunts would come. I would have guest speakers come in as well and there would be a dinner, a luncheon, or a feast of some sort. I would bring somebody in who had a little bit of charisma like a Tai Chi instructor who cooks meals" (Pg 6)
Finding	(Embed cervical cancer screening into community events) Small incentives (U)
Illustration	"It's hard to get people coming unless we have food and incentives." (Pg 6)
Finding	(Health care providers need a trusting rapport with women) Friendly Relationship (U)
Illustration	"The women are, you know, fairly open once they're comfortable." (Pg 6)
Finding	(Health care providers need a trusting rapport with women) Trusting Relationship (U)
Illustration	"It's all about trust and relationships, like if you can't build a relationship you're not gonna get anywhere, you might not even see them again." (Pg 7)
Finding	Integrate cervical cancer screening recruitment into existing health care services (U)
Illustration	"Well, since you're here, you know, we haven't screened for this, and when was your last Pap and what about colon cancer screening?" (Pg 7)
Finding	(Create a dialogue on cervical cancer screening in day-to-day activities) Sustain screening culture with HCP (U)
Illustration	"For our little group of work girls, [the] service team, we always tend to remind each other as well, like, "Hey, is it time for your check-up yet, like?"" (Pg 7)
Finding	(Create a dialogue on cervical cancer screening in day-to-day activities) Word of mouth reminders (U)
Illustration	"I think it's just sort of word of mouth, spreading things around, and just kind of bringing it up, you know, all the time." (p. 7) "I call four people and then each one of the people I call, they call four people and it's something that we're trying to work on." (Pg 7)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Screening as a way for women to re-gain teaching roles (C)

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Illustration	"The regular program for girls meets every second Monday, and it's called the Moon Time Girls." (Pg 7)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Holistic approach for cultural integrity (U)
Illustration	"Well, word it a way to have to do with childbearing, [which] is a sacred gift that's only given to women, let's keep our bodies healthy by getting annual check-ups, and you use some culturally appropriate like pictures, or something." (Pg 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Using non-invasive screening (U)
Illustration	"Informant #14: I think that's the best idea that has come out of anything, like it's not invasive, you do it in your own home, at your own time... I.Z.: Whenever you, you know, are ready for it and... Informant #14: I think that with education along with it that this is what it is, I think that the success rate will be phenomenal with it because it's in their own home." (Pg 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Multi-purpose screening (U)
Illustration	"If it [other STI testing] can be done on the same swab [as self-sampling], I would say, the value of that would be huge." (Pg 8)
Finding	(Incorporate First Nations gender perspective on body and preferred screening techniques to enhance privacy and comfort) Preference of female HCP (U)
Illustration	"The girls are always looking for female doctors" (Pg 8)
Study: Allen-Leigh B et al 2017	
Finding	Beliefs and knowledge about cervical cancer (Correct knowledge) (U)
Illustration	"If you detect it in time, you can take that sickness out so it won't progress any more and with a treatment you are fine." (Pg 4)
Finding	Beliefs and knowledge about cervical cancer (Incorrect knowledge) (U)
Illustration	"Between these two sicknesses [HPV and cervical cancer] we're in danger, we should go to the clinic or a doctor. If we feel pain in the womb, go to a doctor." (Pg 4)
Finding	Perceptions of and knowledge about HPV (Correct knowledge) (U)

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Illustration	[Cervical cancer is caused] "by infections, because later they become like a tumor in the womb." (Pg 4) "There is a vaccine for human papillomavirus. I heard it on the radio, that there are vaccines." (Pg 4)
Finding	Perceptions of and knowledge about HPV (Combined correct and incorrect knowledge) (U)
Illustration	"Well, as far as I know, the virus is transmitted through sexual contact. Then this human papillomavirus, it begins without any warning. Then later it progresses, then the discomfort in our parts [genitals] begins and that's when the discharge starts and it progresses to the cervix and when it gets to the cervix it goes into the uterus and that is when the doctor sends us for an operation." (Pg 4)
Finding	Perceptions, knowledge and experiences related to cytology (Combined correct and incorrect knowledge) (U)
Illustration	"Moderator: What is cervical cancer, what do you know about cervical cancer? Huichol woman 1: They say you can die of cancer, if you don't detect it early. Moderator: And how do you detect it? Huichol woman 1: With the Papanicolaou, doing it periodically. Moderator: And what's periodically? Huichol woman 1: Every three months. Moderator: Everyone, how often do you think you need to get a Papanicolaou? Huichol woman 2: Once a year. Huichol woman 3: Depends on how you feel, once a year or every two years, I get it every two years. Moderator: What do you mean, how you feel? Huichol woman 3: If you feel burning." (Pg 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Pain and fear of unsterile equipment as a barrier) (U)
Illustration	"[When they did the Papanicolaou test], maybe they did it wrong. I don't know, but I felt something like a scrape. Then I thought maybe the equipment wasn't disinfected. ... I thought about it a lot before doing it again, because I was afraid." (Pg 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Not receiving test results as a barrier) (U)
Illustration	"A lot of us say, and we've talked about this before, when we get a Papanicolaou, the results don't arrive, and we don't know what it is that's going on. There we are with the doctor, asking why the results don't arrive. ... Whatever it [the result] is, they should give it to me." (Pg 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Unequal gender relations as a barrier) (U)

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Illustration	"The first time I went to check myself, with the Papanicolaou tests, I had problems. I got beat-up. My husband hit me because he said I had gone to do things with the [male] doctor. When it wasn't even a doctor who examined me, the [female] nurse examined me! She took the sample, but at home my husband didn't believe that." (Pg 4)
Finding	Barriers to cervical cancer screening using cytology or the HPV self-sampled test (Lack of gender-related barriers) (U)
Illustration	"Because with all these [health education] talks they give us, women are more secure in themselves. So, we don't really ask men for permission anymore, because it's something that's good for us." (Pg 4) "I decided it [to perform the HPV test] myself, alone. I don't ask anyone's permission. ... How am I going to ask him if he [her husband] wants it or not? It's not for him, it's for me." (Pg 4)
Finding	Perceived advantages of the self-sampled HPV test (Less embarrassment) (U)
Illustration	"This one [the HPV test] is good because it [cytology or the Papanicolaou test] really does embarrass you, not because your husband doesn't want it or doesn't let us, but because it's embarrassing and because of embarrassment we don't do it, and so this [self-sampled HPV test] is good for us." (Pg 5) "Because you do it yourself, since always, even if there is trust, you feel a little embarrassed to undress in front of someone else" (Pg 5)
Finding	Perceived advantages of the self-sampled HPV test (More comfortable) (U)
Illustration	"Well, this one [the self-sampled HPV test] is better, because it is more comfortable to do it." (Pg 5)
Study: Cerigo H et al 2012	
Finding	Experiences, Attitudes and beliefs about cervical cancer (C)
Illustration	A sizable proportion of the women were unable to identify a cervical cancer risk factor and were unsure if detecting cervical cancer early would affect the chance for a cure. (Pg 5)
Finding	Pap smear history, understanding and experiences (female preference) (U)
Illustration	""Some people don't go [to get Pap smears], maybe they're shy … all of my friends don't go probably because they are shy or sometimes they don't want to [be] checked by a man, if it is a man that is a nurse." (Pg 3)

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Finding	Pap smear history, understanding and experiences (embarrassment) (C)
Illustration	Feelings of embarrassment and pain during Pap smears were common among our study population and others. Among our population, older women reported more feelings of embarrassment than the young women. (Pg 5)
Finding	Pap smear history, understanding and experiences (lack of knowledge) (C)
Illustration	Further, we found that some women did not fully understand the purpose of the Pap smear as a method of cervical cancer screening, as 30% reported that the purpose of the Pap smear test is to screen for STIs. (Pg 5)
Study: Zehbe et al 2017	
Finding	Educational implications (contribution to stereotype) (U)
Illustration	Focus group participants were concerned that any publication of information about HPV rates on reserve could have negative effects, contributing to the stereotype that 'all native people have HPV' (FG6). As one HCP summarised: 'A lot more education has to be done about HPV and cervical cancer together' to alleviate these concerns (Pg 7)
Finding	Educational implications (women blamed for STI's) (U)
Illustration	"'HPV, like, whoa, I don't have that, like, I don't even want to know if I have that" (Pg 7) 'because it's a relationship thing' that implied that someone had 'cheated' (Pg 7) '…. do they [the men] not need to know some of this stuff too?' (Pg 7) '… I know only women get cervical cancer but … they've helping it along in the same sense where they've passing it back and forth' (Pg 7)
Finding	Educational implications (need to explain self-sampling) (U)
Illustration	HCPs felt that education around self-testing might better be done in the context of a well woman healthcare visit, as 'part of a physical, that might get best [results]' (Pg 7) Women would not necessarily understand that they would still need to go to their HCP for other reproductive healthcare: 'They would, they'd probably think, well, I've had that done, so I don't need this done'" (Pg 7) "If you hand me a kit, I won't touch it. Because I wouldn't know what to do, what if I did it wrong, or whatever, right, so? And I think a lot of

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	people would be that way" (pg 7) 'you just give a test kit to them at home; the majority of them are just going to throw them out.'(Pg 7)
Finding	Convenience factor self-sampling (no appointment needed) (U)
Illustration	'It is more comfortable to do it at home … it's simple'. (Pg 6) I think the self-test is beneficial for them all because sometimes people don't have time for appointments to take off work, it [the Pap] is kind of an inconvenience' (Pg 6) 'would go for it' (Pg 6) older woman who had fewer constraints 'were used to getting Paps' (Pg 6) "Even hearing [about self-testing], people are just, What? Oh, I'd do that, for sure, instead of me going to the doctor' " (Pg 6) "Doing a self-test doesn't take very long … it's something that can be dealt with, done, gone' (Pg 6) " I really strongly believe that … because I was the one that was doing it [self-sampling], I was the one that was in control … and this way it gave me the ability to do it myself and I got all the results, they were fine; … it was also self-empowering, great, I like that." (Pg 6)
Finding	Convenience factor self-sampling (more privacy) (U)
Illustration	"I think more people would monitor it that way and test it themselves, like 'well maybe not today, but … eventually I'm going to try it' (Pg 6) option of taking it home might make women 'feel more empowered' and contribute to a better relationship with HCPs. (Pg 6) An option, they can either do it themselves, or you can offer to do it for them' (Pg 6)
Finding	Psychological comfort (high comfort with self-sampling) (U)
Illustration	"doing it yourself would be better" "would feel more comfortable with that [self-test] than a male doctor" (Pg 6) 'shy' of doctors or nurses who visited the communities. (Pg 6) Self-testing would also address the 'trust issues' that discouraged women from seeking care from non-indigenous HCPs. (Pg 6)
Finding	Psychological comfort (male providers) (U)
Illustration	"feel more comfortable [with a female physician or nurse practitioner] because maybe she's going through the same thing that you're going through." (Pg 6)

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	“embarrassed anyway, no matter who did it (Pg 6) “I only got a Pap test once and I got it from a woman … and she was trying to make me feel comfortable, but it’s still really awkward” (Pg 6)
Finding	Physical comfort level (self-sampling is less painful) (U)
Illustration	"Women wouldn’t be so agitated and nervous about having the [self-] test’"(Pg 5) "It’s a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference.’ ‘…I think the prevalence [participation] rate would go up’ (Pg 5) "It’s a lot less clinical … stripping down and allowing someone else to do the scraping of the cervix, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference.’ ‘…I think the prevalence [participation] rate would go up’" (Pg 5)
Finding	Physical comfort level (paps hurt) (U)
Illustration	“it’s kind of a routine after a while … they open you up and they swab and then you’re done” (Pg 5) “the Pap is uncomfortable but it has to be done” (Pg 5) “it doesn’t get any easier, like the first time and then the next year, it didn’t get any easier for me” (Pg 5) “I didn’t like the way it felt, that’s why I didn’t want to go back there’ They don’t want their Paps [be]cause it hurt.” (Pg 5) The test also could be exceptionally painful if providers were ‘in a rush’ or ‘rough’ and did the procedure ‘real quick’ (Pg 5) “Some women, when they come to the appointment, they decide they don't want to get a Pap, because it's uncomfortable, they're just afraid” (Pg 5)
Finding	Cervical Screening practices in partner communities (privacy concerns) (U)
Illustration	"health centres where staff ‘phoning people, telling them, ‘we’ve got your results … you need to make another appointment’ (FG2) was upsetting.” (Pg 5) “I don’t want nobody else to know’ ‘a lot more private, at home, if you do it by yourself”(Pg 5)

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	"Women would prefer to be discreet, do it themselves and get their own results and not have their results shared with others" (Pg 5)
Finding	Cervical Screening practices in partner communities (no efficient follow up) (U)
Illustration	"It's crazy here, how long you have to wait for anything and half the time, you don't even get a call … you go to your next doctor's appointment for something else and [have to ask] … by the way, how about my Pap two months ago?" (Pg 5)
Finding	Cervical Screening practices in partner communities (transportation and child care issues) (U)
Illustration	"If the van was full, [you would] have to get a ride … some of us don't have cars, you know" (Pg 5) Women with children and work commitments also had difficulty arranging childcare and time off work to keep their appointments. As a HCP commented: 'It certainly does pose challenges with regards to babysitting care and the mother being away for an entire day' (Pg 5)
Finding	Cervical Screening practices in partner communities (lack of provider flexibility) (U)
Illustration	"I don't know how many times they've cancelled Pap tests and [when I asked] "okay well can I get it on Friday? …. [heard back] "No, we don't do them on Friday"' (Pg 5)
Finding	Cervical Screening practices in partner communities (no regular screening service) (U)
Illustration	"'Most of us do not have doctors here … you have to drive up to [nearby town]'" (Pg 4)
Study: Adcock A 2019	
Finding	Engagement with health services (C)
Illustration	One in seven (14.36%) had not screened in ten years or more (including those never screened), and one in six (16.37%) were unsure how long it had been since their last cervical screen. (Pg 2)
Finding	Barriers to screening (shyness, embarrassment) (C)
Illustration	Desire for bodily autonomy (retaining privacy, control over one's body) as a reason for not attending regular cervical screening. (Pg 3) encompasses concepts of whakamā (embarrassment/shyness/reticence), (Pg 3) tapu (sacred/taboo/forbidden) (Pg 3)

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	desire for bodily autonomy was often related to negative health experiences (eg, painful pelvic examinations, inappropriate actions/comments by HCPs) (Pg 3)
Finding	Barriers to screening (cost) (C)
Illustration	Older women were more likely to mention a previous bad experience and were less likely than younger women to mention cost or other financial barriers (Pg 3) Opinions varied about cost being a barrier (as many clinics offer free cervical screening). (Pg 3) Some highlighted hidden costs (transport, parking, childcare) (Pg 3)
Finding	Barriers to screening (education) (C)
Illustration	A lack of health literacy about HPV and cervical cancer and a lack of appropriate/empathetic services were also raised as barriers. (Pg 3) relationship-building between communities and health promoters; including whānau/family in HPV education; and ensuring clear information about HPV vaccination (Pg 3) HCPs agreed that with appropriate support and education, HPV self-testing will benefit never/under-screened Māori. (Pg 4)
Finding	Acceptability (U)
Illustration	positive, with participants using terms such as ‘easier’, ‘more comfortable’, ‘less intrusive’ and ‘brilliant’. (Pg 3) Nearly two-thirds (61.21%) said they would prefer an HPV self-test to a clinician-collected vaginal swab or a cervical sample collected with a speculum, or to other options. (Pg 4)
Finding	Implementation (C)
Illustration	picking up the HPV self-test from a clinic, a pharmacy or other community venue, or receiving it by post/mail (Pg 4) concerns were raised about the reliability of post. Participants discussed the value of providing multiple options (Pg 4) Flexibility to cater to diverse populations, such as through community outreach services, was suggested for optimum engagement. (Pg 5)
Finding	Doing the self-test? (U)
Illustration	most frequently said they would be happy to do the HPV self-test in a clinic or their own home, and emphasised having good support and education to increase their confidence about properly doing it. (Pg 5) need for a flexible program, with different options. (Pg 5)

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	Urban women were more likely (than rural) to say they would self-test at a community centre. (Pg 5) One in five said they would like whānau/family or friends to be around in case they needed help, while some (<15%) said they would like either a community health worker or health practitioner to be nearby, available to help. (Pg 5) importance of delivering empathetic cervical screening services, ie 'the process' – taking the time to put mind and body at ease. They emphasised that this process will be important for any new HPV-based cervical screening program. (Pg 5)
Finding	HPV self-test results (C)
Illustration	if negative a text is fine, but if positive it is better to find out in person or by phone. (Pg 5) One women emphasised the importance of results being delivered in an empathetic way.(Pg 5) Older women were less likely than younger women to want to receive a text asking them to call their health practitioner.(Pg 5) Non-PHO-enrolled/unsure women were less likely to choose to receive results by email and more likely to choose a visit with a health practitioner need for clear communication (Pg 5) appropriate/empathetic support for women. (Pg 5) Concerns were raised about who would be responsible for HPV self-test results and making sure that women are added to the screening register when doing a self-test.(Pg 5)
Finding	Positive test results (C)
Illustration	mostly positive response to the idea of seeking further screening (cytology) or diagnosis (colposcopy) if a HPV test was positive – because knowing that something needs follow up would be a good motivator.(Pg 5) HCPs wanted assurance that women would be supported appropriately and empathetically to have cytology or colposcopy if they had a positive HPV self-test result, and many supported going straight to colposcopy.(Pg 5) Eliminating multiple clinic visits was stressed. (g 5)
Study: Henderson R et al 2018	
Finding	(Cancer & HPV Experiences) Disease Progression (U)
Illustration	"Once they corrected the cervical cancer you would end up with cancer somewhere else" (Pg 96)

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Finding	(Cancer & HPV Experiences) Isolating Effects (U)
Illustration	“It took time for one family to realize that it was ‘not just happening to them, it's the whole community’” (Pg 96)
Finding	(Cancer & HPV Experiences) Mistrust of external influences (U)
Illustration	“It must be the pesticides that are being sprayed on our cultivated land...You can see our area and there is a layer of dust in spring and harvest” (Pg 96)
Finding	(Cancer & HPV Experiences) Residential school disruption of community relationships (U)
Illustration	“Opened a whole can of worms; when all the illness and substance abuse went on a skyrocket” (Pg 96) “Speaking of a sister in the end-stage of cervical cancer, one middle-aged participant attributed the illness to chronic sexual abuse: “There were a couple times she had STIs [sexually transmitted infections] and didn't really know, may have been HPV or may have been another” (Pg 96) “In both sharing circles, participants argued that residential school litigation broke students' capacity for intimacy throughout their lives, playing out in self-destructive coping strategies (e.g., substance dependence, early sexualisation) and increased risk of victimization)” (Pg 96)
Finding	(Trauma-informed lens) Contributions of higher Indigenous HPV burden (U)
Illustration	“Our people are starving for affection, support, respect and love.” (Pg 96)
Finding	(Trauma-informed lens) Residential schools (U)
Illustration	“The nuns make sure to remind you no matter what you do, you're going to hell” (Pg 96) “We were taught to be quiet about our private parts...a lot of sexual abuse went on, and spiritual, physical, you name it” (Pg 96) “One speaker brought these lingering effects of residential schools together by describing a shift in her society from monogamous relation- ships to relaxed sexual boundaries, noting that with her own daughters she “didn't lay down the law like our parents did” (Pg 96)
Finding	(Trauma-informed lens) Normalization of substance abuse among youth (U)
Illustration	“She got an STI because she was taken advantage of” (SC2) by someone other than her partner. The mother connected her daughter's

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	exposure to loss of identity and heavy drinking at parties in her community. Lowered inhibitions brought on by alcohol, and illicit drug use were seen by several other speakers to have become normalized among younger generations, who "see the suffering of their parents and grandparents, and they are running to the substance abuse" (Pg 96) "In both circles, speakers expressed compassion for younger generations today: "It was a lot easier in my time because we would go to ceremonies...but today, our grandchildren are exposed to drugs, alcohol, everything" (Pg 96)
Finding	(Trauma-informed lens) Isolation (U)
Illustration	He observed that such statistics reflect disrupted connections between partners, between Elders and youth, as well as with nature and spirituality: "We are not islands, we need to be connected to people and that is what is missing" (Pg 96) "Therefore, the burden of HPV in FN people is rooted, at least in part, in efforts by people across the lifespan to cope with the violent disruption of family and community connectedness, as well as connection to land and spirituality. This burden is manifested not only in risk-taking behaviour, but also in avoidance of wider health systems: "I think First Nations don't get checked when they are supposed to, to be honest. They just wait until it is too late to help them" (Pg 97) Coming from a communal society, another participant observed that today many struggle "to fit into an individualistic society, and we don't fit" (Pg 98)
Finding	(Trauma-Informed lens) Child welfare (U)
Illustration	"A lot of those kids when they come back to the reserve they are very angry, there are a lot of things that happened [to them] that we don't know about" (Pg 97) "It's sad to say that lots of babies have been taken because their mothers were on opium, and the mothers are in treatment and are so sorry" (Pg 97)
Finding	(Family & Community) Communication (U)
Illustration	"I am learning a lot in these workshops. My mother died of stomach cancer, my sister of stomach cancer. I had 5 girls, and 4 of them went through breast cancer. My oldest daughter, her cancer spread. When they were younger, I made sure they all got their needles. But, you know, I have never had a workshop like this. If I get a cold, I can fight it off. When I got those needles, I was told I was able to fight the sicknesses; it won't kill you—that is what I was told. This is really good for my grandchildren; I will take this message home to my

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	family. I have two nurses in my family, they probably know about it, but this is a really good thing I am still learning...[*speaking in Cree] *I was worried the white people would not take care of us, but they have so far [group laughs]. We need to talk to young ladies about how to take care of themselves." (Pg 97)
Finding	(Family & Community) Family (U)
Illustration	Many of the Elders emphasized that health education within Indigenous contexts is anchored in the love and care for children. For one speaker, this affection involves "hugging your children and telling them 'I love you', you don't say goodbye to anyone" (Pg 98) "If I'm drinking and drugging, then I'm going to feel guilty about taking my children to the clinic to help them live a healthier life" (Pg 98)
Finding	(Family & Community) Role of elders (U)
Illustration	"[My husband] goes and sits with the men and teaches them; for me, I can go out and explain things to the mothers and the children, and out of respect...he's getting that message across [in his sweat lodge ceremonies with men]" (Pg 98)
Finding	(Family & Community) Mistrust in HCP (U)
Illustration	"We need to talk to our people in a way they understand, no disrespect to any organization, but people don't look at that material [brochures], it's obvious that the way we deliver that message has to be different" (Pg 98) "The health centre sent out a notification and a consent form, and they listed the benefits and risks...and I paid more attention to the risks, and I decided not to allow her to be vaccinated, because as a parent I needed to do what was best for my children" (Pg 98)
Finding	(Family & Community) Youth (U)
Illustration	"There are anti-bullying programs...and there are cultural programs; some kids are brave and some are afraid of getting immunized, but all the children support each other" (Pg 98)
Finding	(Family & community) Parents & grandparents (U)
Illustration	"The best teachers are your parents" (Pg 98)
Finding	(Changing information landscapes) Lack of knowledge (U)
Illustration	"We never see a doctor. The health department has to get a hold of us, the ones that are never home. There are a lot of us in the community... We hear suddenly of these workshops and information centres in the community. We wanted to go but we had other commitments so

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	needed others to go. My niece, she died about 3 years ago because she had cancer. They brought her to Edmonton, they started her on chemo...left her kids at home." (Pg 97) “One of the girls I used to visit got cervical cancer. She is very angry; she will not talk to me even though I give her support. She was told 3 years ago that she should get the tests and she didn't because she had problems for a long time.” (Pg 97) One health director who was herself FN had not approved the HPV vaccine for her own daughter, believing at the time that vaccines are perhaps “not natural, that they are more chemicals given by the government to hurt us” (Pg 98)
Finding	(Changing information landscapes) new knowledge (U)
Illustration	“These young people are lucky to get these different resources...its scary when you think about it, you didn't think about [the health risk] before, because you didn't know anything about it” (Pg 98)
Study: Wakewich P et al 2015	
Finding	Negative body perceptions (stigma of sexually transmitted infections) (U)
Illustration	"Definitely there’s always going to be a stigma about any kind of thing that’s an STD [STI], because people think it’s dirty or whatever. Like I’ve had people come in, like, even the girl I just saw with genital warts and she just couldn’t believe it, and couldn’t like fathom who she would have got it from because everyone she’s been with has only been with her. Right? " (Pg 373) "Actually in the last 6 months that I’ve been there, I’ve had uh, a minimum of 5 people who have come to me who um, have been concerned about contracting STIs and did not want me writing it into their chart for fear that it would leak into the community … they’re afraid of the stigma surrounding that so and a lot of times too I wasn’t able to do any [cervical] screening." (Pg 373) "Health centres on reserves, don’t want to spread it out too much how many people have HPV. I know how the stereotype works if we pass on the information in non-native communities they will say all native people have HPV." (Pg 373)
Finding	Negative body perceptions (body shyness) (U)
Illustration	"I remember feeling so vulnerable, just so extremely vulnerable and, and I remember at one point he [the male physician] was talking on his phone while I was up in the stirrups and I thought uh, you know, I wonder how many other patients feel like this …. There’s got to be a better, kinder, gentler, more humane way to do it." (Pg 373)

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Finding	Negative body perceptions (sexual abuse) (U)
Illustration	"They have been sexually abused, too, and I know like, in the past residential schools, that kind of thing, those people are just not comfortable because of their experiences in the past …. I will be here for a long time and whenever you need to see me, to come see me, so that even just that little thing and then when they do come I do see them, hopefully that trust builds up and I think that's a big piece with the First Nations." (Pg 374) "I think one of the biggest issues um, [short pause] the barriers that prevents people from going to and maybe it's cause it's taboo is because they've been sexually abused." (Pg 374)
Finding	Negative experience with governmental health service (distrust of health authorities) (U)
Illustration	"She [HCP] thinks it's her [that they don't like], 'like, why is no one coming [for screening]'… 'it's not you, it took me 20 years to get to where I'm at now, and I still get people that don't trust me.'" (Pg 374) "I guess they were told about some kind of vaccine or something, like years ago, and it was something just, to try to get rid of the Native people." (Pg 374) "My doctor who I had since my oldest was born, she just up and quit, so when my next appointment, I had this guy looking at me, it's like [pause] right, so it's not like they stick around here." (Pg 375)
Finding	Negative experience with government health service (lack of confidentiality and privacy) (U)
Illustration	"You know what, just sitting in the waiting room, the receptionists are phoning people telling them, we've got your results, you need an, you know, make another appointment, they say the name right out loud, they say what the test is." (Pg 375) "Privacy, a small town, I mean, you can hear through the walls, you know, walls talk because there's, it's all in one building and coming in here and everybody sees the first person coming in here, they come to see me, they come to see welfare, they come to see housing, they're going and everybody knows everything."(Pg 375) "Being in reserves, a lot of people know people's business and a lot of people get worried about that when you're trying to keep something personal. I mean the teddy bears talk, the leaves talk, the hydro lines talk." (Pg 375)
Finding	Role of family (mother - daughter relations) (U)
Illustration	I just taught them that sex isn't, it's not what you call dirty, you know, but there's a certain way you got to go about having, like for sex with

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	a partner, you know, you got to explain that part to them, but to talk openly about sex, it's not what you call a dirty subject, yeah. If you want to know something just feel free to ask." (Pg 376) "Like my mother, you know, was the kind of woman that … when I first got my period, I didn't know what it was because that was something you don't talk about, your body like that. It's you know that's your own personal, private, so when it happened, I had absolutely no knowledge what was going on in my body … And with my daughter, I didn't want her to have that feeling, so I mean, of course I changed and I explained everything but, and then that's the way things are now, women are more informed than they have been in the past." (Pg 376)
Finding	Self-collected sampling as a way of increasing women's control of cervical screening (U)
Illustration	"think it [self-sampling] would be, I think it would be great, because it provides them with some autonomy and allows them to take control of the situation. That would be really great actually …. I think it would increase [education opportunities] actually, yup, because it's a lot less clinical right, cause like you say, it's a lot less you know, stripping down and you know, allowing someone else to do the scraping of the cervix, you know, the whole uncomfortable procedure of going through a Pap opposed to doing it privately in the bathroom on your own is a huge difference." (Pg 376) "I think it would definitely be more private … They wouldn't be so embarrassed … all kinds of people are easily embarrassed." (Pg 376) "It's simple, it's not like you are having forceps in you." (Pg 376)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page – Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract – Page 2,3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Background - Page 3,4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Background – Page 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Materials and methods - Page 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Methods – Page 5,6,7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Data sources and Searches in Methods section - page 5 and 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Data sources and

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			Searches in Methods section - page 5 and 6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Inclusion and Exclusion criteria – Page 7 Critical Appraisal – Page 8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Critical Appraisal – Page 8 Data extraction and quality assessment in methods section page 7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and synthesis in methods section page 8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	NA
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Themes and

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			illustrations – table 4 Pages 9-16
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Pages 9-16 and SEM model figure 2

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA for qualitative reviews
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Meta regression and socioecological model on page 16
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	NA in qualitative systematic review
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	All forest plots and study data summarised in tables and supplementary files provided
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results pages 9-16, socioecological model figure 2
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA in qualitative systematic review

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Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Socioecological model figure 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion page 16, 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion ending page 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Declaration page 20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Declaration page 13

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